

Towards Multicomponent Diafiltration: Mathematical Modeling and Regression Notes

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Abstract

This document develops and applies a model-based data-analytics workflow for characterizing surface-charged nanofiltration (NF270) membranes from single-salt diafiltration data (NaCl, CaCl₂, LaCl₃). Section 1 states, with mathematical precision, both the *membrane transport model* — an extended Nernst–Planck / Donnan-equilibrium description whose regressed quantities are the membrane fixed charge χ and a thermodynamic partition factor δ^* — and the *statistical methods* (nonlinear least squares, linearized and nonlinear confidence regions, profile likelihood, bootstrap, and multi-start optimization). Section 2 applies the workflow to the NaCl (1:1) data: reproducing and diagnosing the collaborators’ MATLAB regression; quantifying parameter uncertainty and **correcting it for the autocorrelation** inherent to dense time-series data (i.i.d. confidence regions are $\sim 2.5\text{--}2.7\times$ too narrow, concentration polarization (CP)-corrected); developing a validated raw→processed preprocessing pipeline and regressing all nine available NaCl experiments (with a full per-dataset appendix); and **reformulating the estimation directly on raw laboratory measurements**, which shifts the fixed-charge estimate by +140% (CP-corrected) and exposes a formulation-sensitivity question left open for the group. CP shifts NaCl’s headline $|\chi|$ from $\approx 50\text{ mM}$ to $\approx 57\text{ mM}$ — the same order of magnitude, not a change in scientific conclusion. Sections 3 and 4 extend to CaCl₂ (2:1) and LaCl₃ (3:1): both still drive the fitted charge toward zero (or, for one CaCl₂ dataset, to a small but statistically nonzero value $\ll$ NaCl’s) with strong, systematic residual structure, consistent with concentration-dependent counter-ion adsorption that a constant-charge model cannot capture — the valence-dependent trend survives CP. Section 5 synthesizes the three salts and the recurring methodological lessons; Section 1.2 consolidates every open decision into a single discussion agenda. Section 7 records how the analysis was carried out with Claude Code. Items for collaborative discussion are flagged in **red** and **blue**.

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1 Mathematical model and study goals

1.1 Scope and goals

We characterize NF270 membranes (negatively charged above pH 3) from diafiltration experiments on single salts — NaCl (1:1), CaCl_2 (2:1), LaCl_3 (3:1) — and aim to extend the approach to multicomponent (multi-salt) feeds. The central regressed parameters are the membrane **fixed charge density** χ and a thermodynamic **partition factor** δ^* , which enter a Donnan-equilibrium model of ion partitioning between the external solution and the membrane phase. The workflow builds directly on the DATA [1] and DATA 2.0 [2] frameworks, replacing DATA 2.0’s *empirical* concentration-dependent partition coefficient with a *mechanistic* Donnan model. (The same Dowling/Phillip DATA3 program has also produced a soft-sensor conductivity map [3], §2.4, and related manufacturing work on reactive-ink membrane functionalization [4].)

Physical primer. In diafiltration, a salt solution is stirred against one side of a thin polymeric membrane under applied pressure; water and ions pass through to the permeate side at different rates, driven by the pressure (convection) and by concentration differences (diffusion) across the membrane. NF270 carries a negative surface charge, so it electrostatically excludes *co-ions* (same sign as the membrane, e.g. Cl^-) more than it excludes/attracts *counter-ions* (opposite sign, e.g. Na^+ , Ca^{2+} , La^{3+}) — a Donnan-equilibrium effect. What is physically measured, per experiment over time: the cumulative permeate mass (its rate gives the water flux J_w), the retentate- and permeate-side conductivities κ_r, κ_p (converted to concentrations via a calibration), and per-vial ICP-OES elemental concentrations (a direct, high-accuracy measurement used to calibrate/validate the conductivity route). From the feed- and permeate-side concentrations at the membrane interface (c_{int}, c_p) we regress two membrane properties that do not depend on which salt or concentration was used: the **fixed charge density** χ (how strongly and densely the membrane’s own surface groups are charged — more charge means stronger co-ion exclusion and better salt rejection) and a **thermodynamic partition factor** δ^* (a non-ideality correction for how favorably or unfavorably each ion partitions between the external solution and the membrane’s internal nanoporous phase, beyond the ideal Donnan-equilibrium prediction). These two numbers, once estimated from a single-salt run, are what let the model predict rejection for other salts and concentrations without new experiments — the point of this workflow. A handful of statistical terms recur throughout and are used precisely, so we define them once here: a parameter is **identifiable** (or the data “pin it down”) if the data actually constrain it to a narrow range, as opposed to many very different values fitting equally well; two parameters are **confounded** if they enter the model only in combination, so no amount of data from this experiment can separate their individual values (a different problem from noise — it is a structural limit of the model/data combination); **autocorrelation** means successive measurements in a time series are statistically related to their recent neighbors rather than independent, which is expected here since each run is one continuous, densely-sampled time series, not repeated independent trials, and it makes naive uncertainty estimates too narrow; the **effective sample size** n_{eff} is the number of *independent* data points the autocorrelated series is worth, which can be far smaller than the raw count n ; and **concentration polarization (CP)** is the local buildup or depletion of solute right at the membrane surface caused by the permeation flux itself, so the concentration actually seen by the membrane differs from the bulk solution concentration.

Data provenance. The raw experimental campaigns analyzed throughout this report were run by two group members: **Laurianne Estrada** (NaCl MC5 05.27.26 and CaCl_2 MC5

05.27.26/06.27.26; also author of the multi-salt Donnan MATLAB script, `Donnan_Multiple_Salts.m`) and **Jonathan Ouimet** (NaCl MC2 05.07.24, CaCl₂ MC3 07.11.24, and the sole LaCl₃ run, MC2 05.21.24). Below, “Estrada’s”/“Ouimet’s” processed sheets and calibrations refer to their respective datasets; the soft-sensor conductivity models of §1.7 are joint work by Lilonfe, Estrada, Singh, Ouimet, Phillip, and Dowling [3]. “Bill’s MATLAB” (Phillip’s original NaCl regression script) is named throughout where reproduced.

The measured per-vial data (Box: `Analyses/BoE Analysis.xlsx`; twin `Rejection_Analysis.xlsx`) are, by column:

Col	Symbol	Quantity
B	c_{int}	retentate (feed-side) interfacial concentration [mM]
D	I	ionic strength [mM]
F	R	observed rejection [-]
H	c_p	permeate-side interfacial concentration [mM]
J	J_w	water (volumetric) flux [$\text{m}^3 \text{m}^{-2} \text{s}^{-1}$]
K	B	solute permeability coefficient [$\mu\text{m s}^{-1}$]
M	t	experiment time [s]
N	κ_r	retentate conductivity [$\mu\text{S cm}^{-1}$]
O	κ_p	permeate conductivity [$\mu\text{S cm}^{-1}$]

The interfacial concentrations c_{int}, c_p are *derived* (via a concentration-polarization model, §1.5) from bulk retentate/permeate concentrations, which in turn come from ICP-OES and/or the conductivity soft-sensor. This processing layering is precisely what §2.4 unwinds (regressing directly on the raw measurements instead).

1.2 Open decisions for the collaborative meeting

This is a single, walk-down-able agenda consolidating *every* open decision raised anywhere in this document — modeling conventions, the preprocessing pipeline, and the newer statistical/methodological questions from the AR(1) and raw-measurement work (§2.3–2.4). Every item is also flagged in place in **red** ([**T****O****D****O****:**]) or **blue** ([**()**]) where it arises; this list exists so a reader does not have to hunt for them. For each item: *where* it is flagged, *what* the question is, and a *candidate resolution or option set* (not a decision — these are for the meeting).

A. Modeling conventions.

1. **Sign / notation conventions** (§1.3). The source documents disagree ($\chi \leq 0$; δ as exponent vs. exponential). *Candidate:* we adopt $\chi < 0$, $|\chi|$, $\delta_i^\circ = \exp[\cdot]$ (Table 2) to proceed — confirm or override.
2. **The factor of 2** in the MATLAB Donnan root (§2.1). *Resolved as a preliminary-code bug:* it is an exact reparameterization ($|\chi| \rightarrow 2\times$, $\delta^* \rightarrow 4\times$, SSE unchanged) and the corrected form is now primary throughout — confirm the group agrees before this becomes the permanent convention.
3. **Parameterization of non-ideality.** Is δ^* a single fitted parameter per salt, or reconstructed from per-ion δ_i° ? How does δ° enter for $z \neq 1$ (Eqs. 12, 13)? *Current working choice:* CaCl₂/LaCl₃ lump $\delta^* \delta_{\text{co}}^\circ / \delta^*(\delta_{\text{co}}^\circ)^2$ into a single constant K (§3.1, §4.1) — biases absolute K but not the diagnostic shapes; confirm the split with the group (§1.8).

4. **Fixed vs. estimated** auxiliary quantities: membrane thickness ℓ , water fraction ϕ_w , membrane diffusivities D_m — *confounded* with δ^* (i.e. they enter the model only combined together, so the data alone cannot separate their individual values) through $B = D_m H / \ell$ (7) (flagged concretely in §2.1). *Options*: keep fixed at literature/assumed values (current choice) or co-estimate a subset.
5. **Transferability**: does a single-salt χ (and B) carry over to multi-salt mixtures, where H depends on the full external composition (§1.5)? *Unresolved* — this is the central barrier to the multicomponent goal (§5).
6. **Which transport mechanisms** (diffusion / convection / electromigration) are warranted by the data (model reduction M1–M6, §1.5)? *Not yet attempted* — candidate approach is AIC/estimability selection, as in DATA 2.0.

B. Preprocessing pipeline (§1.6).

7. **Membrane area/density**. Is $A_m \approx 3.95 \text{ cm}^2$ (still *inferred*, not confirmed against the stirred-cell spec sheet) and $\rho \approx 1000 \text{ g/L}$ correct? *Still open*.
8. **J_w smoothing**. Is the 51-row centered rolling-window slope (tuned to 3.8% median error against the one reference column) acceptable, or should a finer/different smoother be used?
9. **Calibration source, and whether vial and inline conductivity share an instrument/scale**. The vial-fit and embedded inline conductivity-to-concentration calibrations disagree in both slope and intercept, worse for $\text{CaCl}_2/\text{LaCl}_3$ than NaCl , and a first-principles physical check (Shedlovsky/MSA) sides with the vial-ICP calibration as the more defensible convention — the full numbers, the physical argument, and the **data request** (asking the team to share the $\text{CaCl}_2/\text{LaCl}_3$ inline calibration curves/procedure so this can be settled directly) are given once, in full, in §1.6’s validation paragraph and §1.7 — not repeated here. *Open question for the team*: are the vial “Conductivity @ Temp” measurement and the inline probe the same instrument/scale? **Not relitigated here**: the regressions in §2–4 deliberately use the processed-sheet bulk c_{int} (not vial-ICP) for comparability with prior results and to isolate the CP effect cleanly (item 11); this calibration-source question is therefore a magnitude-only question (it does not change the shape-based qualitative conclusions, §5.1) and is the **third sensitivity axis** in this report alongside CP-vs-non-CP (item 11) and transformed-vs-raw (item 14).
10. **ICP molar mass**. Elemental Na^+ (22.99 g/mol), validated against stated Feed/Diafiltrate concentrations — *resolved*, confirm the group agrees.
11. **CP vs. non-CP – RESOLVED**. CP is now the **standing convention throughout**: computed from raw signals during preprocessing (§1.6 follow-up item 5) and applied to every single-salt point-estimate/uncertainty regression (§2–2.5, §3, §4; `analysis/cp_authors_bulk.py`). The c_{int} shift (median +10–14% NaCl , +6.6–14.9% CaCl_2 , +28.7% LaCl_3) moves parameter point estimates by a comparable order of magnitude ($\text{NaCl } |\chi|: 50.36 \rightarrow 56.74 \text{ mM}$) but **does not change any qualitative conclusion** — NaCl remains well-identified and nonzero, $\text{CaCl}_2/\text{LaCl}_3$ still mostly drive $|\chi| \rightarrow 0$ (one CaCl_2 dataset now resolves to a small but nonzero value, still $\approx 19\times$ below NaCl), and the valence trend survives (§5.1). Non-CP is retained in each results section for comparison. *Primary analysis uses the processed-sheet bulk c_{int}* (not the raw/vial-ICP reconstruction) for comparability with prior DATA3 results and to isolate

the CP effect cleanly – see item 9 for why this is a deliberate choice, not an oversight. The autocorrelation (AR(1)) analysis and the raw-measurement reformulation have **also** been re-run CP-corrected (§2.3, §2.4).

12. **The “(2)”-sheet mostly-pasted column.** Only 22 of 747 rows are a live calibration formula; the rest are hardcoded pasted values that satisfy it only approximately (median 1%, max 38% relative deviation). Does this change how authoritative we treat the sheet (§1.6 follow-up item 4)?

C. Statistical/methodological (from the AR(1) and raw-measurement work).

13. **AR(1) correction convention** (§2.3). Method A (GLS/whitening, actual $n - p$ df) and Method B (n_{eff} heuristic) agree closely ($\sim 2.5\text{--}2.9\times$ i.i.d. CI widening; $2.5\text{--}2.7\times$ CP-corrected, $2.7\text{--}2.9\times$ non-CP) — *which should be the report’s headline convention?* Separately, AR(1) is *not fully sufficient* (the whitened ACF still shows structure, including a negative lag-1, essentially unchanged by CP) — is AR(p)/ARMA worth the added complexity, or is AR(1) good enough for now?
14. **Raw-measurement reformulation** (§2.4). The estimates shift $\sim 2\text{--}3\times$ ($|\chi|$: $57 \rightarrow 136$ mM, CP-corrected, +140%; $50 \rightarrow 115$ mM non-CP, +128% — CP *widens* this gap) between the transformed and raw-measurement (i.i.d.) fits — *is the transformed $|\chi| \approx 57$ mM robust to this errors-in-variables issue, or should the raw-measurement number be taken more seriously?* Neither is fully trustworthy yet: CP substantially improves the raw fit’s AR(1) behavior ($\hat{\phi}$: $0.999 \rightarrow 0.900$, $n_{\text{eff}} :< 1 \rightarrow 36.5$, no longer at the unit-root edge) but does not resolve the point-estimate discrepancy with the transformed fit (which it in fact widens), and $n_{\text{eff}}/n = 5.3\%$ is still severe, so a non-stationary/state-space noise model (plus time-varying calibration/temperature corrections) is needed before either estimate can be adopted with confidence.
15. **Concentration-dependent surface charge, $\chi(c)$.** Both CaCl_2 (§3.3) and LaCl_3 (§4.3) show effective- χ drift with concentration that a single constant cannot capture. *Proposed model extension:* a data-driven $\chi(c)$ (or $\chi(t)$) spline, replacing the constant- χ Donnan closure for multivalent counter-ions.
16. **Multi-salt identifiability.** Conductivity is a single scalar signal, but a multi-salt feed has multiple ionic contributors — separating them needs an MSA multi-salt conductivity model plus per-ion ICP, not yet built. This is the concrete barrier standing between the current single-salt results and the multicomponent goal (§5).

1.3 Notation and conventions (working assumptions)

To proceed we *adopt* the following (matching Bill’s MATLAB and the physical NF270 case); the choice is itself discussion point 1 above:

- NF270 is negatively charged: $\chi < 0$; we use $|\chi|$ and absolute-value valences $|z_i|$ (the group SI instead writes $\chi > 0$).
- *co-ion* = same sign as the membrane (anion Cl^- , *excluded*, partition $H_{\text{co}} < 1$); *counter-ion* = opposite sign (cations Na^+ , Ca^{2+} , La^{3+} , *enriched*, $H_{\text{ct}} > 1$).
- Partition coefficient $H_i \equiv c_i^{\text{m}}/c_i^{\text{s}}$.

- Superscripts s/m = solution/membrane phase; subscripts f/int = feed-side interface, p = permeate.

The per-ion partition factor (Born/dielectric + steric exclusion beyond ideal Donnan) and its salt-level product are

$$\delta_i^\circ = \exp\left[\frac{\mu_i^{s,0} - \mu_i^{m,0}}{RT}\right], \quad \delta^* := \delta_{co}^\circ \delta_{ct}^\circ. \quad (1)$$

Ideal Donnan is $\delta_i^\circ = 1$ ($\gamma_i = 1$).

1.4 Nomenclature reconciliation across source documents

The model was assembled from several documents that use *different* conventions. Table 1 lists the sources (with locations) and Table 2 reconciles the notational conflicts, so team discussions can refer to a single mapping. Box paths are relative to the shared-folder root **Diafiltration Modeling and Experiments (Phillip and Dowling groups)/**; repo paths are relative to **data3_regression/**.

Key	Document	Location	Ref
[S]	Scanned Donnan notes (1:1/2:1/3:1)	repo prior_analysis/Donnan *.jpg	[5, 6, 7]
[M]	Bill's NaCl regression (MATLAB)	repo prior_analysis/B_regression_concentration_NAClanal.m	[8]
[SI]	Supplementary Information (two-salt ENP)	Box .../Manuscript/Supplementary Information.docx	[9]
[Dn]	Thermodynamics, MembranePhase — Donnan	Box .../Manuscript/JAO/Thermodynamics, MembranePhase - Donnan.docx	[10]
[Tr]	Transport, MembranePhase	Box .../Manuscript/JAO/Transport, MembranePhase - Diffusion, Convection, Electromigration.docx	[11]
[BoE]	Back of the envelope calculations for SI	Box .../Manuscript/Back of the envelope calculations for SI.docx	[12]
[MS]	Multi-salt Donnan (MATLAB)	Box .../Analyses/MATLAB codes/Donnan.Multiple.Salts.m	[13]

Table 1: Source documents for the membrane model and their locations.

Quantity	Representation across sources	Adopted here
Fixed-charge sign	$\chi > 0$ [SI]; $\chi < 0$ via $- \chi $ [Tr]; $\chi = -44$ mM [BoE]; $ \chi = \text{abs}(-44)$ [M]; $ \chi $ [S]	$\chi < 0$, use $ \chi $
Partition factor	$\delta_i = \exp[(\mu_i^{s,0} - \mu_i^{m,0})/RT]$ [SI, Eq. 46]; $\delta_i^\circ = (\mu_i^{s,0} - \mu_i^{m,0})/RT$ (the <i>exponent</i>), then $\delta^* = \delta_1^\circ \delta_3^\circ$ [BoE]	$\delta_i^\circ = \exp[\cdot]$, $\delta^* = \delta_{co}^\circ \delta_{ct}^\circ$ (1)
Partitioning variable	$H_i = c_i^m/c_i^s$ [SI, Dn, BoE, MS]; c^m with fit params β_1, β_2 [S, M]	both: H for theory, c^m for regression
Regression parameters	β_1, β_2 [M]	$\beta_1 \leftrightarrow \chi $, $\beta_2 \leftrightarrow \delta^*$
Valence	signed z_i [SI]; $ z_i $ [Tr, MS]	$ z_i $
Factor of $\frac{1}{2}$ (co-ion root)	present [S, Donnan root]; omitted [M]	present (corrected; §2.1)
Ion indexing	1, 2 cations / 3 anion [SI, MS]; co-/counter-ion [S]	co-/counter-ion (single salt); 1, 2, 3 (multi)

Table 2: Nomenclature reconciliation. Bracketed keys refer to Table 1.

1.5 Membrane transport model

This section follows the extended Nernst–Planck / Donnan-equilibrium transport framework underlying the DATA and DATA 2.0 workflows [1, 2], specialized here with the membrane-phase fixed-charge closure of the group’s multicomponent Donnan theory [14].

Water flux. Spiegler–Kedem / solution-diffusion,

$$J_w = L_p(\Delta P - \sigma \Delta \pi) = L_p(\Delta P - \sigma nRT \Delta c), \quad (2)$$

with hydraulic permeability L_p , reflection coefficient σ , van ’t Hoff osmotic pressure.

Extended Nernst–Planck ion flux. Each ion moves by diffusion, convection, and electromigration,

$$j_i^s = -D_i^s \nabla c_i^s + c_i^s J_w - \frac{z_i D_i^s F}{RT} c_i^s \nabla \phi, \quad (3)$$

closed by zero current $\sum_i z_i j_i^s = 0$ and electroneutrality $\sum_i z_i c_i^s = 0$. Eliminating $\nabla \phi$ for a two-cation/common-anion system (cations 1, 2; anion 3) gives diffusion-matrix fluxes

$$j_1^s = c_1^s J_w + D_{11}^s \nabla c_1^s + D_{12}^s \nabla c_2^s, \quad j_2^s = c_2^s J_w + D_{21}^s \nabla c_1^s + D_{22}^s \nabla c_2^s, \quad (4)$$

the anion from $j_3^s = -(z_1/z_3)j_1^s - (z_2/z_3)j_2^s$. In the single-salt limit (4) reduces to the ambipolar (Nernst–Hartley) result

$$j_1^s = c_1^s J_w + \frac{D_1^s D_3^s (z_3 - z_1)}{z_1 D_1^s - z_3 D_3^s} \nabla c_1^s. \quad (5)$$

Membrane-phase flux with fixed charge. The same form holds inside the membrane, with electroneutrality carrying χ ($|z_1|c_1^m + |z_2|c_2^m - |z_3|c_3^m - |\chi| = 0$). Eliminating $\nabla \phi$ (group “Transport” doc, Eq. 44 [11]),

$$j_1^m = J_w c_1^m \left(1 - \frac{|z_1| D_1^m |\chi|}{\Theta}\right) + D_{11}^m \nabla c_1^m + D_{12}^m \nabla c_2^m, \quad (6)$$

$\Theta = c_1^m(|z_1|^2 D_1^m + |z_1||z_3| D_3^m) + c_2^m(|z_2|^2 D_2^m + |z_2||z_3| D_3^m) - |z_3| D_3^m |\chi|$; $\chi = 0$ recovers the solution-phase form. Membrane diffusivities follow Mackie–Meares, $D_i^m/D_i^s = (\phi_w/(2 - \phi_w))^2$.

Model reduction. The SI [9] enumerates nested models by which mechanisms are retained (M1 all three; M2 diff+conv; M3 diff+electromig; M4 conv+electromig; M5 diff only; M6 conv only); AIC/estimability selects the parsimonious set, as in DATA 2.0 [2].

Diffusion limit, B , and the multi-salt caveat. When diffusion dominates ($\text{Pe} = J_w \ell / D_m \rightarrow 0$), solving Fick’s law across the membrane gives

$$J_s \approx \frac{D_m H}{\ell} (c_{\text{in},f} - c_{\text{in},p}) \equiv B \Delta c, \quad B := \frac{D_m H}{\ell}. \quad (7)$$

Because $H_i = H_i(c_1^s, c_2^s, \chi)$ depends on the *full* external composition, the single-salt B_{11} is *not* the multi-salt B_{11} (group Docs 2–3 [10, 11]) — a central transferability question (discussion point 5).

Concentration polarization. Interfacial and bulk concentrations are linked by a boundary layer of thickness $\delta_i = D_i^s/k_i$, with the stirred-cell mass-transfer coefficient $k_i = 0.23(v^0)^{0.57}(D_i^s)^{0.67}\nu^{-0.24}b^{-0.43}$. Single salt: thin-film $(c_{\text{in},f} - c_h)/(c_f - c_h) = \exp(J_w/k)$. Multicomponent: integrate the boundary-layer ODE built from (4) from bulk to $x = \delta$; a closed form follows from eigendecomposition of the diffusion matrix and agrees with a direct `ode15s` solve to $\sim 1.5\%$.

1.6 Data preprocessing: from raw signals to the analysis columns

The per-vial analysis columns (§1.1) are not measured directly; they are reduced from the raw instrument time series — time t , cumulative permeate mass $m(t)$, applied pressure, and retentate/permeate conductivities $\kappa_r(t), \kappa_p(t)$ — together with per-vial ICP-OES concentrations. The pipeline (`analysis/preprocessing/preprocess_nacl.py`) was built and validated for NaCl and has since been generalized to CaCl_2 (2:1) and LaCl_3 (3:1), with concentration polarization now computed for *every* single-salt dataset in this one layer rather than left to the regression stage (this raw-signal reprocessing effort moved CP upstream into preprocessing; see the salt-general paragraph and validation below). The reduction is:

1. **Water flux** from a *smoothed* slope of the permeate mass:

$$J_w = \frac{1}{\rho A_m} \frac{dm}{dt}. \quad (8)$$

2. **Bulk concentration** from conductivity via a linear calibration,

$$c_{\text{bulk}} = \frac{\kappa - a}{s}, \quad (9)$$

intercept a and slope s fit *per experiment* from that run's vial (conductivity, ICP) pairs.

3. **Interfacial concentrations** c_{int}, c_p from the bulk values via the thin-film concentration-polarization model of §1.5, $c_{\text{int}} = c_p + (c_{\text{bulk}} - c_p) \exp(J_w/k)$.
4. **Derived quantities:** ionic strength $I = \frac{1}{2} \sum_i z_i^2 c_i$ (salt-general: $I = c$ for NaCl, $3c$ for CaCl_2 , $6c$ for LaCl_3), observed rejection $R = 1 - c_p/c_{\text{int}}$, and solute permeability $B = J_w c_p/(c_{\text{int}} - c_p)$. Preprocessing emits *salt* concentrations throughout; the co-ion (Cl^-) stoichiometry is a regression-model concern, applied downstream in §1.8, not here.

Constants: $\rho \approx 1000 \text{ g/L}$, $A_m \approx 3.95 \text{ cm}^2$ (salt-independent, same stirred cell), with the mass-transfer correlation of §1.5 ($b = 2.54 \text{ cm}$, $\nu = 1.003 \times 10^{-6} \text{ m}^2/\text{s}$, 350 rpm) evaluated at the *solution*-phase ambipolar diffusivity D^s of the salt being processed: $D_{\text{Na}} = 1.33 \times 10^{-9} \text{ m}^2/\text{s}$, $D_{\text{Ca}} = 0.79 \times 10^{-9} \text{ m}^2/\text{s}$, $D_{\text{La}} = 0.626 \times 10^{-9} \text{ m}^2/\text{s}$, $D_{\text{Cl}} = 2.03 \times 10^{-9} \text{ m}^2/\text{s}$ (shared anion) — see the salt-generalization paragraph below for the resulting per-salt D^s and k . These are solution-phase values, distinct from the *membrane*-phase D 's (§2.1, §3, §4) used for B in (7); both draw from the same underlying ion-diffusivity numbers (one source of truth, flagged in the code).

Data-flow map. Table 3 lists every processed column (§1.1) against how it is produced and which raw input(s) it consumes, so the whole pipeline is visible at a glance without opening the code (`analysis/preprocessing/preprocess_nacl.py`).

Col	Quantity	How produced	Raw input(s)
M	t	direct copy	raw time column
N	κ_r	direct copy	raw retentate conductivity
O	κ_p	direct copy (+ light smoothing, item 5 below)	raw permeate conductivity
J	J_w	<i>smoothed</i> slope of cumulative mass, Eq. 8	permeate mass $m(t)$ vs. t
—	c_{bulk} (retentate/permeate)	linear calibration, Eq. 9 (intermediate, not a report column)	κ_r, κ_p ; per-experiment vial (conductivity, ICP) pairs
B	c_{int}	CP correction (or bulk directly if <code>apply_cp=False</code>), §1.5	c_{bulk} (retentate), c_p, J_w
H	c_p	bulk permeate concentration (unpolarized, item 5 below)	c_{bulk} (permeate)
D	I	$\frac{1}{2} \sum_i z_i^2 c_i = c_{\text{int}}$ for 1:1	c_{int}
F	R	$1 - c_p/c_{\text{int}}$	c_{int}, c_p
K	B	$J_w c_p / (c_{\text{int}} - c_p)$	J_w, c_{int}, c_p

Table 3: Raw→processed data-flow: how each analysis column (§1.1) is produced by `preprocess_nacl.py`.

Illustrative figures. Figure 1 shows (a) a representative per-experiment calibration fit (MC5 07.23.24_NaCl, vial conductivity vs. dilution-corrected ICP concentration) and (b) a parity plot of our reconstructed c_{int} against the “(2)” sheet’s B column for its own raw source (05.27.26_NaCl): the 22 live-formula rows sit exactly on the parity line, while the 725 hardcoded/pasted rows scatter around it (the finding in follow-up item 4 below, shown honestly rather than hidden).

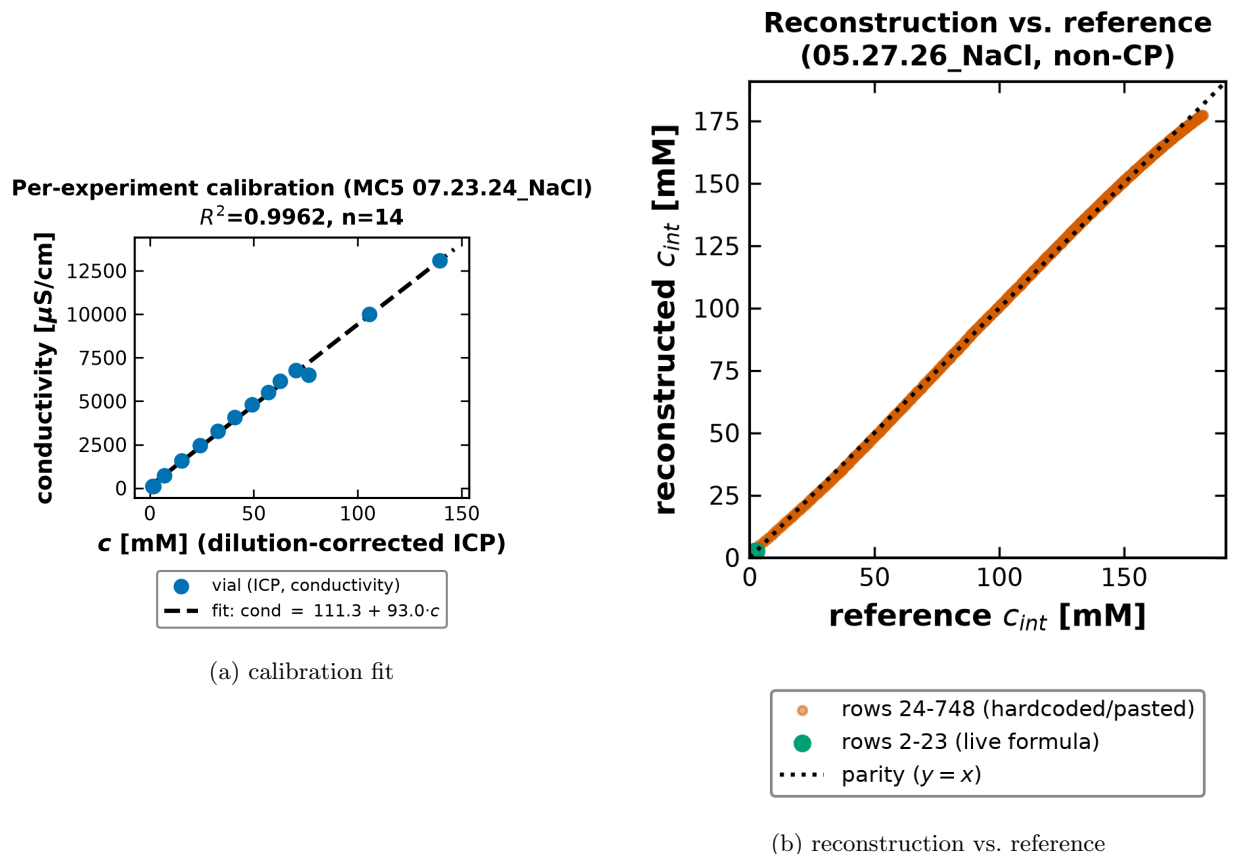


Figure 1: (a) Per-experiment conductivity calibration (one representative raw-only run). (b) Parity plot of the reconstructed vs. reference c_{int} for 05.27.26_NaCl, colored by whether the reference cell is a live formula or a pasted value.

Salt generalization. A single `Salt` record (elemental ICP molar mass, cation valence z) parametrizes the whole pipeline for a $1:z$ salt (NaCl $z = 1$, CaCl_2 $z = 2$, LaCl_3 $z = 3$; anion always Cl^-): the ionic-strength multiplier is $\frac{1}{2}z(z+1)$ and the CP solution diffusivity is $D^s = (1+z)D_{\text{cation}}D_{\text{Cl}}/(zD_{\text{cation}} + D_{\text{Cl}})$, both of which collapse to the familiar NaCl forms at $z = 1$. ICP-OES measures the elemental metal (moles metal = moles salt), so the dilution-corrected $\text{mg}/\text{L} \rightarrow \text{mM}$ conversion of item 3 below extends unchanged with $M_{\text{Ca}^{2+}} = 40.08 \text{ g/mol}$, $M_{\text{La}^{3+}} = 138.91 \text{ g/mol}$ in place of M_{Na^+} . Resulting constants: NaCl $D^s = 1.607 \times 10^{-9} \text{ m}^2/\text{s}$, $k = 2.547 \times 10^{-5} \text{ m/s}$; CaCl_2 $D^s = 1.333 \times 10^{-9} \text{ m}^2/\text{s}$, $k = 2.247 \times 10^{-5} \text{ m/s}$; LaCl_3 $D^s = 1.301 \times 10^{-9} \text{ m}^2/\text{s}$, $k = 2.211 \times 10^{-5} \text{ m/s}$. Every $\text{CaCl}_2/\text{LaCl}_3$ single-salt dataset (the 3 CaCl_2 runs of §3, the 1 LaCl_3 run of §4, plus the two NaCl “validation twin” runs) has now been reprocessed from its raw campaign workbook with CP computed both on and off; full calibrations, provenance, and CSV inventory in `analysis/preprocessing/README.md`. **The refactor is a strict generalization:** with no salt argument supplied, every existing NaCl caller reproduces its pre-refactor output byte-for-byte (regression-tested against all 6 raw-only-run CSVs).

Raw-reprocessing validation (CP-off vs. non-CP processed sheets; CP-on vs. the one already-CP-corrected sheet). NaCl’s CP-off reconstruction reproduces Estrada’s/Quimet’s non-CP sheets to $\sim 1\%$ median relative error on c_{int} (as before); the CP-on reconstruction reproduces the one already-CP-corrected NaCl sheet (05.27.26_NaCl main) to 11.9% median (consistent

with the pre-existing Validation 2 numbers, confirming the refactor changed nothing about the NaCl CP path). $\text{CaCl}_2/\text{LaCl}_3$ CP-off reconstruction, however, does **not** reach the same quality: median relative error on c_{int} ranges 1.5–23.4% across the three CaCl_2 runs and is 26.4% for LaCl_3 — despite LaCl_3 ’s calibration R^2 (0.999) being *better* than a typical NaCl fit. Recovering Ouimet’s *own* implied calibration from his processed sheets (his c_{int} vs. his inline retentate conductivity) shows it is in fact **linear**: $\kappa = -132.6 + 152.1 c_{\text{int}}$ for CaCl_2 MC3 and $\kappa = -302.6 + 216.3 c_{\text{int}}$ for LaCl_3 , both $R^2 \approx 0.999$ (a quadratic term barely improves the fit). The discrepancy is therefore **not** nonlinearity but a *different calibration*: our vial-ICP fit ($\kappa = +96.4 + 187.6 c_{\text{int}}$ for the same CaCl_2 run) differs from Ouimet’s inline calibration in slope ($\sim 20\%$) and has an opposite-sign intercept. The very signature that first suggested nonlinearity — the ratio of his c_{int} to ours shrinking from $\sim 3.1\times$ at low concentration to $\sim 1.3\times$ at high — is exactly what two *linear* calibrations with different intercepts produce (intercept-dominated at low c_{int} ; approaching the slope ratio $187.6/152.1 \approx 1.23$ at high c_{int}), not a curved relationship. Unlike NaCl, the raw $\text{CaCl}_2/\text{LaCl}_3$ workbooks contain *no* embedded inline conductivity calibration, so Ouimet’s calibration is external to the data we have and is not reconstructible from raw; no vial-fit convention (elemental/molecular molar mass, with/without dilution) reproduces it. This is precisely the vial-vs.-inline instrument question already raised as discussion point 9 (§1.2), now *quantified* as a 20–26% concentration effect for the multivalent salts. It is flagged, not silently accepted (row-level evidence in `analysis/preprocessing/README.md`). §1.7 independently confirms this with first-principles physics: model $\kappa(c)$ (Shedlovsky/MSA) is linear for all three salts (rebutting an earlier working guess that the gap was nonlinearity), and the sensor’s purely-multiplicative cell constant means Ouimet’s large negative intercepts are physically inadmissible while the vial-ICP fits are close to the physically expected slope and intercept. **Consequence for the CP study**: because the goal is to isolate the effect of concentration polarization, the $\text{CaCl}_2/\text{LaCl}_3$ CP analysis applies the CP correction *on top of the processed-sheet bulk* c_{int} (the original linear calibration), which both reproduces prior work and cleanly separates the CP shift from the calibration question; the raw-derived vial-ICP CSVs are retained as a labeled calibration-sensitivity artifact. The CP shift itself (median % change in c_{int} , non-CP→CP) is NaCl +10–14%, CaCl_2 +6.6–14.9%, LaCl_3 +28.7% — all single-digit-to-tens-of-percent, i.e. not negligible for any salt.

Follow-up items (data reduction — for team discussion).

1. (A_m and ρ remain *not documented* in the source materials; $A_m \approx 3.95 \text{ cm}^2$ is still *inferred* (unchanged from the earlier estimate) rather than confirmed against the stirred-cell spec sheet. Still open.)
2. (J_w smoothing is now a concrete, implemented choice: a centered 51-row rolling-window linear-regression slope of $m(t)$, tuned by minimizing the error against the one reference J column we have (the “(2)” sheet) — median relative error 3.8%. This is noticeably worse than the near-machine-precision achievable for the calibration/derived columns below: J_w is fundamentally a derivative of a noisy signal, so this is a genuine approximation, not a bug, and a finer/different smoother could still improve it.)
3. (Conductivity calibration, resolved with one real bug found and fixed: the vial block’s “ICP Salt 1 (mg/L)” is the concentration of the *diluted* ICP sample, not the original vial — it must be scaled by the sample/acid dilution factor before converting to mM; skipping this understates concentration 40–200 $\times$ and gives an unphysical calibration slope ($\sim 12\,000 \mu\text{S}/\text{cm}$ per mM instead of $O(100)$). After the fix, per-experiment calibrations fit from 6 raw-only NaCl runs’ vial (conductivity, ICP) pairs give slopes 93–98 $\mu\text{S cm}^{-1}/\text{mM}$ ($R^2 = 0.995\text{--}0.9995$), consistent with each other

across 3 membrane cuts and landing within $\sim 25\%$ of the “(2)” sheet’s embedded formula slope ($76.685 \mu\text{S cm}^{-1}/\text{mM}$) — a reasonable cross-probe/session agreement. The ICP $\text{mg/L} \rightarrow \text{mM}$ conversion uses the **elemental Na^+ molar mass** (22.99 g/mol), confirmed because it (not NaCl ’s 58.44 g/mol) reproduces each run’s stated Feed/Diafiltrate concentrations (e.g. “1 mM Feed, 150 mM Diafiltrate”). **Caveat:** the two runs already used in §2/§2.5 (“05.27.26.NaCl” and “05.07.24.NaCl”) turn out to have *no per-vial conductivity recorded at all*, so this per-experiment fit could not be independently checked against *their* original calibrations — only against the 6 newly-processed raw-only runs. **A further nuance (likely important):** the vial-fit line (e.g. MC5 07.23.24: $\kappa = 111 + 93 c$) differs from the inline-applicable embedded curve ($\kappa = 76.7 c - 64$) not only in slope ($\sim 20\%$) but in *intercept* ($\sim +175 \mu\text{S/cm}$). Because this vial-fit calibration is applied to the *inline* retentate/permeate conductivity, the intercept offset biases the low-concentration end (where it dominates the $\sim 100 \mu\text{S/cm}$ early readings) — a plausible driver of the raw-only runs’ reconstruction noise. *Open question for the team:* are the vial “Conductivity @ Temp” and the inline probe on the same instrument/scale? If not, a run’s vials should not be used to calibrate its inline sensor. Full validation tables: [analysis/preprocessing/README.md](#).)

4. (A second, unexpected finding while validating against the “(2)” sheet: its interfacial-concentration column is a *live* calibration formula for only the first 22 of 747 rows; the remaining 97% are hardcoded pasted values that do *not* exactly satisfy that formula ($R^2 = 0.999$ against retentate conductivity, but not exact – median 1% / max 38% relative deviation). Most of the “(2)” sheet was therefore produced by some undocumented downstream step, not by the formula visible in the cell; §2’s fit to that sheet is still valid as a fit to whatever values are actually in the column, but the column is not as simply reproducible/auditable as the visible formula suggests.)
5. (**Concentration polarization: now implemented for every single-salt dataset, and now the primary results-level convention throughout.** CP is computed in this preprocessing layer (both on and off) for all $\text{NaCl}/\text{CaCl}_2/\text{LaCl}_3$ datasets (salt-generalization paragraph above); the main processed sheet applies the thin-film CP correction, the “(2)” sheet does not (bulk = interfacial), and **the results in §2/§2.5/§3/§4 (and the AR(1) and raw-measurement work that build on the NaCl fit, §2.3/§2.4) are now CP-corrected**, with non-CP retained alongside for comparison. The shift was not negligible ($+10$ – 29% median in c_{int} , salt-dependent; raw-reprocessing validation paragraph above) — see discussion point 11, §1.2, for the resolved outcome.)
6. (Permeate-side smoothing (cols H, O): the flattening ($\approx 0.04 \text{ mM}$ for the first ~ 14 – 25 rows in the reference sheets) now has a plausible physical explanation — column O (permeate conductivity) is itself blank over that same window, consistent with sensor dead-volume before permeate first reaches the conductivity cell, rather than arbitrary smoothing. We approximate it (floor + rolling median) but do not reproduce it exactly.)

1.7 First-principles conductivity models (Shedlovsky & MSA)

The preceding paragraph’s “linear calibration, different source” conclusion for $\text{CaCl}_2/\text{LaCl}_3$ was reached empirically (comparing two fitted lines). This subsection puts a first-principles, peer-reviewed number on it, using the two validated conductivity models from the group’s soft-sensor manuscript [3]: the **same three salts** (NaCl , CaCl_2 , LaCl_3), the **same conductivity sensor** (Innovative Sensor Technology LFS 1107, the same model as the DATA3 inline probes), co-authored by Phillip and Dowling. The **Shedlovsky** model (manuscript Eqs. 4–9; single salt, validated to 0.1N ionic strength) and the **mean spherical approximation (MSA)** (manuscript Eqs. 11–16; validated to 4 M, and the model that extends to binary/ternary mixtures, Eqs. 20–23) both reproduce the manuscript’s own measured $\kappa(c)$ to $\text{MAPE} = 1.0$ – 2.3% against a

probe with $\sim 0.93\%$ intrinsic precision (manuscript Table 3, Fig. 3). Code is vendored verbatim (`analysis/conductivity_models/conductivity.py`, run unmodified) with parameters from the manuscript’s Tables 1–2; all DATA3-specific usage (ranges, plotting, linearity/adjudication analysis) is in `analysis/conductivity_models/calibration_curves.py`. NaCl ($\rightarrow 180\text{ mM} = 0.18\text{ N}$) and CaCl_2 ($\rightarrow 108\text{ mM} = 0.216\text{ N}$) exceed Shedlovsky’s 0.1 N validity at the high end of the DATA3 range, so MSA (valid throughout) is the primary model here; Shedlovsky is reported alongside and flagged where extrapolated.

Implementation check. Reproducing the manuscript’s Table 3 MAPEs against its own experimental data (`salts-conductivity-modelling/Data/{NaCl,CaCl2,LaCl3}.csv`, at $T = 298.15\text{ K}$) confirms the vendored code and parameters before drawing any conclusions: Shedlovsky/MSA MAPE = $0.96\%/1.16\%$ for NaCl (manuscript 1.0% both), $2.21\%/2.27\%$ for CaCl_2 (manuscript $2.2\%/2.3\%$), and $1.89\%/2.20\%$ for LaCl_3 (manuscript $1.9\%/2.2\%$) — all within 0.2 percentage points of the published values (NaCl’s MSA number is the largest gap, 1.16% vs. 1.0% , plausibly from the rounded diffusivities in Table 2 vs. the notebook’s unrounded values; immaterial to the conclusions below).

Linearity verdict (rebutts the nonlinearity guess). Fitting linear and quadratic $\kappa(c)$ to the MSA/Shedlovsky curves over each salt’s DATA3 c_{int} range ($T = 298.15\text{ K}$) gives $R^2 \geq 0.9971$ for the linear fit in every case (MSA: NaCl 0.9994 , CaCl_2 0.9976 , LaCl_3 0.9971), a small negative quadratic curvature (concave-down, as expected physically), and a maximum linear-fit deviation of only $2.1\text{--}4.6\%$ of the κ range (NaCl 2.1% , CaCl_2 4.3% , LaCl_3 4.6%) — concentrated at the range extremes, not a systematic curve. Combined with the manuscript’s own Fig. 3 and Table 3 for these same salts and sensor, this is the definitive, first-principles rebuttal of the “nonlinear conductivity–concentration relationship” guess (superseded working hypothesis, corrected above): $\kappa(c)$ is linear to good approximation over the full DATA3 range for all three salts, so the $\text{CaCl}_2/\text{LaCl}_3$ reconstruction gap is not a modeling artifact of using a linear calibration.

Adjudicating discussion point 9 (calibration source). The LFS 1107 sensor obeys $\chi = \Phi \cdot (I/V)$ with cell constant Φ a *purely multiplicative* scale (manuscript Eq. 3) — a probe/cell miscalibration is a slope error with **no additive offset**, and physical $\kappa(c) \rightarrow 0$ as $c \rightarrow 0$. The MSA linear fit’s intercept over each DATA3 range is small and **positive** (NaCl $+394\text{ }\mu\text{S/cm}$, CaCl_2 $+831\text{ }\mu\text{S/cm}$, LaCl_3 $+371\text{ }\mu\text{S/cm}$ — each $< 5\%$ of that salt’s κ range), exactly the small-positive-intercept signature a concave-down curve produces when linearized over a high- c window. Against this physical anchor: the **processed-sheet inline** calibrations (Estrada’s for NaCl, Ouimet’s for $\text{CaCl}_2/\text{LaCl}_3$; §1.6 raw-reprocessing validation paragraph) have slopes $24\text{--}13\text{--}23\%$ *below* the MSA-implied slope (NaCl, CaCl_2 , LaCl_3 respectively) *and* large **negative** intercepts (-63.7 , -132.6 , $-302.6\text{ }\mu\text{S/cm}$) — physically inadmissible for a purely multiplicative-scale sensor. The **vial-ICP fits** (§1.6) are markedly closer to the physical prediction in both respects: slopes within 8% of MSA (NaCl -7.9% , CaCl_2 $+7.1\%$, LaCl_3 -7.8%) and **positive** intercepts of the right order of magnitude ($+111$, $+96$, $+94\text{ }\mu\text{S/cm}$). **The physics therefore supports the vial-ICP calibration over the processed-sheet inline formula** as the more physically defensible convention for all three salts, corroborating (from an independent, first-principles direction) the calibration-source conclusion already reached empirically above. **Caveat:** the manuscript itself reports the models slightly *underpredict* CaCl_2 and (more) LaCl_3 conductivity by $\sim 2\%$ MAPE (dielectric decrement/hydration effects on multivalent ions), so this is a *shape/linearity/intercept* argument, not a to-the-percent absolute calibration authority for the multivalent salts — it does not, by itself,

resolve which slope is exactly correct, only that the processed-sheet large negative intercepts are not physically justifiable and the vial-ICP fits are closer on both counts. Figure 2 shows the full picture.

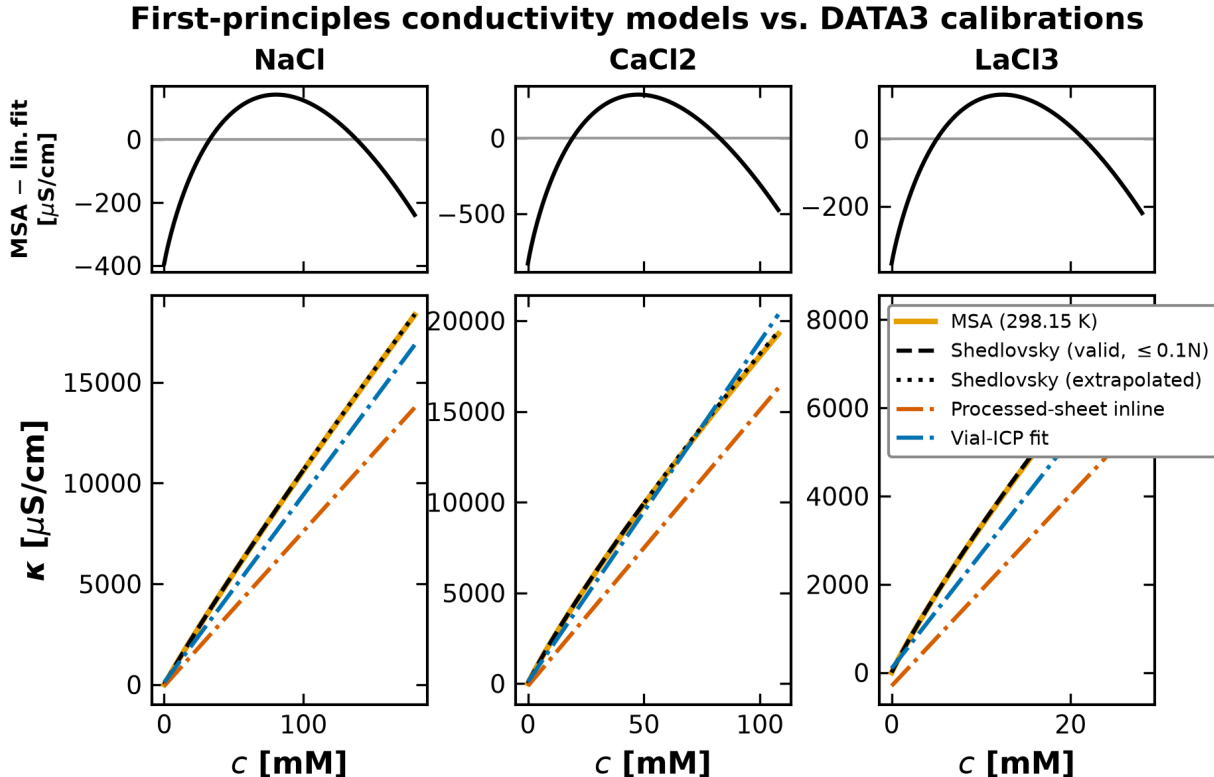


Figure 2: First-principles NaCl/CaCl₂/LaCl₃ conductivity models (MSA, primary; Shedlovsky, secondary, dotted where extrapolated beyond 0.1 N) at $T = 298.15$ K over each salt’s DATA3 c_{int} range, with the processed-sheet inline and vial-ICP linear calibrations overlaid (bottom row) and the MSA curve’s deviation from its own best linear fit (top row, showing the small, concave-down curvature).

Looking ahead: multicomponent conductivity. The **MSA multi-salt model** (manuscript Eqs. 11–16, extended to binary/ternary mixtures via Eqs. 20–23) is vended here as a building block for the eventual multicomponent extension (discussion point 16, §1.2): no new model is needed to go from single-salt to multi-salt conductivity, only per-ion parameters (already in Table 2) and the additional per-ion ICP concentrations discussed there. The manuscript’s Hunter–Reiner model-discrimination experimental design (Eqs. 20–23, its Fig. 5) is directly reusable for planning which multi-salt diafiltration compositions would most sharply distinguish candidate conductivity models before committing bench time to them.

1.8 Donnan partitioning

Ion partitioning is set by equal electrochemical potential across the interface (a common Donnan potential) with membrane electroneutrality including χ ; only the real, positive root is physical,

following the group's multicomponent Donnan exclusion theory [14]. In the ideal limit,

$$\left(\frac{c_1^s}{c_1^m}\right)^{1/|z_1|} = \left(\frac{c_2^s}{c_2^m}\right)^{1/|z_2|} = \left(\frac{c_3^m}{c_3^s}\right)^{1/|z_3|}. \quad (10)$$

Single-salt polynomials appear in two equivalent forms — the group's partition coefficient H form, and the membrane-concentration c^m form used in the scanned notes [5, 6, 7] and Bill's MATLAB [8] (divide by $(c^s)^n$ to convert):

$$1:1 \text{ NaCl, co-ion: } c_{\text{co}}^m{}^2 + |\chi| c_{\text{co}}^m - \delta^* c_{\text{co}}^s{}^2 = 0, \quad c_{\text{co}}^m = \frac{1}{2} \left(-|\chi| + \sqrt{|\chi|^2 + 4\delta^* c_{\text{co}}^s{}^2} \right), \quad (11)$$

$$2:1 \text{ CaCl}_2, \text{ co-ion: } c_{\text{co}}^m{}^3 + |\chi| c_{\text{co}}^m{}^2 - \delta^* \delta_{\text{co}}^{\circ} c_{\text{co}}^s{}^3 = 0, \quad (12)$$

$$3:1 \text{ LaCl}_3, \text{ co-ion: } c_{\text{co}}^m{}^4 + |\chi| c_{\text{co}}^m{}^3 - \delta^* (\delta_{\text{co}}^{\circ})^2 c_{\text{co}}^s{}^4 = 0. \quad (13)$$

The 2:1 cubic has a Cardano closed form (denominators 27,729 for the co-ion; 216,46656 for the counter-ion); the 3:1 quartic has no compact closed form, motivating a nonlinear-constraint treatment (§4). **(Confirm the $z \neq 1$ non-ideal factors: co-ion carries $\delta_{\text{co}}^{\circ}$ (2:1) and $(\delta_{\text{co}}^{\circ})^2$ (3:1) beyond δ^* .)** For mixtures the SI [9] derives a single polynomial in $H_1 = c_1^m/c_1^s$ (e.g. $H_1^3 - \frac{|\chi|}{c_1^s} H_1^2 - (1 + 2\rho)H_1 + 2\rho = 0$ for NaCl+CaCl₂, $\rho = c_2^s/c_1^s$), making the composition dependence of H explicit.

1.9 Statistical methods

Let θ collect the parameters (here $(|\chi|, \delta^*)$), y_i the response for observation i , and $\hat{y}_i(\theta)$ the model prediction. We use the following, in order of increasing rigor. In plain terms: we first find the single best-fitting θ , then ask how uncertain that estimate is by several methods of increasing computational cost and increasing trustworthiness when the model's response surface is strongly curved.

Point estimation (nonlinear least squares). *What it does:* finds the single parameter vector $\hat{\theta}$ that best fits the data, by minimizing the total squared mismatch.

$$\hat{\theta} = \arg \min_{\theta} \text{SSE}(\theta), \quad \text{SSE}(\theta) = \sum_{i=1}^n w_i (y_i - \hat{y}_i(\theta))^2, \quad (14)$$

with weights w_i (unity for OLS; inverse measurement variance for the heteroscedastic WLS used in DATA 2.0 [2], following the same model-based data-analytics workflow as DATA [1]). Solved by trust-region / Levenberg–Marquardt (scipy) or interior point (Ipopt, via ParmEst).

Linearized (Wald) covariance and intervals. *What it does and how to read it:* the simplest error bars — it approximates the model as a flat (linear) surface right around the best fit and reports how far θ could move before the fit visibly worsens. Cheap to compute, but *optimistic* (too narrow) if the true response surface is noticeably curved near the optimum.

$$\mathbf{V}_{\hat{\theta}} = \hat{\sigma}^2 (\mathbf{J}^{\top} \mathbf{J})^{-1}, \quad \hat{\sigma}^2 = \frac{\text{SSE}(\hat{\theta})}{n-p}, \quad J_{ik} = \left. \frac{\partial \hat{y}_i}{\partial \theta_k} \right|_{\hat{\theta}}, \quad (15)$$

giving marginal intervals $\hat{\theta}_k \pm t_{n-p, 1-\alpha/2} \sqrt{(\mathbf{V}_{\hat{\theta}})_{kk}}$. These assume the model is locally linear in θ (an ellipsoidal region).

Nonlinear (likelihood-ratio / F -test) confidence region. *What it does and how to read it:* the rigorous version of the Wald region — instead of assuming the surface is flat, it directly maps out every θ whose fit is not statistically distinguishable from the best one, so it can come out curved or lopsided where the model is nonlinear, rather than a symmetric ellipse. Exact for Gaussian errors,

$$\left\{ \theta : \text{SSE}(\theta) \leq \text{SSE}(\hat{\theta}) \left[1 + \frac{p}{n-p} F_{p, n-p, 1-\alpha} \right] \right\}, \quad (16)$$

equivalently, asymptotically, $\text{SSE}(\hat{\theta})[1 + \chi^2_{p, 1-\alpha}/(n-p)]$ (the χ^2 form ParmEst uses). This region can be curved/asymmetric where the model is nonlinear.

Profile likelihood. *What it does and how to read it:* the one-parameter-at-a-time slice through the nonlinear region above — for each candidate value of one parameter, it re-optimizes every other parameter and asks whether the resulting fit is still acceptable, so the reported interval already accounts for how the parameters trade off against each other. For a single θ_k , the profile $\text{SSE}_k(\theta_k) = \min_{\theta_{j \neq k}} \text{SSE}(\theta)$ yields the nonlinear interval $\{\theta_k : \text{SSE}_k(\theta_k) \leq \text{SSE}(\hat{\theta})[1 + F_{1, n-p, 1-\alpha}/(n-p)]\}$.

Bootstrap. *What it does:* a model-free resampling check — repeatedly draw a new dataset (with replacement) from the observed data, refit on each, and see how much the resulting estimates scatter. It makes no assumption about the shape of the response surface, so agreement with the analytical (Wald/profile) intervals is reassuring, and disagreement flags that those assumptions may be breaking down. Resample the n cases with replacement, refit B times, and take percentile intervals of the resulting $\{\hat{\theta}^{(b)}\}$.

Multi-start optimization. *What it does:* a convergence check, not an uncertainty method — optimize from many initial guesses sampled over a parameter box; the spread of converged optima diagnoses local minima and optimizer reliability (used in §2.2 to explain the MATLAB convergence issue).

Independence caveat. All SSE-based intervals above assume independent errors. Diafiltration data are dense, autocorrelated time series (one run), so the effective sample size is $\ll n$ and reported intervals are *optimistic*; this is flagged throughout and is a target of future work.

2 Analysis of NaCl data

The analyzed dataset is sheet NF270_MC5 05.27.26_NaCl (2) of the analysis workbook [15] (Box `.../Analyses/BoE Analysis.xlsx`; repo twin `data/RejectionAnalysis.xlsx`); the full inventory of available NaCl experiments is in §2.5. (The “(2)” sheet uses *non-concentration-polarization interfacial concentrations* (bulk = interfacial); see §1.6, follow-up item 4. This section below reports the CP-corrected refit as primary (with non-CP retained for comparison), resolving what was earlier an open decision.) The Python analysis (`analysis/step2_nacl/`, `analysis/step3_nacl_uncertainty/`) runs in the documented `data3-regression` conda environment.

2.1 Reproduction of the MATLAB analysis

Regression formulation. Bill’s MATLAB (`prior_analysis/B_regression_concentration_NaClanal.m`) forms the solute flux $J_s = c_p J_w$ and the transformed “response”

$$\Delta c_m := \frac{J_s \ell}{D_m}, \quad D_m = \frac{2D_{\text{Na}}D_{\text{Cl}}}{D_{\text{Na}} + D_{\text{Cl}}}, \quad \ell = 80 \text{ nm}, \quad (17)$$

then fits the difference of Donnan co-ion roots (11):

$$f(\beta; c_{\text{int}}, c_p) = \sqrt{\beta_1^2 + 4\beta_2 c_{\text{int}}^2} - \sqrt{\beta_1^2 + 4\beta_2 c_p^2}, \quad \beta_1 \leftrightarrow |\chi|, \quad \beta_2 \leftrightarrow \delta^*, \quad (18)$$

by nonlinear least squares (guess $\beta_0 = (22.73, 0.09)$), with a brute-force $|\chi| \times \delta^*$ residual contour. We ported this to Python/scipy and reproduced it (Table 4, Fig. 3).

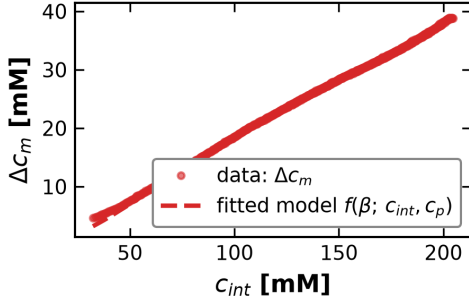
Potential modeling issue (factor of 2). The Donnan root (11) carries a $\frac{1}{2}$: $c_{\text{co},f}^m - c_{\text{co},p}^m = \frac{1}{2}[\sqrt{\dots c_{\text{int}} \dots} - \sqrt{\dots c_p \dots}]$, which (18) omits — so the code fits $2(c_{\text{co},f}^m - c_{\text{co},p}^m)$. Bill notes the MATLAB was preliminary, so we treat the missing $\frac{1}{2}$ as a bug. It is an *exact reparameterization*: since $f(2\beta_1, 4\beta_2) = 2f(\beta_1, \beta_2)$ (pull 4 from each square root), fitting the corrected $f_{\text{corr}} = \frac{1}{2}f$ returns the identical SSE surface with $|\chi|$ doubled and δ^* quadrupled (verified: ratios 2.000000, 4.000000; SSE unchanged to machine precision). Fit quality is therefore unaffected; only the parameter values change. Reassuringly the corrected $|\chi| \approx 50 \text{ mM}$ is closer to the $\chi \approx -44 \text{ mM}$ used independently in the BoE calculations [12] than the uncorrected $\approx 25 \text{ mM}$ (weak physical support for the corrected form). *We adopt the corrected form f_{corr} as primary hereafter. (ℓ and D_m are held fixed here, but they are confounded with δ^* through $B = D_m H / \ell$ (7) — a fixed-vs.-estimated choice (discussion point 4, §1.2) that this section’s point estimate does not resolve.)*

Fit	$ \chi $ [mM]	δ^* [–]	SSE	n
MATLAB reproduction (rows 57–748, uncorrected f)	25.18 ± 0.14	0.0816 ± 0.0001	56.52	692
Full valid range (rows 2–748, uncorrected f)	24.78 ± 0.19	0.0813 ± 0.0002	129.62	747
Corrected (rows 57–748, $f_{\text{corr}} = \frac{1}{2}f$)	50.36 ± 0.27	0.3263 ± 0.0005	56.52	692
Corrected (rows 2–748, $f_{\text{corr}} = \frac{1}{2}f$)	49.55 ± 0.39	0.3251 ± 0.0007	129.62	747

Table 4: NaCl regression estimates (scipy). The uncorrected rows reproduce Bill’s setup; “corrected” restores the $\frac{1}{2}$ of (11). Bill’s row window (57–748) skips the low-concentration startup transient; the full valid data span rows 2–748.

Recommendation for the MATLAB code. See §2.2: switch `lsqcurvefit` to `algorithm='levenberg-marquardt'` (its default trust-region-reflective is prone to boundary-trapping at the $|\chi| \rightarrow 0$ local minimum), and restore the $\frac{1}{2}$ in (18).

NaCl Donnan regression: data vs. fitted model
(Bill's MATLAB reproduction, rows 57-748)



log₁₀ (SSE) contour (rows 57-748, uncorrected f)

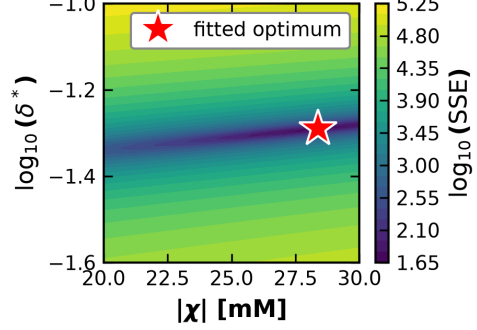


Figure 3: Left: fitted model f vs. transformed data Δc_m across $c_{\text{int}} \in [25, 180]$ mM. Right: $\log_{10}$ SSE over the MATLAB grid ($|\chi| \in [20, 30]$, $\delta^* \in [10^{-1.6}, 10^{-1}]$) with the optimum marked; the elongated valley reflects the $|\chi|$ - δ^* correlation.

2.2 Uncertainty analysis and nonlinear confidence regions

We stress-tested optimizer reliability and quantified uncertainty in both `scipy` and `Pyomo/ParmEst`; the two implementations cross-validate to 6–7 significant figures.

Concentration polarization is now the primary convention. §2.1’s table reproduces Bill’s original (non-CP) MATLAB setup exactly, on the “(2)” sheet’s bulk c_{int} . From here on, **CP is applied on top of that bulk value** (§1.6, `analysis/cp_authors.bulk.py`: $c_{\text{int}}^{\text{CP}} = c_p + (c_{\text{int}}^{\text{bulk}} - c_p) \exp(J_w/k)$, $k_{\text{NaCl}} = 2.55 \times 10^{-5}$ m/s, the same constant used throughout this report) and is the **standing convention** for every table/figure below. Table 5 compares: CP shifts $|\chi|$ from 50.36 to 56.74 mM (+12.7%) and δ^* from 0.3263 to 0.2049 (−37.2%, since CP raises c_{int} enough that δ^* absorbs part of the shift) — the *same order of magnitude*, not a change in the qualitative conclusion that NaCl is well-identified with a nonzero fixed charge of several tens of mM. §2.3 (AR(1)) and §2.4 (raw-measurement) are now also re-run CP-corrected (discussion point 11, §1.2).

Parameter	non-CP (bulk, §2.1)		CP-corrected (primary)	
	point est.	95% profile CI	point est.	95% profile CI
$ \chi $ [mM]	50.36 ± 0.27	(49.68, 51.04)	56.74 ± 0.28	(56.06, 57.44)
δ^* [−]	0.3263 ± 0.0005	(0.3250, 0.3277)	0.2049 ± 0.0004	(0.2039, 0.2059)
SSE	56.52		52.37	

Table 5: NaCl CP-vs-non-CP comparison, corrected model, rows 57–748 ($n = 692$). CP-corrected is now primary; non-CP is retained for comparison and is what §2.3/§2.4 used before their own CP-corrected re-run, still shown there alongside the CP-corrected numbers.

Convergence diagnosis (why the MATLAB fit was unreliable). Across 300 random starts over a deliberately wide box ($|\chi|_0 \in [1, 200]$ mM, δ_0^* log-uniform on $[10^{-3}, 10]$), `scipy lm` and `trf` reached the global optimum **100%** of the time on the CP-corrected data (as on non-CP); `dogbox` was trapped **24.0%** of the time (non-CP: 27.3%) at a *genuine* local minimum at $|\chi| \rightarrow 0$, where its reflective step pins $|\chi|$ to the lower bound. The Jacobian is genuinely ill-conditioned ($\text{cond}(\mathbf{J}^\top \mathbf{J}) \sim 10^6$ – 10^7 ; reparameterizing to $\log_{10} \delta^*$ roughly halves it), but conditioning is *not* the cause of failure — `lm`/`trf` are unbothered by it. `ParmEst`/`Ipopt` likewise hit the global optimum in 100% of 30

starts. **Conclusion:** MATLAB `lsqcurvefit`’s default *trust-region-reflective* is the analogue of the failing `dogbox`; switching to *Levenberg–Marquardt* (or starting from a physical guess) resolves the unreliability.

Confidence regions and intervals. At $n = 692$ the nonlinear F -test region (16) is visually indistinguishable from the linearized Wald ellipse — the NaCl fit is well-identified and nearly linear near the optimum, CP-corrected as on non-CP. Profile-likelihood intervals are close to Wald (Table 6), and a 1000-rep case-resampling bootstrap gives SEs close to Wald ($\text{SE}_{|\chi|} = 0.286$ vs. 0.277 ; $\text{SE}_{\delta^*} = 0.00039$ vs. 0.00040), all pointing the same direction: Wald is slightly optimistic but not badly wrong here.

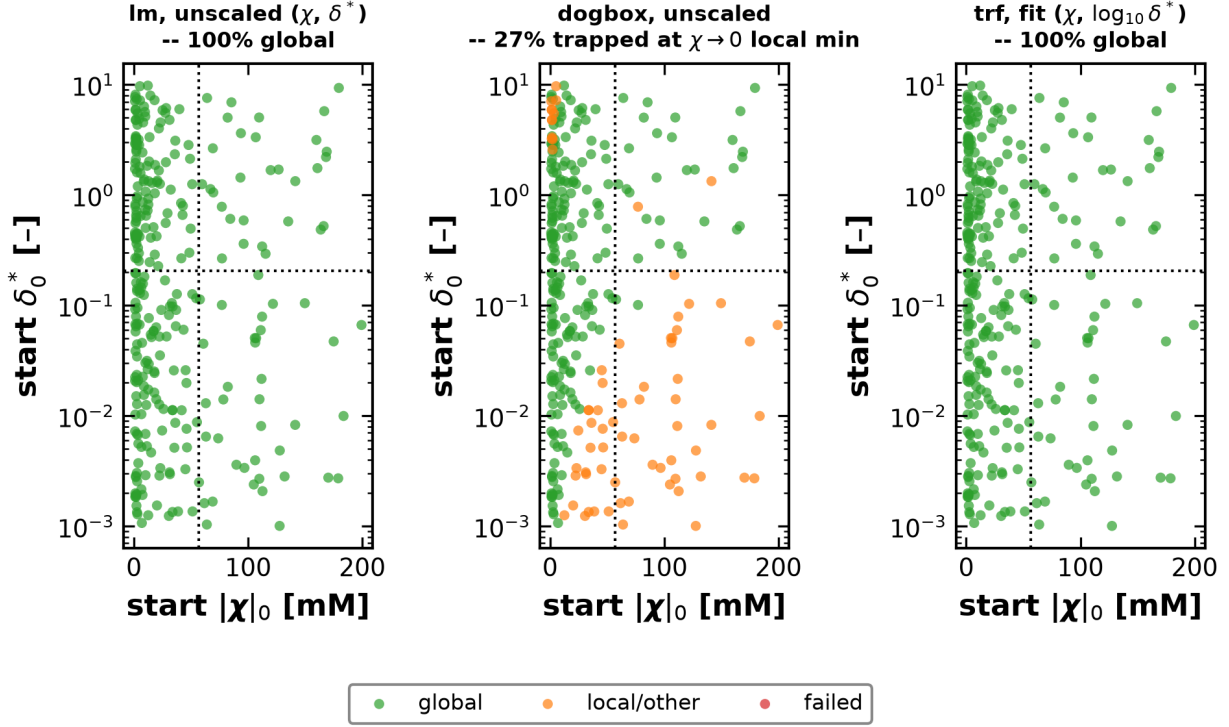
Parameter	point est.	95% profile CI	95% Wald CI
$ \chi $ [mM]	56.74	(56.06, 57.44)	(56.20, 57.29)
δ^* [-]	0.2049	(0.2039, 0.2059)	(0.2041, 0.2057)

Table 6: NaCl 95% confidence intervals, CP-corrected primary result, corrected model, rows 57–748 ($n = 692$). Profile intervals are close to Wald.

ParmEst cross-validation. ParmEst’s extensive-form `theta_est()` (each of the 692 rows as one Experiment) matches scipy’s estimates to ~ 7 significant figures ($|\chi| = 56.7424$, $\delta^* = 0.20489$, $\text{SSE} = 52.374$, CP-corrected) and its `cov_est()` Wald SEs to 6 figures; its χ^2 confidence-region test and bootstrap agree with scipy’s F -test region and bootstrap. (API notes for us: (i) `cov_est()` silently returns $\sim 3.5\times$ inflated SEs unless `measurement_error` is explicitly `None`; (ii) `calc_cov/cov_n` on `theta_est()` are deprecated in favor of `cov_est()`; (iii) `objective_at_theta()` re-solves the full NLP per grid point (~ 450 ms), making fine confidence grids costly.) (Correction: built-in ParmEst multi-start is not yet released – it is an open pull request (Pyomo PR #3575), not a feature of 6.10.1. This session’s manual re-initialization loop (§2.2) remains the right approach until that PR merges.)

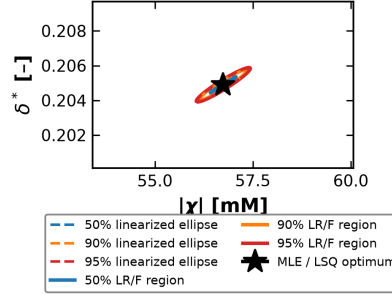
Key caveat. All intervals assume i.i.d. errors; the 692 points are an autocorrelated single-run time series, so the true uncertainty is larger than any figure above. **This is now corrected:** §2.3 models the residuals as an AR(1) process and quantifies how badly the i.i.d. regions were undersized ($n_{\text{eff}} \approx 100$ of $n = 692$, CP-corrected; both a rigorous GLS/whitening correction and an effective-sample-size heuristic widen the CIs by ~ 2.5 – $2.7\times$, with the profile still tracking the Wald — i.e. the correction widens the region but does *not* make it markedly more nonlinear). Extending the correction to CaCl_2 and richer models (ARMA / higher-order AR, or explicitly modeling the missing physics — cf. the CaCl_2 residual structure, §3) remain future work.

Multi-start outcomes by starting point (color = convergence outcome)



(a) scipy multi-start

Nonlinear LR/F-test confidence region vs. linearized ellipse (NaCl corrected model, rows 57-748)



(b) nonlinear region vs. ellipse

Figure 4: (a) Multi-start convergence (CP-corrected primary data): `lm`/`trf` reach the global optimum from all starts, `dogbox` is trapped 24% of the time at $|\chi| \rightarrow 0$. (b) The nonlinear F -test confidence region coincides with the linearized Wald ellipse at $n = 692$. Companion figures (`nacl_profile_vs_wald`, `nacl_bootstrap_scipy`, `nacl_multistart_parmest`, `nacl_confidence_parmest`, `nacl_bootstrap_parmest`) are in `docs/reports/figures/`.

2.3 Autocorrelation-corrected uncertainty

The §2.2 intervals assume i.i.d. errors, but the $n = 692$ points are a dense single-run time series, so this section corrects for the resulting autocorrelation and quantifies how badly the i.i.d. regions understated uncertainty, using the primary NaCl dataset and corrected model f_{corr} throughout,

CP-corrected (the standing convention, §1.2 item 11; non-CP retained alongside for comparison) (`analysis/step8_autocorrelation/ar1_correction.py`, reusing Step 2/3’s data loading, model, and i.i.d. F-test/profile machinery unmodified – it automatically inherits Step 2’s `APPLY_CP` default).

AR(1) residual model. Order the OLS residuals $r_i = y_i - f_{\text{corr}}(\hat{\theta}; x_i)$ by time and model

$$r_i = \phi r_{i-1} + a_i, \quad a_i \stackrel{\text{iid}}{\sim} N(0, \sigma_a^2), \quad |\phi| < 1, \quad \hat{\phi} = \frac{\sum_{i=2}^n r_i r_{i-1}}{\sum_{i=1}^n r_i^2}. \quad (19)$$

Effective sample size.

$$n_{\text{eff}} = n \frac{1 - \hat{\phi}}{1 + \hat{\phi}}, \quad \text{naive CI-inflation} = \sqrt{n/n_{\text{eff}}} = \sqrt{\frac{1+\hat{\phi}}{1-\hat{\phi}}}. \quad (20)$$

GLS via Prais–Winsten whitening. The AR(1) covariance is $\Sigma_{ij} = \frac{\sigma_a^2}{1-\phi^2} \phi^{|i-j|}$; whitening is the *linear* operator

$$\tilde{v}_1 = \sqrt{1 - \phi^2} v_1, \quad \tilde{v}_i = v_i - \phi v_{i-1} \quad (i \geq 2), \quad (21)$$

applied to any time-ordered sequence v . Because (21) is linear, $\widetilde{(y - f(\theta))} = \tilde{y} - \tilde{f}(\theta)$ exactly, so the whitened residual is simply $\tilde{r}_i(\theta; \phi) = \widetilde{r(\theta)_i}$ — no separate whitening of y and f is needed.

Cochrane–Orcutt iterates $\hat{\phi} \leftrightarrow \hat{\theta}$ to convergence: fit θ by OLS, estimate $\hat{\phi}$ via (19), refit θ by nonlinear least squares on $\tilde{r}(\theta; \hat{\phi})$, re-estimate $\hat{\phi}$ from the new residuals, and repeat (converged here in 7 iterations, tolerance 10^{-10}).

Two consistent corrections (not complementary). Whitening and the n_{eff} deflation of (20) are **alternative** ways to account for the same autocorrelation, not two effects to stack: GLS whitening already encodes the information loss in the whitened Jacobian $\tilde{\mathbf{J}}$, so its region is correctly sized with the *actual* $n - p$ degrees of freedom; separately, one can skip whitening entirely and instead deflate $n \rightarrow n_{\text{eff}}$ directly in the i.i.d. formulas. Using *both* (whitened SSE *and* $n_{\text{eff}} - p$ df) double-counts the correction. We therefore report both, side by side, rather than picking one.

Method A (GLS/whitening). Wald covariance from the whitened Jacobian, $\mathbf{V}_A = \hat{\sigma}_a^2 (\tilde{\mathbf{J}}^\top \tilde{\mathbf{J}})^{-1}$, $\hat{\sigma}_a^2 = \text{SSE}_{\text{whitened}}(\hat{\theta})/(n - p)$; the nonlinear F -test/profile region on $\text{SSE}_{\text{whitened}}(\theta) = \sum_i \tilde{r}_i(\theta; \hat{\phi})^2$ with the **actual** $n - p$ denominator degrees of freedom.

Method B (n_{eff} heuristic). No whitening: keep the OLS fit $\hat{\theta}$ and $\text{SSE}(\theta)$ from §2.2, but replace n by n_{eff} (20) everywhere it enters the i.i.d. formulas:

$$\text{SE}_B = \text{SE}_{\text{OLS}} \cdot \sqrt{n/n_{\text{eff}}}, \quad F\text{-test/profile region on } \text{SSE}(\theta) \text{ with } n_{\text{eff}} - p \text{ df.} \quad (22)$$

Sufficiency of AR(1). Compare the ACF of the OLS residuals and of the whitened residuals $\tilde{r}(\theta_{\text{GLS}}; \hat{\phi})$ against the $\pm 1.96/\sqrt{n}$ white-noise band; if the whitened ACF falls inside the band beyond lag 0, AR(1) suffices.

Results (CP-corrected primary fit). $\hat{\phi} = 0.7430$ (OLS residuals); Cochrane–Orcutt converges in 7 iterations to $\hat{\phi} = 0.7466$. This gives $n_{\text{eff}} = 100.4$ (**only** 14.5% of $n = 692$ are effectively independent) and a naive inflation factor of 2.63. **The point estimate barely moves:** OLS→GLS (Method A) shifts $|\chi|$ by -1.00% and δ^* by -0.46% (56.18 vs. 56.74 mM; 0.2039 vs. 0.2049; Method B does not refit θ , so its point estimate is the OLS value by construction) — the autocorrelation correction changes the *uncertainty*, not the point estimate. **CP vs. non-CP:** CP shifts $\hat{\phi}$ down slightly ($0.7734 \rightarrow 0.7430$ OLS; $0.7758 \rightarrow 0.7466$ Cochrane–Orcutt) and n_{eff} up slightly ($87.4 \rightarrow 100.4$), narrowing the widening folds from $\sim 2.7\text{--}2.9\times$ to $\sim 2.5\text{--}2.7\times$ (Table 7) — **essentially unchanged in character:** the autocorrelation is a property of how the dense time series was collected, not of the CP shift applied to c_{int} , so this mild narrowing is expected rather than surprising.

Param.	point	95% Wald (i.i.d.)	95% Wald (A)	95% Wald (B)	fold A / B
$ \chi $ [mM]	56.74	(56.20, 57.29)	(54.80, 57.55)	(55.32, 58.17)	$2.53\times / 2.63\times$
δ^*	0.2049	(0.2041, 0.2057)	(0.2020, 0.2059)	(0.2028, 0.2069)	$2.52\times / 2.63\times$
Param.	point	95% profile (i.i.d.)	95% profile (A)	95% profile (B)	fold A / B
$ \chi $ [mM]	56.74	(56.06, 57.44)	(54.46, 57.94)	(54.92, 58.62)	$2.53\times / 2.68\times$
δ^*	0.2049	(0.2039, 0.2059)	(0.2015, 0.2065)	(0.2023, 0.2076)	$2.52\times / 2.68\times$
non-CP (pre-CP-correction baseline, retained for comparison), 95% profile:					
$ \chi $ [mM]	50.36	(49.68, 51.04)	(48.05, 51.73)	(48.42, 52.34)	$2.71\times / 2.89\times$
δ^*	0.3263	(0.3250, 0.3277)	(0.3217, 0.3287)	(0.3226, 0.3302)	$2.70\times / 2.89\times$

Table 7: NaCl 95% confidence intervals, i.i.d. vs. Method A (GLS/whitening, $n - p$ df) vs. Method B (n_{eff} heuristic), rows 57–748, $n = 692$, CP-corrected primary fit ($n_{\text{eff}} = 100.4$); non-CP row for comparison ($n_{\text{eff}} = 87.4$). **Methods A and B agree closely** ($2.5\text{--}2.7\times$ CP-corrected, $2.7\text{--}2.9\times$ non-CP, for both Wald and profile), and **profile tracks Wald under each method** (profile/Wald ratio ≈ 1.27 for i.i.d. and A, ≈ 1.30 for B, CP-corrected — all close to the i.i.d. case’s own ratio, i.e. mild, comparable nonlinearity throughout, essentially unchanged from non-CP). This confirms the two corrections are consistent alternatives, and that the region does *not* become dramatically more nonlinear once the correction is applied without double-counting.

Fig. 5 (right) shows all three regions are comparable in size and shape (Method B slightly larger than Method A, both $\sim 2.5\text{--}2.7\times$ the i.i.d. region, CP-corrected) — unlike an earlier draft of this analysis, which combined whitening with the $n_{\text{eff}} - p$ df correction and reported a spurious $\sim 8\times$ profile widening with a visibly more elongated region; that number resulted from double-counting the same correlation twice and should not be used. **(Open question for the PI: Methods A and B are both defensible and agree closely here, but which should be the report’s headline convention going forward? Separately: is the n_{eff} heuristic (Method B) even well justified given AR(1) is not fully sufficient (below) — its derivation assumes the AR(1) model is exact, whereas Method A’s whitened residuals still show some structure. Method A (a full GLS refit) is the more rigorous choice on that basis, but both are reported here rather than silently picking one.)**

Residual ACF: raw vs. whitened (NaCl primary dataset, rows 57-748)

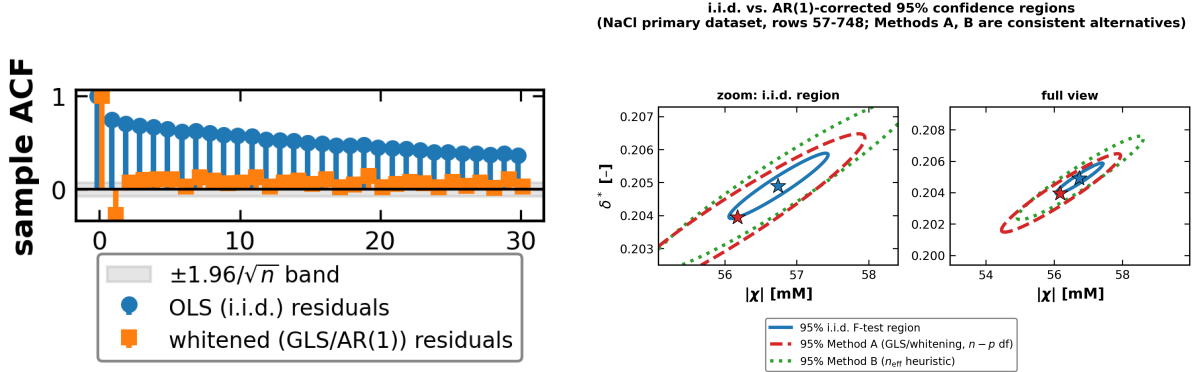


Figure 5: CP-corrected primary fit. **Left:** residual ACF, OLS (i.i.d.) vs. whitened (GLS/AR(1)), with the $\pm 1.96/\sqrt{n}$ band. AR(1) removes most of the raw series’ slow decay but 16 of 30 lags (including a negative lag-1 of -0.28) still exceed the band — AR(1) is a large first-order improvement but not fully sufficient (non-CP: 12 of 30, lag-1 -0.30 , essentially the same picture); higher-order AR(p)/ARMA is future work. **Right:** the i.i.d. vs. Method-A vs. Method-B 95% F -test regions on the $(|x|, \delta^*)$ plane (zoom panel + full view); all three are visible at comparable scale once the correction is applied consistently (contrast the double-counted $\sim 8\times$ region of the earlier draft, which would have dwarfed this axis).

Verdict. The i.i.d. regions understated uncertainty by $\sim 2.5\text{--}2.7\times$ (CP-corrected; $2.7\text{--}2.9\times$ non-CP), consistently across both Wald and profile and across both correction methods — **this, not the earlier $\sim 8\times$ figure, is the number that should be quoted going forward.** That earlier number was an artifact of applying the $n_{\text{eff}} - p$ df deflation on top of the whitened (GLS) SSE surface; whitening and the n_{eff} heuristic are alternative corrections for the same effect; the earlier draft combined them and thereby double-counted the correlation. AR(1) captures the dominant correlation structure and is a necessary correction, but the residual structure remaining in the whitened ACF (Fig. 5, left) shows it is not fully sufficient; richer models (AR(p), ARMA, or explicitly modeling the missing physics, cf. the CaCl_2 residual structure, §3) remain future work, as does extending this correction to CaCl_2 (the raw time-series reformulation, §2.4, is now also CP-corrected).

2.4 Alternative formulation: regress on the raw time-series measurements

The §2.1 fit forms the transformed $\Delta_{\text{cm}} = c_p J_w \ell / D_m$ and regresses *that* — so the noisy *processed* quantities c_p, J_w appear inside the response, and c_p appears on both sides of the estimation (both as a derived predictor, via the calibration, and as part of what is predicted). This mis-states the uncertainty. This section instead poses the regression directly on the **raw measured permeate conductivity** $\kappa_p(t)$, with no transformation of the response, and compares it against §2.1–2.2’s transformed fit (`analysis/step9_raw_measurement/raw_regression.py`, reusing the transformed-fit analysis’s data loading, model, and F-test/profile machinery unmodified). (This is an exploratory **reduced** forward model on a single raw channel (κ_p), not the fully rigorous DATA 2.0-style dynamic DAE / Soft-Sensor ($\kappa \rightarrow c$ via Shedlovsky/MSA [3]) treatment of all raw channels jointly (also $\kappa_r, m(t)$ as responses, not just as predictors) — that remains future work; see the working assumptions and verdict below.)

Forward model. Primary dataset: raw sheet NF270_MC5.xlsx / 05.27.26_NaCl (the raw twin of the “(2)” sheet), restricted to the same row window as §2.1–2.3 (spreadsheet rows 57–748, $n = 692$) for a direct comparison. Known inputs per time t : retentate conductivity $\kappa_r(t) \rightarrow$ feed-bulk concentration $c_f(t)$ via the embedded linear calibration $c = (\kappa - a)/s$ ($a = -63.706$, $s = 76.685$, the same values §1.6 uses for this sheet); permeate mass $m(t) \rightarrow J_w(t)$ via §1.6’s smoothed-slope pipeline (tuned window = 51 rows); feed interface $c_{\text{int}}(t)$ is **CP-corrected** (the standing convention): the same thin-film model as §1.6/§2.3, $c_{\text{int}} = c_p + (c_f - c_p) \exp(J_w/k)$, $k_{\text{NaCl}} = 2.55 \times 10^{-5} \text{ m/s}$, the same constant used throughout (`analysis/cp_authors_bulk.py`), permeate unpolarized. The co-ion mass balance that *defines* Δc_m in (17) is, before rearranging into “ Δc_m as the response,” the **implicit equation for** c_p at fixed $\theta = (|\chi|, \delta^*)$:

$$J_w c_p = \frac{D_m}{\ell} \left[c_{\text{co}}^m(c_{\text{int}}; \theta) - c_{\text{co}}^m(c_p; \theta) \right], \quad c_{\text{co}}^m(c^s; \theta) = \frac{1}{2} \left(-|\chi| + \sqrt{|\chi|^2 + 4\delta^*(c^s)^2} \right) \quad (23)$$

(the Donnan co-ion root, (11)). **The real subtlety:** under CP, c_{int} is no longer fixed but itself a function of c_p via the CP equation above, so (c_{int}, c_p) must be solved *jointly*. Because the CP equation is affine in c_p for fixed (c_f, J_w, k) , substituting $c_{\text{int}}(c_p)$ directly into (23) collapses the 2×2 coupled system into a single equation in c_p alone — an **exact reduction**, not an approximation or a seeded/decoupled shortcut. This reduced equation has a unique root $c_p(\theta, t) \in [0, c_f]$ for $|\chi|, \delta^* \geq 0$ and $J_w, c_f > 0$ (bracketed: at $c_p = 0$, $c_{\text{int}}(0) = c_f \exp(J_w/k)$ and the left side of (23) minus the right side is ≤ 0 ; at $c_p = c_f$, $c_{\text{int}}(c_f) = c_f$ regardless of the exponential factor and the difference is $J_w c_f > 0$ — the same Descartes’-rule-style bracketing argument as the non-CP path), solved via a single `scipy.optimize.brentq` call per time point (same computational cost as the non-CP solve; non-CP recovered by setting $c_{\text{int}} = c_f$ directly, retained for comparison). The predicted measurement is then $\hat{\kappa}_p(\theta, t) = a + s c_p(\theta, t)$ (inverting the calibration); the residual is $\kappa_p(t) - \hat{\kappa}_p(\theta, t)$, with **no smoothing or transformation** of the measured κ_p . At the converged CP-corrected fit, the CP shift in c_{int} (relative to c_f) has median +10.4%, max +28.9% across the 692 time points.

Measurement-error model (working assumption). No replicate conductivity measurements exist in this dataset to calibrate σ_κ directly, so a generic DATA 2.0-style constant-floor + proportional-term form is assumed and stated explicitly: $\sigma_\kappa(\kappa) = 5.0 \mu\text{S/cm} + 0.01 |\kappa|$. The weighted residual $[\kappa_p(t) - \hat{\kappa}_p(\theta, t)]/\sigma_\kappa(\kappa_p(t))$ is fit by nonlinear least squares, bounded $|\chi|, \delta^* \geq 0$ (unlike the Δc_m *difference* form, the standalone c_{co}^m is not invariant to the sign of χ , so bounding matters here, as for CaCl_2 , §3.1). **Robustness check** (non-CP fit, not re-run under CP): refitting with uniform (unweighted) residuals gives $|\chi| = 140.3 \text{ mM}$, $\delta^* = 1.114$ — the same qualitative (large) shift from the transformed fit persists regardless of the weighting scheme, confirming this is a genuine effect of the raw-measurement formulation and not an artifact of the error-model weighting choice. AR(1) correction (where applied) reuses §2.3’s Method A (GLS/whitening, actual $n - p$ df) exclusively.

Results (CP-corrected primary fit). Table 8 compares the point estimates.

Formulation	$ \chi $ [mM]	95% profile CI	δ^*	95% profile CI
Transformed (§2.1)	56.74	(56.06, 57.44)	0.2049	(0.2039, 0.2059)
Raw measurement (i.i.d.)	136.39	(133.82, 139.03)	0.6692	(0.6604, 0.6783)
Raw measurement (AR(1))	135.73	(125.60, 146.98)	0.6626	(0.6274, 0.7015)
non-CP (pre-CP-correction baseline, retained for comparison):				
Transformed (non-CP)	50.36	(49.68, 51.04)	0.3263	(0.3250, 0.3277)
Raw measurement (non-CP, i.i.d.)	114.89	(112.25, 117.60)	1.0143	(0.9994, 1.0295)
Raw measurement (non-CP, AR(1))	4.92	(1.80, 60.27)	0.00157	(0.00055, 0.01155)

Table 8: NaCl point estimates, transformed vs. raw-measurement formulation (rows 57–748, $n = 692$), CP-corrected primary rows with the non-CP pre-CP-correction baseline retained below for comparison. **The point estimate shifts dramatically:** $|\chi|$ more than doubles (+140% CP-corrected, vs. +128% non-CP), δ^* more than triples (+227% CP-corrected, vs. +211% non-CP), transformed→raw (i.i.d.) — **the CP correction widens, rather than closes, this gap. Unlike the non-CP case, the CP-corrected AR(1) row is physically well-behaved** (see verdict below) — this is the more notable finding of re-running this analysis CP-corrected.

Raw-measurement fit: NaCl primary dataset (rows 57-748)

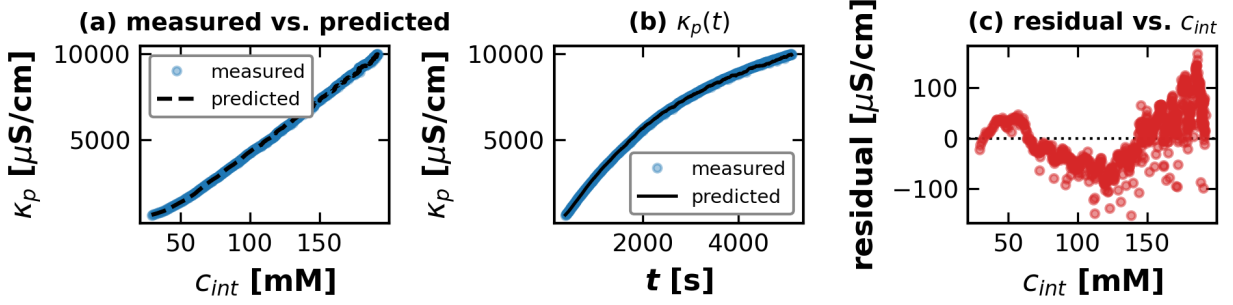


Figure 6: CP-corrected raw-measurement fit (i.i.d.): (a) measured vs. predicted κ_p vs. the solved c_{int} — the absolute curve is tracked closely; (b) $\kappa_p(t)$ time series; (c) residual vs. c_{int} , showing clear systematic wave structure — the raw conductivity signal carries more persistent structure than the differenced/transformed response of §2.1 did, essentially the same picture as the non-CP fit.

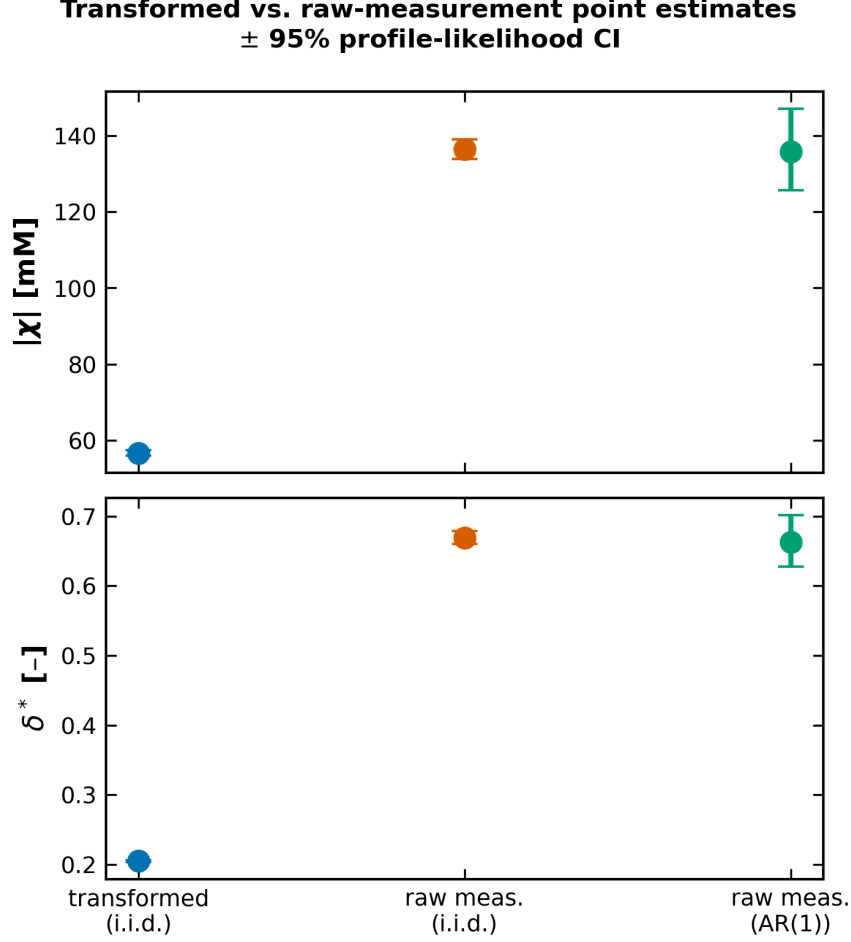


Figure 7: Point estimates $\pm$ 95% profile-likelihood CI, CP-corrected: transformed vs. raw-measurement (i.i.d.) vs. raw-measurement (AR(1)). Unlike the non-CP version of this figure, the AR(1) panel’s error bar is now of comparable size to the i.i.d. one (see verdict) — the near-unit-root breakdown does not recur here.

Verdict. The errors-in-variables-free formulation moves the estimates dramatically, and CP widens the gap further: $|\chi|$ shifts +140% transformed $\rightarrow$ raw (i.i.d.), CP-corrected, vs. +128% non-CP — this is a materially different conclusion from §2.1, not a minor refinement, and the size of the discrepancy (not just its existence, nor its direction under CP) is the headline result: it suggests the transformed fit’s $|\chi| \approx 57$ mM may not be robust to the very errors-in-variables issue this reformulation was designed to test. **The surprise: CP substantially, but only partially, resolves the AR(1) breakdown seen non-CP.** Non-CP, Cochrane–Orcutt drove $\hat{\phi}$ to the edge of the unit circle (0.9989, vs. 0.7758 for the transformed fit’s residuals), collapsing n_{eff} to < 1 and producing a non-physical “corrected” CI (Wald interval for δ^* including negative values; profile CI spanning nearly two orders of magnitude). **CP-corrected**, $\hat{\phi}$ drops to 0.8999 (Cochrane–Orcutt) and $n_{\text{eff}} = 36.5$ (5.3% of $n = 692$) — still heavily autocorrelated and far short of stationary-and-sufficient, but no longer at the unit-root edge: the Wald/profile CIs are now finite and physically sensible (Wald $\delta^* \in (0.633, 0.692)$; profile $|\chi| \in (125.6, 147.0)$, Table 8), a qualitatively different outcome from the non-CP near-degenerate result. **Plausible mechanism** (not disentangled by this exploratory script): $c_{\text{int}}(t)$ ’s CP correction is itself a smooth function of $J_w(t)$, which drifts slowly

over the run; folding that smooth, physically-motivated nonlinear structure into the forward model absorbs part of the slowly-varying trend that was previously left in the residuals as spurious near-unit-root autocorrelation. This does *not* mean AR(1) is now fully adequate here ($n_{\text{eff}}/n = 5.3\%$ is still a severe information loss, well below §2.3’s transformed-fit value of 14.5%) — raw permeate conductivity still rises smoothly and almost monotonically over the run, more consistent with a slowly-varying, near-integrated process than a stationary one fluctuating around a fixed curve, so AR(1) whitening remains an approximation, not a resolution. Likely remaining contributors: (i) the reduced forward model still omits time-varying calibration/temperature effects that could explain part of the residual trend; (ii) the measurement-error model is a working assumption, uncalibrated against replicate data; (iii) a genuinely richer noise model (integrated/ARIMA, or a state-space formulation) may still be needed for raw-signal regression — consistent with this being, as flagged above, a reduced model standing in for the fully rigorous DATA 2.0-style dynamic DAE on all raw channels. **Recommendation to the group:** treat this as exploratory, not a corrected replacement for §2.1 — CP materially improves the AR(1) diagnostic’s behavior but does not resolve the transformed-vs-raw point-estimate discrepancy (which CP in fact widens); validate the measurement-error model, add time-varying-calibration/ temperature corrections, and move to a properly stationary or state-space noise treatment before trusting either the point estimate or its uncertainty here; extending this comparison to CaCl_2 remains future work.

2.5 Workflow applied to all available NaCl datasets

We ran the Step-2/Step-3 pipeline across every available single-salt NaCl experiment (multiple membrane cuts / dates) to test whether $|\chi|$ and δ^* are reproducible across runs and membranes. Table 9 inventories the datasets *as they arrived*: three were already processed into the standard analysis schema (columns B, H, J); six existed only as raw campaign data and required the raw→processed preprocessing pipeline of §1.6 (interfacial concentration / J_w / calibration) before they could be regressed. **That pipeline has since been built, validated, and applied to all six raw-only runs — every row of Table 9 is now both processed and regressed;** the last column gives the exact location of each run’s results (§2.5.1 for the summary, the per-dataset appendix §8 for full detail).

Memb.	Date	Sheet / raw tab	Rows	Arrived as	Now analyzed in
MC2	05.07.24	NF270_MC2 05.07.24_NaCl	482	processed [A]	§2.5.1, §8.1
MC5	05.27.26	NF270_MC5 05.27.26_NaCl	747	processed [A]	§2.5.1, §8.2
MC5	05.27.26	NF270_MC5 05.27.26_NaCl (2)	747	processed [A]	§2.1–2.2 (primary), §8.3
MC3	07.09.24	07.09.24_NaCl	404	raw [Rw]	§2.5.1 (ill-cond.), §8.4
MC3	07.22.24	07.22.24_SNaCl (diluting)	1312	raw [Rw]	§2.5.1, §8.5
MC4	07.11.24	07.11.24_SNaCl	423	raw [Rw]	§2.5.1, §8.6
MC5	07.23.24	07.23.24_NaCl	516	raw [Rw]	§2.5.1, §8.7
MC5	07.23.24	07.23.24_SNaCl	520	raw [Rw]	§2.5.1, §8.8
MC5	07.23.24	07.23.24_S2NaCl (diluting)	1324	raw [Rw]	§2.5.1, §8.9

Table 9: All nine single-salt NaCl diafiltration experiments: **initial state** (“arrived as”) vs. **current status** (all nine are now processed and regressed). [A] = arrived pre-processed in the analysis workbook [15], Box .../Analyses/BoE Analysis.xlsx (repo twin data/Rejection Analysis.xlsx); the three [A] sheets are identical across both workbooks. [Rw] = arrived as raw campaign data only, Box .../Diafiltration Campaign and Raw Data/NF270_MC{2,3,4,5}.xlsx [16, 17, 18, 19] (28-col raw schema); these six were run through the preprocessing pipeline (§1.6, analysis/preprocessing/) to obtain c_{int}, c_p, J_w before regression. “Rows” is each source’s declared row count; the row count actually regressed (after validity filtering) is given per-dataset in §2.5.1/8. *Exclude* the mislabeled tab NF270_MC5 05.26.26_NaCl — its own note identifies it as a CaCl₂ run. S/S2 prefixes denote sequential/reverse (diluting) runs.

2.5.1 Results: all nine datasets

All nine NaCl runs were regressed with the corrected model (f_{corr} , full valid range, **CP-corrected, primary**). The **primary CP set is uniformly *our* thin-film CP**: MC2 (Ouimet’s run) and MC5(2) (Estrada’s run) (applied on top of the processed-sheet bulk c_{int} – both are the non-CP convention, confirmed by the $c_{\text{int}}/\kappa_{\text{ret}}$ coefficient-of-variation diagnostic, CV= 0.096 and 0.088 respectively) and the six newly preprocessed raw-only runs (reconstructed via §1.6’s pipeline, using the vial-ICP+CP *_CP.csv twins). **MC5-main is the lone exception**: its column B is already CP-corrected *by Estrada* (CV= 0.255, clearly not the non-CP cluster) – rather than double-applying our CP on top of hers (which would be wrong), it is kept as an explicitly-labeled *comparison* row, and **MC5(2)+our-CP is the primary “MC5 (05.27.26)” entry** in the table, figures, and cross-dataset comparisons below. Table 10 reports all nine together, marked by provenance; Figure 9 shows the point estimates and nonlinear confidence regions (primary datasets only – MC5-main comparison excluded from the plots to avoid a confusing second MC5 point), Figure 10 the per-dataset fits. (The detailed per-dataset appendix, §8, still shows the non-CP write-up (predating the CP-corrected regression re-run) for each run; refreshing all nine individual write-ups to CP-corrected is flagged as remaining work, not silently stale.)

Dataset	n	Provenance	$ \chi $ [mM] (95% CI)	δ^* (95% CI)	SSE
MC2 (05.07.24)	482	orig. proc. (our-CP)	21.42 (21.02, 21.82)	0.1900 (0.1879, 0.1921)	2.56
MC5 (05.27.26)	747	orig. proc. (our-CP, primary)	56.07 (55.06, 57.09)	0.2040 (0.2026, 0.2055)	125.04
<i>MC5 (05.27.26)[‡]</i>	<i>747</i>	<i>orig. proc. (Estrada’s CP, comparison)</i>	<i>49.56 (48.58, 50.55)</i>	<i>0.3251 (0.3232, 0.3270)</i>	<i>130.07</i>
MC3 (07.09.24)	391	reconstructed	<i>ill-conditioned (diverges)</i>		44192
MC3 (07.22.24, S)	1312	reconstructed	26.21 [†] (17.96, 37.89)	0.3025 (0.2583, 0.3596)	71798
MC4 (07.11.24, S)	317	reconstructed	122.23 (112.67, 132.73)	1.1768 (1.1298, 1.2289)	225.3
MC5 (07.23.24)	496	reconstructed	74.50 (72.83, 76.21)	0.4812 (0.4753, 0.4872)	43.8
MC5 (07.23.24, S)	500	reconstructed	134.79 (74.14, 365.73)	0.4001 (0.2568, 0.9462)	11734
MC5 (07.23.24, S2)	1324	reconstructed	77.71 (53.75, 121.04)	0.4484 (0.3451, 0.6306)	44830

Table 10: NaCl regression, **all nine datasets, CP-corrected (primary)**, corrected model, full valid range; 95% CIs are profile-likelihood. “orig. proc.” = arrived pre-processed; the primary CP set (MC2, MC5(2)→“MC5 (05.27.26)”, and the six reconstructed runs) all carry *our* thin-film CP applied to a non-CP bulk value; the *italicized* row (‡: the “(2)”-suffixed sheet’s twin, NF270.MC5 05.27.26.NaCl without the “(2)”) is Estrada’s own already-CP-corrected sheet, kept for comparison only (not double-CP’d, not part of the primary set/plots). “reconstructed” = built from raw data via §1.6’s vial-ICP+CP pipeline (a noisier, preliminary tier – see the Reconstruction-quality paragraph below). All non-ill-conditioned runs reached the global optimum in 100% of multi-starts. †MC3 07.22.24’s unconstrained fit converged to the mirror sign (−26.21 mM); harmless since f_{corr} depends only on χ^2 , shown here as $|\chi|$ for consistency with the other rows.

Reproducibility (originally-processed runs). Our-CP vs. Estrada’s CP: a third CP-related sensitivity axis. MC5(2)+our-CP ($|\chi| = 56.07$ mM) and MC5-main’s own CP (Estrada’s) ($|\chi| = 49.56$ mM) disagree by 13% (CIs do not overlap). This is *not* the non-CP-vs-CP question (point 11) or the calibration-source question (point 9, §1.7) – both MC5 entries already carry *some* CP correction. It is instead an **implementation-level discrepancy**: our Leveque-type thin-film mass-transfer coefficient k almost certainly differs from whatever CP procedure Estrada used (raw reconstruction already found our-CP reproduces Estrada’s CP MC5 sheet only to $\sim 12\%$ median – consistent with this $\sim 13\%$ parameter-level gap via the bulk-based mechanism too). (Flagged as a minor open item, not a correctness bug: the *qualitative conclusions* (NaCl well-identified, $|\chi|$ in the tens of mM) are unaffected either way.) **MC2, now correctly CP-corrected (it was mistakenly treated as already-CP in an earlier pass – its $c_{\text{int}}/\kappa_{\text{ret}}$ CV of 0.096 places it firmly in the non-CP cluster, not MC5-main’s 0.255), still disagrees sharply with MC5** ($|\chi| = 21.4$ vs. 56.1 mM, non-overlapping CIs; non-CP values were 18.5 vs. 49.6 mM – CP moves both but does not close the gap). We attribute this primarily to MC2’s much *narrower concentration range* ($c_{\text{int}} \lesssim 44$ mM vs. MC5’s $\lesssim 182$ mM): the Donnan curvature that identifies $|\chi|$ is weak at low concentration, so MC2 is a less informative experiment rather than necessarily a different membrane. (Neither cut’s CI contains the independent BoE reference $|\chi| \approx 44$ mM; MC5(2)+our-CP is now $\sim 27\%$ high (MC5-main, Estrada’s CP, is $\sim 13\%$ high), MC2+our-CP is $\sim 2\times$ low. Worth discussing whether the discrepancy is concentration-range identifiability, genuine cut-to-cut variation, or the missing-physics issues that also affect CaCl₂ (§3).)

Reconstruction quality: the six newly preprocessed raw-only runs. The six reconstructed runs (Table 10, now vial-ICP+CP) scatter widely and do *not* agree even among themselves — the three 07.23.24 MC5 runs alone give $|\chi| = 75, 135,$ and 78 mM — and one (MC3 07.09.24) is ill-conditioned (diverges to nonphysical $|\chi|$; excluded from Fig. 9, which cannot render a divergent optimum on a finite axis). (This scatter is almost certainly dominated by *reconstruction uncertainty*, not membrane physics: these runs require the full raw→processed chain ($\sim 3.8\%$ J_w smoothing error, per-

experiment calibration, vial-ICP (not the processed-sheet inline calibration), heuristic permeate handling — §1.6). The S/S2 reverse (diluting) runs are worst behaved (barely identified, extreme trim-sensitivity, and a visible high- c_{int} divergence in Fig. 10 – see the per-dataset appendix, §8, for exactly where each fit breaks down – *the appendix write-ups are not yet refreshed to these CP-corrected numbers*.) **The reliable reproducibility conclusion therefore still rests on the three originally-processed runs above;** the reconstructed tier does not sharpen identifiability and is preliminary pending the open preprocessing decisions (§1.6).

**NaCl point estimates $\pm$ 95% profile-likelihood CI
(corrected model, full range)**

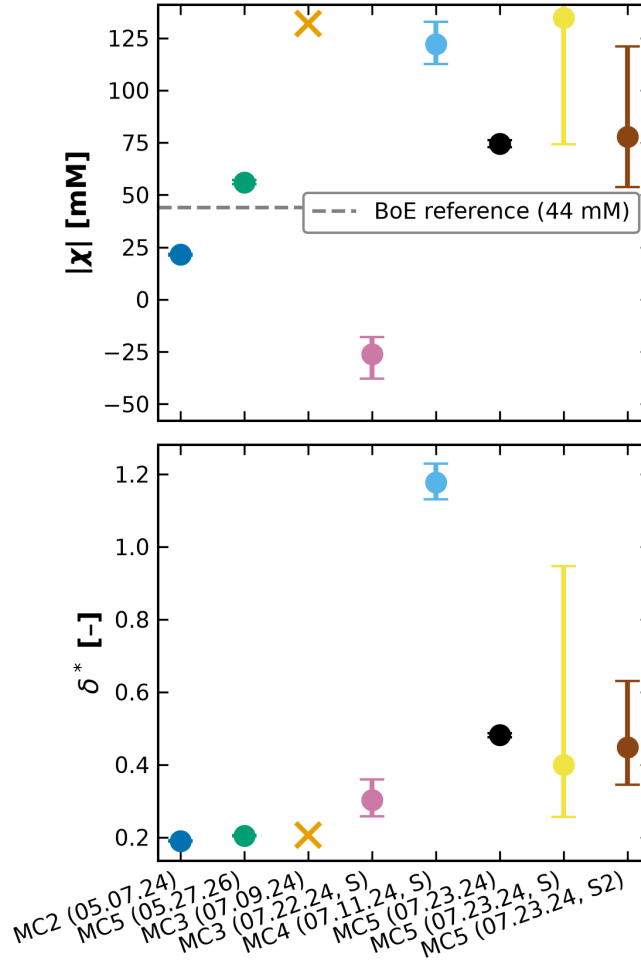


Figure 8: NaCl point estimates $\pm$ 95% profile-likelihood CI across the eight **primary** (uniformly our-CP) datasets (corrected model, full range, **CP-corrected**; MC3 07.09.24 omitted – ill-conditioned, see text; the MC5-main comparison row (Estrada’s CP) is excluded here to avoid a confusing second MC5 point, see Table 10). MC3 07.22.24 (S) plots at its raw fitted value (-26.21 mM, the unconstrained fit’s mirror-sign convergence, Table 10); read as $|\chi| = 26.21$ mM. Dashed line: BoE reference $|\chi| \approx 44$ mM.

95% nonlinear confidence regions across NaCl datasets
(top: per-dataset zoom on the 4 well-identified runs, each panel independently scaled)

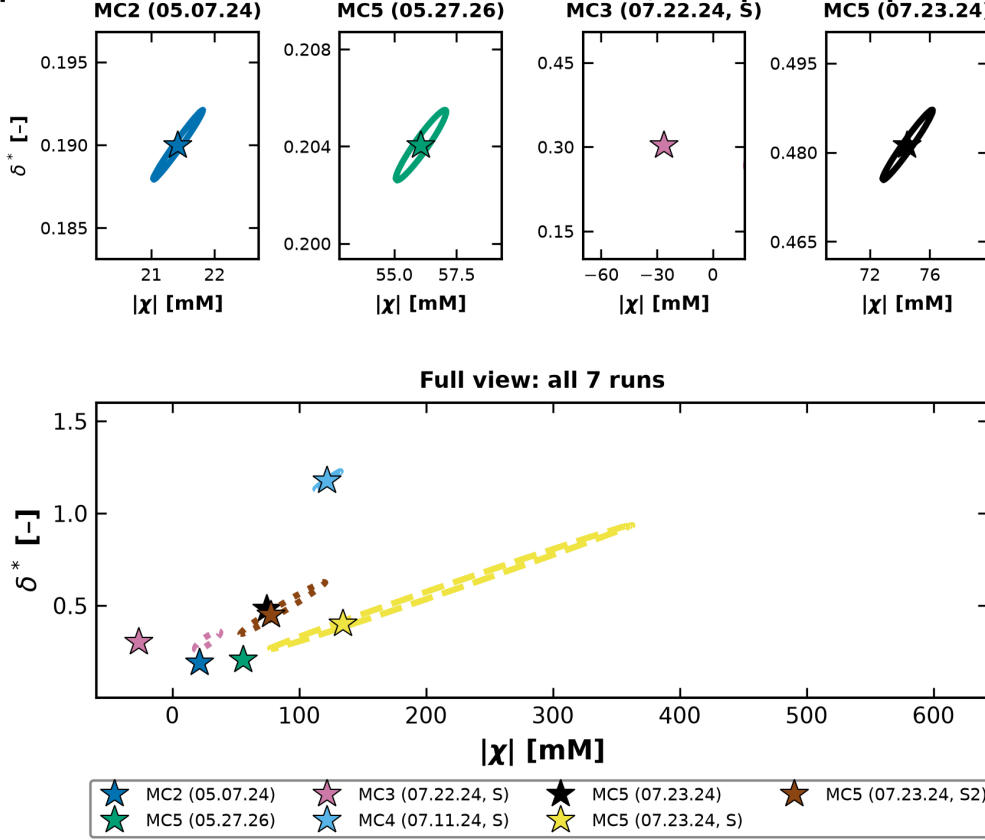


Figure 9: 95% nonlinear (F-test) confidence regions across the seven non-ill-conditioned **primary** NaCl datasets (uniformly our-CP; the the MC5-main comparison row (Estrada’s CP) is excluded, Table 10). **Top row**: a per-dataset zoom on the four *well-identified* runs (95% CI width < 15% on $|\chi|$) – each panel independently scaled to its own optimum ± 8 SE, so the region actually renders as a visible ellipse rather than collapsing to a dot. **Bottom**: the full view across all seven runs, for context, where the poorly-identified runs’ wide axis range *does* make the tight regions look like points – this is a visualization effect, not a numerical one (verified: the top-row zoom uses the identical §1.9 F-test region computation on the identical fits). *Caution*: a tight region reflects internal *precision*, not accuracy — MC5 (07.23.24) and MC3 (07.22.24, S) appear among the well-identified runs yet are *reconstructed*/CP-shifted runs whose estimates ($|\chi| \approx 75$ mM, 26 mM) are biased away from the primary MC5 (≈ 56 mM); only MC2 and MC5(2) (both our-CP, originally-processed) are both precise *and* trustworthy. Every dataset has a fixed, distinct color, reused across Figs. 8–10.

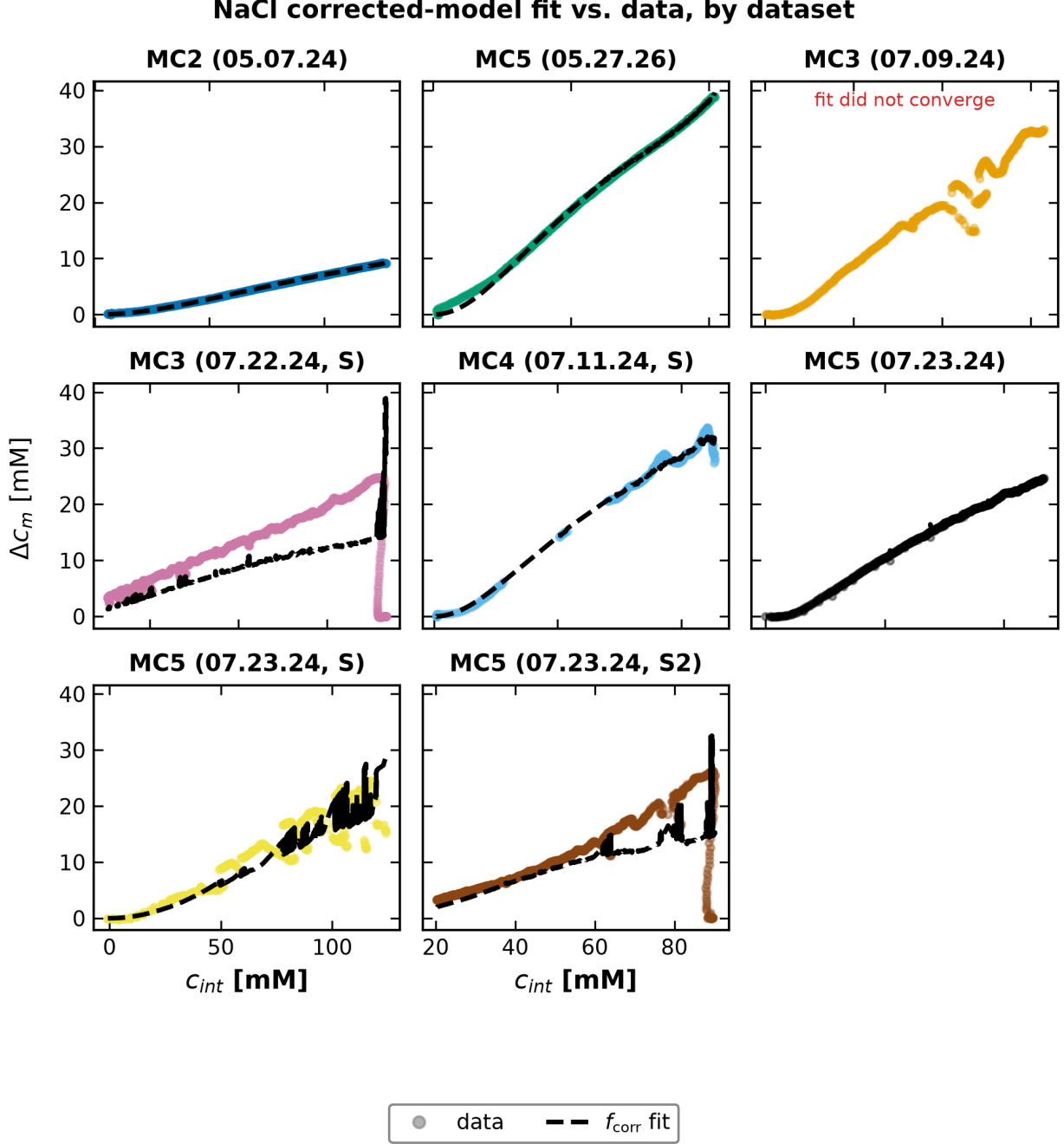


Figure 10: Per-dataset fits of the corrected Donnan model to Δc_m for the eight **primary** (uniformly our-CP) NaCl runs (grid; the MC5-main comparison row, Estrada’s CP, is excluded, Table 10). The two originally-processed runs (MC2, MC5(2)) fit the 1:1 model cleanly; the six reconstructed raw-only runs are visibly noisier.

2.6 Recommendations and discussion

- **MATLAB fix:** restore the $\frac{1}{2}$ in (18) and use Levenberg–Marquardt; the current estimates are otherwise a faithful (if reparameterized) optimum.

- **Uncertainty:** NaCl is well-identified, but the i.i.d. regions *were* optimistic as flagged here — **this is now quantified:** §2.3 shows the true (AR(1)-corrected) regions are $\sim 2.5\text{--}2.7\times$ wider (CP-corrected), and §2.4 shows the point estimate itself is sensitive to whether the regression is posed on transformed or raw measurements (+140% shift, CP-corrected). Both remain open decisions (§1.2, items 13–14), not resolved ones.
- **Physical check:** corrected, our-CP primary $|\chi| \approx 57$ mM (MC5(2)+our-CP; the MC5-main comparison value, Estrada’s own CP, is ≈ 50 mM) is the value to compare against zeta-potential / literature estimates.
- **Reproducibility (§2.5): our-CP vs. Estrada’s-CP is a real, if minor, implementation discrepancy:** MC5(2)’s our-CP $|\chi| = 56.07$ mM is 13% above MC5-main’s (Estrada’s-CP) 49.56 mM value (a third CP-related sensitivity axis, §2.5) — not a correctness issue, since both already carry *some* CP correction. MC2 (now correctly CP-corrected) vs. MC5(2) still diverge regardless of CP — most plausibly a concentration-range identifiability effect; prefer wide-range experiments for parameter estimation.
- **Current roadmap** (updated — several items of the original list are now **done**, see §5 for the cross-salt view): (i) tighten the raw-only NaCl reconstruction pipeline (Am/ρ , J_w -smoothing, §1.2 items 7–10, 12; CP-vs-non-CP, item 11, is now **resolved** — CP is the standing convention throughout, including the AR(1) and raw-measurement re-runs, §2.3/§2.4); (ii) a first-cut concentration-dependent $\chi(c)$ spline has now been tried on CaCl_2 ’s wide-range data (§6) — the constant-charge model is clearly inadequate, but the fitted shape is not yet safely interpretable (a follow-up null check found the confound is not resolved by a simple AR(1) correction either); **extending the AR(1) correction itself to $\text{CaCl}_2/\text{LaCl}_3$ remains pending**; (iii) the multicomponent extension remains deferred pending group feedback on the open decisions in §1.2.

3 Analysis of CaCl_2 data

The 2:1 co-ion cubic (12) replaces the NaCl quadratic; rather than its (known but unwieldy) Cardano closed form, we solve it directly (a fully vectorized closed-form root, §3.1) and fit it to every available processed CaCl_2 dataset, following the same workflow as §2.5. Uncertainty analysis proceeds as in §2.2 (profile-likelihood CIs, multi-start).

Working hypothesis: concentration-dependent charge from Ca^{2+} adsorption. We *expected the CaCl_2 fits to be poorer than NaCl*. The leading hypothesis is that Ca^{2+} adsorbs onto the membrane surface and alters its effective fixed charge χ , and that this adsorption is **concentration dependent**. The present model treats χ as a single constant per dataset and does not represent adsorption, so we anticipated (i) a worse overall fit and (ii) *systematic residual structure versus concentration* (equivalently, an apparent χ that drifts with c_{int}). Both are borne out below, with a more specific finding than anticipated: the global fit does not merely land on “some smaller χ than NaCl” – it drives $|\chi|$ to (numerically) zero for all three datasets, deferring all of the partitioning behavior to the lumped constant K (see (24)).

3.1 Model, datasets, and implementation

Co-ion and stoichiometry. Co-ion = Cl^- ; its solution concentration is set by stoichiometry, $c_{\text{co}}^s = 2 c_{\text{CaCl}_2}$ (2 Cl^- per CaCl_2), applied at both the feed face (c_{int}) and permeate face (c_p).

Governing equation. We fit the 2:1 cubic (12) with a single **lumped partition constant** $K \geq 0$ absorbing $\delta^* \delta_{\text{co}}^\circ$ and constant stoichiometric factors:

$$c_{\text{co}}^{\text{m}^3} + |\chi| c_{\text{co}}^{\text{m}^2} - K c_{\text{co}}^{\text{s}^3} = 0, \quad \text{prediction} = c_{\text{co}}^{\text{m}}(c_{\text{int}}) - c_{\text{co}}^{\text{m}}(c_{\text{p}}), \quad (24)$$

solved for the unique positive real root (Descartes’ rule of signs guarantees exactly one, for any real $|\chi|$). The transformed response mirrors the corrected NaCl form, but on the Cl^- flux (twice the CaCl_2 molar flux, by stoichiometry):

$$\Delta c_{\text{m}} = \frac{J_{\text{s,Cl}} \ell}{D_{\text{CaCl}_2, \text{m}}}, \quad J_{\text{s,Cl}} = 2 c_{\text{p}} J_{\text{w}}, \quad D_{\text{CaCl}_2, \text{m}} = \frac{3 D_{\text{Ca}, \text{m}} D_{\text{Cl}, \text{m}}}{2 D_{\text{Ca}, \text{m}} + D_{\text{Cl}, \text{m}}}, \quad (25)$$

with membrane-phase diffusivities $D_{\text{Ca}, \text{m}} = 0.79 \times 10^{-12} \text{ m}^2/\text{s}$, $D_{\text{Cl}, \text{m}} = 2.03 \times 10^{-12} \text{ m}^2/\text{s}$ (same $\sim 1000\times$ solution-to-membrane scaling Bill used for NaCl) and $\ell = 80 \text{ nm}$ as before. We fit ($|\chi|, K$) to Δc_{m} by `scipy.optimize.least_squares`, with $|\chi|, K \geq 0$ **bounded** (`method="trf"`) rather than unconstrained `lm`: unlike the NaCl model, where β_1 only ever appears squared inside a square root (so its sign cannot matter), $|\chi|$ enters (24) unsquared and asymmetrically, so an unconstrained fit can drift to a spurious negative “ χ ” with a negligibly lower SSE and no physical meaning (confirmed empirically: -0.06% SSE change for a sign flip on MC5 05.27.26). These are Opus’s working assumptions (per the task prompt to Sonnet), chosen to mirror the NaCl treatment closely enough to be directly comparable; a lumped-constant prefactor mismatch in K would bias absolute $|\chi|/K$ values but *not* the residual-shape/drift diagnostics that are the point of this section. ([Confirm the 2:1 non-ideal factor split \(\$\delta^* \delta_{\text{co}}^\circ\$ vs. \$K\$ alone\) with the group – discussion point in §1.8.](#))

Root-solve implementation. Rather than a per-point `numpy.roots` call (correct but far too slow at these array sizes, n up to ~ 1900 , called repeatedly inside multi-start and the sliding-window diagnostic below), `analysis/cac12/regress_cac12.py` solves (24) with a fully vectorized closed-form root in two regimes, selected by how small the true root is relative to $|\chi|$: an asymptotic-balance-plus-Newton-polish branch (avoiding the catastrophic cancellation a general Cardano solve suffers there) and a general complex Cardano branch otherwise. Verified against `numpy.roots` on a 5000-point random parameter scan to $< 10^{-9}$ relative error with zero non-finite outputs (`analysis/cac12/README.md`).

Datasets. All three processed CaCl_2 sheets, read from `data/BoE Analysis.xlsx` (the 27-sheet superset; *not* `data/Rejection Analysis.xlsx`, which is missing the 06.27.26 run), same 15-column schema as NaCl (columns B/H/J):

Membr.	Sheet	Valid rows	c_{int} range
MC3	NF270_MC3 07.11.24_SCaCl2	609	1.0–47 mM
MC5	NF270_MC5 05.27.26_CaCl2	1645	1.6–108 mM
MC5	NF270_MC5 06.27.26_CaCl2	1885	1.3–103 mM

Table 11: The three processed CaCl_2 datasets analyzed in this section. The MC5 05.27.26 vs. 06.27.26 pair is a near-repeat (reproducibility check, see below). **Not yet processed** (raw-only, out of scope here): MC2 05.07.24, MC3 07.11.24_CaCl2, MC3 07.12.24_S2CaCl2, and the mislabeled tab `NF270_MC5 05.26.26_NaCl` (its own note identifies it as CaCl_2 , not NaCl – also excluded from the NaCl accounting, Table 9).

3.2 Results: all three datasets

All three processed CaCl_2 runs were regressed with the bounded ($|\chi|, K$) model, full valid range, **CP applied on top of the processed-sheet bulk** c_{int} (all three sheets are the non-CP convention, §1.6 raw-reprocessing validation; $k_{\text{CaCl}_2} = 2.25 \times 10^{-5} \text{ m/s}$). Table 12 reports all three together; Figure 11 shows the per-dataset fits and Figure 12 the point estimates with 95% profile-likelihood CIs. *Full fit/residual/drift detail for every individual run is in the appendix, §9 (not yet refreshed to these CP-corrected numbers).*

Dataset	n	$ \chi $ [mM] (95% CI)	K (95% CI)	SSE	pseudo- R^2	MS %
MC3 (07.11.24, S)	609	0.0000 (0.000, 34.4)	0.01116 (0.01100, 0.01130)	147.1	0.979	100%
MC5 (05.27.26)	1645	2.9466 (2.288, 3.596)	0.00057 (0.00056, 0.00058)	418.3	0.959	99%
MC5 (06.27.26)	1885	0.0494 (0.000, 0.617)	0.00023 (0.00022, 0.00023)	666.5	0.892	96%

Table 12: CaCl_2 regression, all three datasets, **CP-corrected (primary)**, bounded 2:1 cubic model, full valid range; 95% CIs are profile-likelihood; MS = multi-start % reaching the global optimum, 100 random starts. **CP sharpens, but does not overturn, the headline finding:** $|\chi|$ is still pinned at (or extremely close to) zero for two of the three datasets; MC5 (05.27.26) — the widest-range dataset — now resolves to a small but statistically *nonzero* $|\chi| \approx 2.9 \text{ mM}$ (CI excludes zero), still $\approx 19\times$ smaller than NaCl’s $\approx 57 \text{ mM}$. Fit quality improves under CP across the board (pseudo- R^2 : 0.84–0.95 $\rightarrow$ 0.89–0.96). MC3’s much wider CI reflects its narrower concentration range ($\lesssim 47 \text{ mM}$ vs. MC5’s $\gtrsim 100 \text{ mM}$), the same range-identifiability effect noted for NaCl’s MC2 vs. MC5 (§2.5.1).

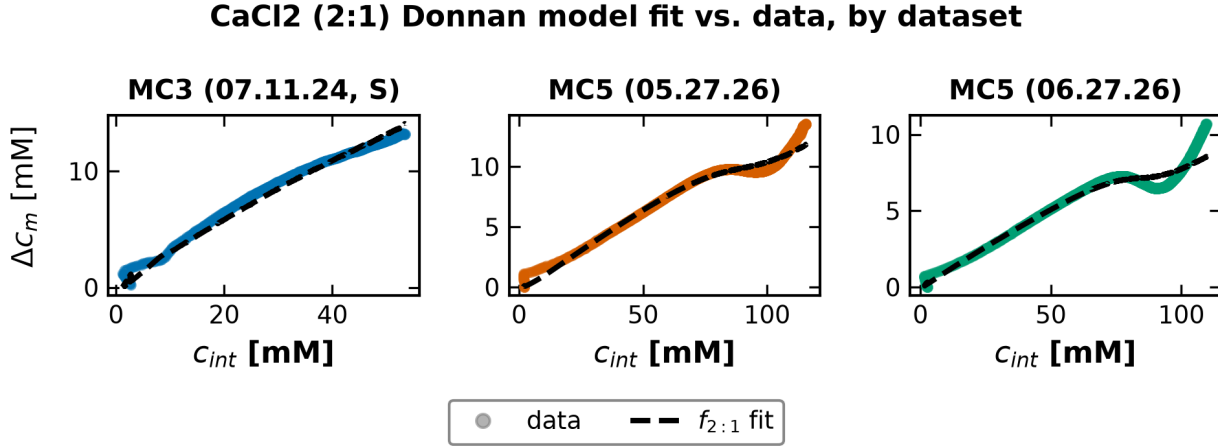


Figure 11: CaCl_2 corrected-model (2:1 cubic) fit vs. data, by dataset. The model tracks the overall trend, but see Figure 13 for the systematic residual structure the point estimate above does not reveal.

**CaCl₂ point estimates $\pm$ 95% profile-likelihood CI
(2:1 cubic model, full range)**

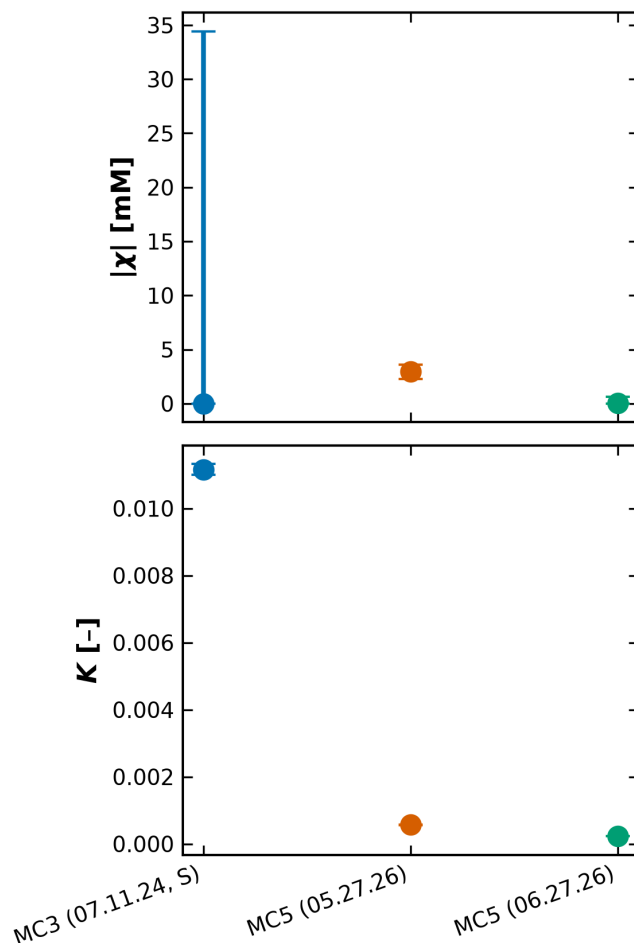
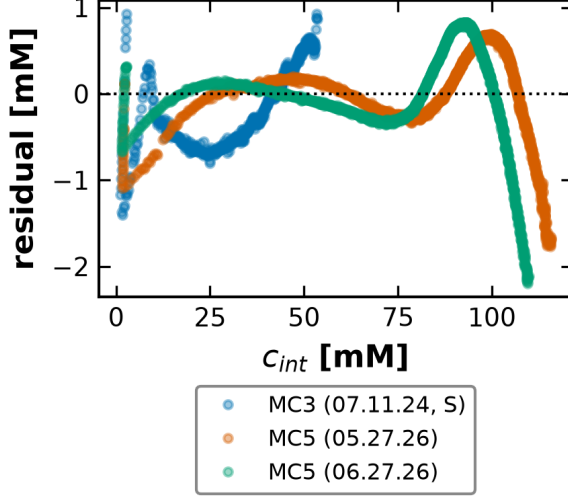


Figure 12: CaCl₂ point estimates $\pm$ 95% profile-likelihood CI, all three datasets, CP-corrected. The two MC5 runs no longer fully agree under CP (05.27.26's CI excludes zero, 06.27.26's does not); MC3's much wider $|\chi|$ CI reflects weaker identifiability at its narrower concentration range.

3.3 Adsorption-hypothesis diagnostics

Residuals vs. concentration (Figure 13). All three datasets show a clear, smooth, *non-random* wave pattern (CP-corrected, as on non-CP). A sign-runs test (fewer sign changes than random residuals would produce) is strongly significant for all three ($z = -23.1, -39.2, -42.3$ for MC3, MC5 05.27.26, MC5 06.27.26 respectively – z this far from 0 simply confirms the visually obvious structure is not noise). This matches the anticipated signature: the single- $|\chi|$ model cannot capture whatever concentration-dependent physics is really at play.

Residuals vs. concentration (adsorption-hypothesis diagnostic)



Effective $|\chi|$ vs. concentration (adsorption drift diagnostic; x = unreliable window)

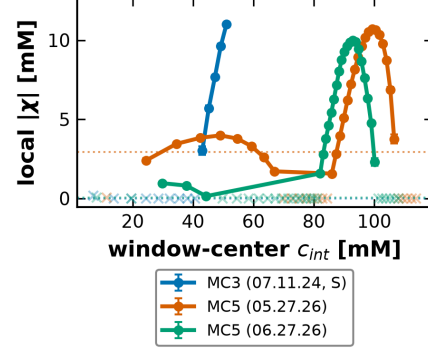


Figure 13: **Left:** residuals vs. c_{int} – the key adsorption-hypothesis diagnostic; all three datasets show the same smooth non-random wave shape. **Right:** sliding-window (80 points/window) local $|\chi|$ refits (K fixed at the global estimate); unreliable windows (large SE or pinned at the window search-box bound, mostly at very low c_{int}) shown as faint $\times$ and excluded from the trend read below.

Effective $|\chi|$ vs. concentration (Figure 13, right). *Method* (used here, in §4.3, and as the model-free cross-check in §6): sort the run by c_{int} and slide a fixed-size window along it (80 consecutive points, stride 40); within each window W_j refit a *single local* charge, holding the partition constant at its global value $\hat{K}$,

$$\hat{\chi}_j = \arg \min_{\chi \in [\varepsilon, \chi_{\max}]} \sum_{i \in W_j} \left[\Delta c_{\text{m}}^{(i)} - (c_{\text{co}}^{\text{m}}(c_{\text{int}}^{(i)}; \chi, \hat{K}) - c_{\text{co}}^{\text{m}}(c_{\text{p}}^{(i)}; \chi, \hat{K})) \right]^2, \quad (26)$$

and plot $\hat{\chi}_j$ against the window-mean concentration $\bar{c}_j = \frac{1}{|W_j|} \sum_{i \in W_j} c_{\text{int}}^{(i)}$. A window is flagged *unreliable* (faint $\times$, excluded from the trend) when $\hat{\chi}_j$ pins at the search bound $\chi_{\max}$ or its local Wald standard error is large relative to $\hat{\chi}_j$ — i.e. its narrow c_{int} sub-range does not identify a local charge. Holding $\hat{K}$ fixed makes each $\hat{\chi}_j$ a one-parameter read of how much fixed charge that concentration slice *alone* implies; it is a *diagnostic* of whether a single constant charge suffices, not a fitted $\chi(c)$ model (that is §6). Both MC5 runs still show a **non-monotonic (inverted-U) drift** under CP: local $|\chi|$ rises from near 0 at low c_{int} to a peak of ~ 10 – 11 mM around $c_{\text{int}} \approx 90$ – 100 mM, then falls back down at the highest concentrations. MC3’s few reliable windows (5 of 14; the rest are too weakly identified over its narrow range to trust and are excluded from the trend) rise from ~ 3 to ~ 11 mM over its range – consistent with sitting on the same *rising* leg MC5 shows over a wider range. (This is drift, but not a simple monotonic one: a naive “adsorption increases with concentration, monotonically suppressing $|\chi|$ ” story predicts a monotonic *decrease*; what is observed is a rise-then-fall. Two non-exclusive readings: (i) Ca^{2+} adsorption may already be saturated/screening the charge even

at low c_{int} , with the mid-range rise-then-fall reflecting an unaccounted effect such as concentration polarization (not corrected for CaCl_2 here, §1.5); or (ii) the windowed local-refit diagnostic's shape may be partly an artifact of how each window's c_{int} sub-range interacts with the K held fixed at the global estimate, rather than a direct read of the physics. Either way, $|\chi|$ is clearly not flat with concentration for any of the three datasets – worth resolving with the group before committing to a specific $|\chi|(c)$ functional form.)

Reproducibility: MC5 05.27.26 vs. 06.27.26. CP separates these two runs that previously agreed: under CP, 05.27.26 resolves to a small but nonzero $|\chi| = 2.95 \text{ mM}$ (CI excludes 0) while 06.27.26 stays pinned near zero (0.049 mM, CI includes 0) – their CIs no longer overlap. 06.27.26 still has a visibly worse pseudo- R^2 (0.892 vs. 0.959) – worth a follow-up look at whether 06.27.26 has more noise or a genuinely different high-concentration behavior (e.g. a different membrane fouling state on that date), and now also whether the CP shift itself is partly responsible for the split.

Verdict. The adsorption hypothesis is supported, with a specific and unexpected refinement – and CP sharpens rather than overturns it. The fit is materially different from NaCl's (chi pinned at/near 0 for two of three datasets vs. NaCl's clean 18–57 mM), residuals show strong, consistent, non-random structure vs. concentration in all three datasets, and the local effective $|\chi|$ demonstrably drifts with concentration rather than staying flat – all as anticipated. That the *global* fit drives $|\chi|$ to (near) zero for two of three datasets, rather than landing on “some smaller nonzero value than NaCl,” is a sharper piece of evidence than expected; MC5 (05.27.26)'s CP-corrected $|\chi| \approx 2.9 \text{ mM}$ is the exception (small but statistically significant, still $\approx 19\times$ below NaCl), consistent with the local drift diagnostic already showing nonzero effective $|\chi|$ over parts of that same dataset's range – a single global constant necessarily averages that concentration-dependence away. The non-monotonic shape of that drift is the open question for follow-up – most likely resolved by either a concentration-dependent $|\chi|(c)$ model extension or ruling out concentration-polarization/window-diagnostic artifacts first (see the note above) – concentration polarization is no longer uncorrected-for at the global-fit level (this section), but the sliding-window diagnostic itself does not re-derive CP locally per window.

4 Analysis of LaCl_3 data

The 3:1 co-ion quartic (13) has **no compact closed form** (unlike NaCl's quadratic or CaCl_2 's cubic). Rather than coding an explicit (numerical) quartic solution as the only route, this section fits it **two independent ways and cross-checks** them, mirroring §2.2's `scipy-vs-ParmEst` validation: (A) a bracketed root-find per point, driving a `scipy.optimize.least_squares` fit; (B) the membrane co-ion concentration introduced as a Pyomo decision *variable*, **constrained** by the quartic (positivity-bounded to select the physical root), estimated via ParmEst/Ipopt. Method B needs no closed-form root or explicit residual function at all — it is the formulation that **generalizes directly to the eventual multicomponent extension**, where no closed form exists for any salt combination, so validating it here is directly useful beyond LaCl_3 itself.

4.1 Model, dataset, and implementation

Co-ion, stoichiometry, and governing equation. Co-ion = Cl^- ; $c_{\text{co}}^s = 3 c_{\text{LaCl}_3}$ (3 Cl^- per LaCl_3), at both c_{int} and c_p . We fit the 3:1 quartic (13) with a lumped constant $K \geq 0$ absorbing

$\delta^*(\delta_{\text{co}}^\circ)^2$ and stoichiometric factors:

$$c_{\text{co}}^{\text{m}4} + |\chi| c_{\text{co}}^{\text{m}3} - K c_{\text{co}}^{\text{s}4} = 0, \quad \text{prediction} = c_{\text{co}}^{\text{m}}(c_{\text{int}}) - c_{\text{co}}^{\text{m}}(c_{\text{p}}), \quad (27)$$

with a unique positive real root (Descartes: sign pattern $+, +, 0, 0, -$). The transformed response mirrors CaCl₂'s form, on the $3 \times \text{Cl}^-$ flux: $\Delta c_{\text{m}} = J_{\text{s,Cl}} \ell / D_{\text{LaCl}_3, \text{m}}$, $J_{\text{s,Cl}} = 3c_{\text{p}} J_{\text{w}}$, $D_{\text{LaCl}_3, \text{m}} = \frac{4D_{\text{La}, \text{m}} D_{\text{Cl}, \text{m}}}{3D_{\text{La}, \text{m}} + D_{\text{Cl}, \text{m}}}$, $D_{\text{La}, \text{m}} = 0.626 \times 10^{-12} \text{ m}^2/\text{s}$, $D_{\text{Cl}, \text{m}} = 2.03 \times 10^{-12} \text{ m}^2/\text{s}$, $\ell = 80 \text{ nm}$. As for CaCl₂, $(|\chi|, K)$ are fit **bounded** ≥ 0 : $|\chi|$ enters (27) unsquared/asymmetrically, so an unconstrained fit can drift to a spurious negative value with no physical meaning. These are Opus's working assumptions (mirroring §3.1), sufficient for the hypothesis test; a lumped-constant prefactor mismatch would bias absolute $|\chi|/K$ but not the residual-shape/drift diagnostics. (Confirm the 3:1 non-ideal factor $(\delta_{\text{co}}^\circ)^2$ vs. K alone with the group, as for CaCl₂ – §1.8.)

Method A: bracketed root-find. A per-point Brent search (expanding-bracket for the upper endpoint), verified against `numpy.roots` on a 3000-point random parameter scan to $< 10^{-6}$ relative error, $\sim 2.5 \times$ faster than a per-point `numpy.roots` call at this dataset's scale ($n = 623$).

Method B: ParmEst nonlinear-constraint quartic. Each experiment introduces auxiliary Vars $c_{\text{co}, \text{f}}^{\text{m}}, c_{\text{co}, \text{p}}^{\text{m}}$ (membrane co-ion concentration at the feed/permeate face), each bounded positive and constrained by (27) at that face's c_{co}^{s} ; the response constraint is $\Delta c_{\text{m}} = c_{\text{co}, \text{f}}^{\text{m}} - c_{\text{co}, \text{p}}^{\text{m}}$. ParmEst/Ipopt solves the joint extensive-form NLP over all experiments, tying $(|\chi|, K)$ to shared decision variables — no explicit residual function or root formula appears anywhere in this formulation.

Dataset. One processed LaCl₃ run: NF270_MC2 05.21.24.LaCl3 in `data/BoE Analysis.xlsx` (also in `Rejection Analysis.xlsx`), $n = 623$, standard 15-column schema. **Narrow concentration range**, $c_{\text{int}} \approx 2\text{--}28 \text{ mM}$. A raw-only MC4 07.11.24.SLaCl3 run exists in the raw campaign workbook but is **out of scope here** (would need the preprocessing pipeline of §1.6; noted as future work, as the raw-only NaCl runs were before §2.5's pipeline was built). **Anticipated headwinds** (both realized below): (i) the narrow range gives weak $|\chi|$ identifiability, the same effect flagged for NaCl's MC2 and CaCl₂'s MC3 (07.11.24, S); (ii) low La³⁺ rejection may ill-condition the fit.

4.2 Results

Dataset	n	$ \chi $ [mM] (95% CI)	K (95% CI)	SSE	pseudo- R^2
MC2 (05.21.24)	623	0.0000 (0.000, 34.48)	3.281×10^{-8} (3.163, 3.404×10^{-8})	4.633	0.906

Table 13: LaCl₃ regression, **CP-corrected (primary)**, bounded 3:1 quartic model, Method A, full valid range ($k_{\text{LaCl}_3} = 2.21 \times 10^{-5} \text{ m/s}$, CP applied on top of Ouimet's processed-sheet bulk c_{int} – this sheet is the non-CP convention, §1.6 raw-reprocessing validation); 95% CI is profile-likelihood; 100% multi-start convergence, 100 starts. **One processed LaCl₃ dataset is analyzed here** – the raw-only MC4 run (§4.1) is future work. $|\chi|$ is **pinned at its lower bound** (≥ 0), even more extremely than CaCl₂'s near-zero estimates (Table 12), and **CP narrows the CI and improves fit quality** (pseudo- R^2 : $0.803 \rightarrow 0.906$; non-CP CI was (0, 94 mM)).

(This is a *boundary* solution, not an interior optimum: the CI (0, 34 mM) is one-sided against the $|\chi| \geq 0$ bound. $|\chi|$ is therefore essentially **unidentified** from this dataset, not “measured to be 0” — the data are simply uninformative about it in either direction.)

Method A vs. Method B agreement. ParmEst (warm-started at Method A’s converged root, for a fair, strong comparison) gives $|\chi| = 1.196 \times 10^{-3}$ mM, $K = 3.28128 \times 10^{-8}$, SSE = 4.6337, vs. Method A’s $|\chi| = 2.41 \times 10^{-18}$ mM, $K = 3.28117 \times 10^{-8}$, SSE = 4.6332 (CP-corrected). K **matches to 5 significant figures and SSE to $< 0.01\%$** between the two independent numerical routes (bracketed-Brent scipy fit vs. Pyomo/Ipopt nonlinear-constraint NLP) — a clean cross-validation, unchanged in character by CP. $|\chi|$ ’s raw relative difference is large only because *both* methods independently converge to a value indistinguishable from zero; dividing two near-zero numbers is not a meaningful relative comparison. The substantive, cross-validated result is that $|\chi|$ **is not identifiable from this dataset, and both methods agree it is effectively zero.**

4.3 Adsorption-hypothesis diagnostics

Residuals vs. concentration (Figure 14). A strong, smooth, systematic wave persists under CP (as on non-CP). Sign-runs test: **2 runs observed vs. ≈ 295 expected under randomness**, $z = -24.9$ — about as extreme a non-random signature as is observable on $n = 623$ residuals (two long monotonic-sign stretches, not the alternating pattern white noise would produce).

Effective $|\chi|$ vs. concentration. Of 19 sliding windows, only 7 (all at the high-concentration end) are reliably identified; low/mid-range windows are unidentified (pinned at/near the search-box bound) and excluded from the trend. Among the reliable windows, local $|\chi|$ rises monotonically from ≈ 0.07 to ≈ 0.59 mM with concentration (CP-corrected; non-CP was 0.06–0.85 mM) — i.e. even where a local estimate is possible at all, $|\chi|$ stays two orders of magnitude below NaCl’s 18–57 mM and below CaCl₂’s already-small estimates, with a simple monotonic (not CaCl₂’s non-monotonic inverted-U) trend.

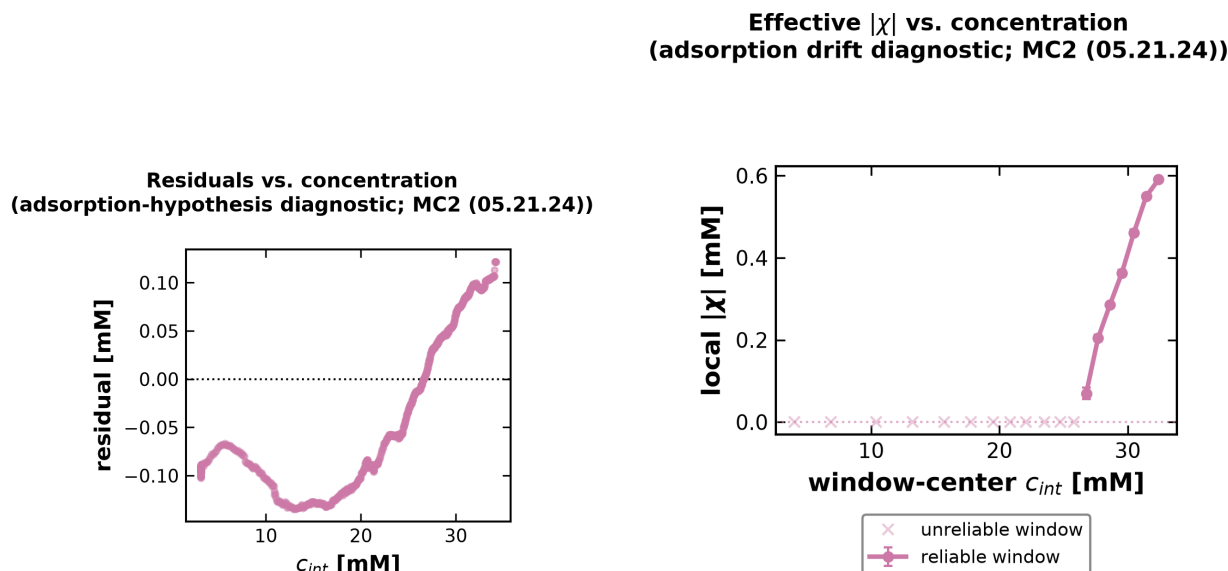


Figure 14: **Left:** residuals vs. c_{int} — the key adsorption-hypothesis diagnostic; a smooth, strongly non-random wave ($z = -24.9$). **Right:** sliding-window local $|\chi|$ refits (K fixed at the global estimate); unreliable windows (12 of 19, mostly at low/mid c_{int}) shown as faint $\times$ and excluded from the trend.

**LaCl₃ (3:1) Donnan model fit vs. data
MC2 (05.21.24)**

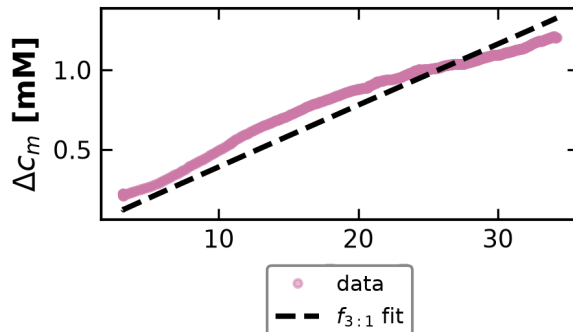


Figure 15: LaCl₃ corrected-model (3:1 quartic) fit vs. data, MC2 (05.21.24). The data’s S-shaped curvature versus the fit’s near-linear trend (a consequence of $|\chi| \rightarrow 0$, leaving only the K -scaling term) is exactly the residual structure quantified in Figure 14.

Comparison across salts and verdict. $|\chi| \rightarrow 0$ was already CaCl₂’s headline finding for two of three datasets (§3.3); LaCl₃ pushes this further — the global estimate is pinned *exactly* at the $|\chi| \geq 0$ bound (not just small), and only a narrow high-concentration slice of even the local windowed estimates ever exceeds numerical noise. Fit quality (pseudo- $R^2 = 0.906$, CP-corrected; was 0.803 non-CP) is now *comparable to or better than* the CaCl₂ datasets (0.89–0.96, Table 12), though the $|\chi| \rightarrow 0$ pinning itself is unchanged and remains the more extreme signature of the three salts. **Verdict: the adsorption hypothesis extends to La³⁺, and more severely than for Ca²⁺ — and this survives CP.** The trend NaCl (well-identified, nonzero $|\chi| \approx 57$ mM, no adsorption signature) $\rightarrow$ CaCl₂ ($|\chi| \rightarrow 0$ for two of three datasets, small but nonzero for the widest-range one, moderate residual structure) $\rightarrow$ LaCl₃ ($|\chi|$ pinned at exactly 0, strongest residual structure of the three) tracks increasing cation charge/adsorption affinity, as expected physically. **Critical caveat for this trend:** LaCl₃’s $|\chi| \rightarrow 0$ is *confounded* with its narrow 2–28 mM range and low La³⁺ rejection (weak identifiability, §4.1) — whereas CaCl₂’s $|\chi| \rightarrow 0$ came from *wide-range* data ($\rightarrow 108$ mM, Table 11). The NaCl $\rightarrow$ CaCl₂ adsorption trend therefore rests firmly on two wide-range comparisons (NaCl $|\chi| \approx 57$ mM vs. CaCl₂ mostly $\rightarrow 0$); the LaCl₃ point is *consistent but range-confounded* — a single narrow-range dataset with no repeat (the raw-only MC4 SLaCl3 run is future work) — so it should be read as *suggestive, not confirmatory* of the trend continuing to 3:1. (The monotonic, not inverted-U, shape of LaCl₃’s effective- χ drift is plausibly explained by the same range effect: a 2–28 mM window samples only the low- c_{int} *rising leg* that CaCl₂ also shows at low concentration, §3.3, not the full rise-then-fall.) A wider-range LaCl₃ dataset would strengthen the conclusion; both the nonlinear-constraint approach and these diagnostics carry directly into the eventual multicomponent extension.

5 Cross-salt synthesis and discussion

§2–4 developed the NaCl, CaCl₂, and LaCl₃ analyses largely independently; this section ties the 1:1/ 2:1/3:1 story together and states what carries across all three, cross-referencing rather than

restating the per-salt numbers.

5.1 The valence trend

A constant-charge Donnan model **fails progressively for multivalent counter-ions**:

- **NaCl (1:1, §2.1–2.2)**: a clean, well-identified fit on the primary dataset, **CP-corrected (primary, §2.2)**, $|\chi| \approx 56.74$ mM, $\delta^* \approx 0.2049$ (non-CP: 50.36 mM, 0.3263 — same order of magnitude), with the nonlinear region visually indistinguishable from the Wald ellipse. Across all nine datasets (§2.5), the primary CP set is uniformly *our* thin-film CP (MC2, MC5(2)→“MC5 (05.27.26)”, and the six reconstructed runs); the lone dataset with its own already-applied CP (MC5-main, Estrada’s own CP) is kept as an explicit comparison, not a primary row, since it disagrees with MC5(2)+our-CP by 13% (49.56 vs. 56.07 mM) — an our-CP-vs-Estrada’s-CP implementation discrepancy (§2.5.1), not a non-CP-vs-CP question. MC2, now correctly identified as needing CP (it was mistakenly classified already-CP in an earlier pass), still disagrees cut-to-cut with MC5(2) (MC2 $|\chi| = 21.42$ mM vs. 56.07 mM) — a concentration-range effect (§5.2), not evidence against the model itself.
- **CaCl₂ (2:1, §3)**: under CP, *two of three* datasets still drive $|\chi| \rightarrow 0$ (0.0000–0.0494 mM), while the widest-range dataset (MC5 05.27.26) now resolves to a small but *statistically nonzero* $|\chi| = 2.95$ mM (CI excludes zero, Table 12) — still $\approx 19\times$ below NaCl. All three show strong non-random residual structure (sign-runs $z = -23.1$ to -42.3) and effective- χ drift with concentration (non-monotonic, inverted-U). This remains *sharper* than “a smaller χ than NaCl” — the co-ion exclusion effect appears mostly masked on average, consistent with concentration-dependent Ca^{2+} adsorption; CP *sharpens* (reveals a small residual signal for one dataset) rather than overturns this.
- **LaCl₃ (3:1, §4)**: pushes this further still — $|\chi|$ is pinned *exactly* at its ≥ 0 bound (a boundary solution, not an interior optimum, unchanged by CP), fit quality *improves under CP* (pseudo- $R^2 = 0.906$, up from 0.803 non-CP — now comparable to or better than CaCl₂), and the residual sign-runs statistic ($z = -24.9$ on $n = 623$) is the most extreme in the report. **But this point is range-confounded** (§4.3): LaCl₃’s one dataset spans only 2–28 mM, while CaCl₂’s $|\chi| \rightarrow 0$ result (for two of three datasets) held over a *wide* range ($\rightarrow 108$ mM). The reliable part of the valence trend is therefore **NaCl (wide-range, $|\chi| \approx 57$ mM) vs. CaCl₂ (wide-range, mostly $\rightarrow 0$)**; LaCl₃ is *consistent with, but does not yet confirm*, the trend continuing to 3:1 — a wider-range LaCl₃ dataset (or the raw-only MC4 run) would settle it.

5.2 Methodological cross-cuts

Several findings recur across salts and are properties of the *method*, not of any one dataset:

1. **Precision $\neq$ accuracy.** A tight confidence region reflects internal precision, not correctness. Reconstructed NaCl run MC5 (07.23.24) is tightly identified ($|\chi| \approx 75$ mM, Fig. 9) yet biased away from the primary MC5’s ≈ 56 mM (both our-CP) — low-information or reconstruction-limited runs can look precise while being wrong.
2. **Identifiability needs concentration range.** The Donnan curvature that pins $|\chi|$ is a higher-concentration feature. NaCl’s MC2 ($\lesssim 44$ mM), CaCl₂’s MC3 07.11.24 ($\lesssim 47$ mM), and LaCl₃’s sole dataset ($\lesssim 28$ mM) are all narrow-range and correspondingly weakly identified (or, for LaCl₃, entirely unidentified); MC3 07.09.24-NaCl’s low rejection compounds

the same issue into outright divergence. **Design wide-range experiments for parameter estimation.**

3. **Reconstruction-dominated $\neq$ physics.** The six raw $\rightarrow$ processed reconstructed NaCl runs (§2.5.1, now vial-ICP+CP) scatter widely among themselves ($|\chi| = 75, 135, 78$ mM for the three 07.23.24 MC5 runs alone) — this is reconstruction uncertainty (calibration, J_w smoothing, vial-ICP-vs-processed-sheet-inline calibration source, §1.6), not membrane physics. The reliable reproducibility conclusion rests on the three originally-processed runs.
4. **Working assumptions bias absolute values, not diagnostic shapes.** The lumped K ($\text{CaCl}_2/\text{LaCl}_3$), Cl^- stoichiometry, ambipolar D_m , and the δ° treatment for $z \neq 1$ (§1.2, item 3) all shift the absolute $|\chi|/K$ values but *not* the residual-vs-concentration or effective- χ adsorption diagnostics — so the qualitative conclusions (clean 1:1; adsorption-driven 2:1/3:1) are robust even while the absolute parameter conventions await group consensus.
5. **Autocorrelation is pervasive.** The dense, single-run time series structure that inflates NaCl’s i.i.d. confidence regions by $\sim 2.5\text{--}2.7\times$ (CP-corrected; §2.3) is a property of *how the data are collected*, not of NaCl specifically — the same correction is needed for CaCl_2 and LaCl_3 (not yet run; §1.2, item 13).
6. **Formulation / errors-in-variables sensitivity.** Transforming measurements into the response (Δc_m) vs. regressing directly on the raw signal (κ_p) shifts NaCl’s $|\chi|$ by +140% (CP-corrected; §2.4) — which estimate is trustworthy depends on where the measurement error actually lives, and neither is fully validated yet: CP substantially improves the raw fit’s AR(1) behavior (no longer near-unit-root, $\hat{\phi} : 0.999 \rightarrow 0.900$) but does not resolve the point-estimate gap, which it in fact widens.
7. **Cross-validating independent implementations catches real bugs.** `scipy-vs-ParmEst` agreement (NaCl to 6–7 significant figures, §2.2; LaCl_3 ’s K to 5 figures via the nonlinear-constraint formulation, §4.2) is how several subtle errors were actually caught: the MATLAB `dogbox` local minimum, ParmEst’s `measurement_error` covariance trap, and the AR(1) profile **double-counting** (§2.3 – combining GLS whitening with an n_{eff} degrees-of-freedom correction inflated the profile CI to a spurious $\sim 8\times$, caught by an internal audit before it reached collaborators). This is a practice worth continuing, not a one-off.

5.3 Implications and next steps

The valence trend and cross-cuts above point to the same two next steps:

- **A concentration-dependent $\chi(c)$ model.** A constant χ cannot fit $\text{CaCl}_2/\text{LaCl}_3$; both show effective- χ drift with concentration (§3.3, §4.3). The proposed extension is a data-driven $\chi(c)$ (or $\chi(t)$) spline replacing the constant- χ closure for multivalent counter-ions (§1.2, item 15) — CaCl_2 ’s wide range makes it the better dataset to develop this on first.
- **The multicomponent barrier.** Extending to multi-salt feeds needs (i) an MSA multi-salt conductivity model plus per-ion ICP to resolve multiple ionic contributors from one scalar conductivity signal (§1.2, item 16), and (ii) validating single-salt χ/B transferability to mixtures (§1.5). The 3:1 nonlinear-constraint formulation (§4.1) — no closed form, membrane concentrations as Pyomo variables constrained by the Donnan polynomial — is exactly the approach that scales to this eventual multicomponent case.

- **Uncertainty quantification catching up across salts.** AR(1) (or better) for $\text{CaCl}_2/\text{LaCl}_3$, and resolving the raw-measurement-vs-transformed question for at least one multivalent salt, remain open (§1.2, items 13–14).

6 Toward a concentration-dependent membrane charge $\chi(c)$ (preliminary)

Both CaCl_2 (§3.3) and LaCl_3 (§4.3) resist a *single constant* fixed charge: the residuals carry smooth, concentration-ordered structure and a sliding-window “effective $|\chi|$ ” drifts with concentration, consistent with concentration-dependent counter-ion adsorption. This section foreshadows a possible next step — replacing the scalar $|\chi|$ with a smooth function $\chi(c)$ regressed *jointly* with the partition constant — and reports a deliberately simple first cut. It is **exploratory**: it establishes what the data demand, not a settled model.

6.1 A varying-coefficient (spline) model

The idea is a *varying-coefficient* regression (a model where one parameter, instead of being a single fixed number, is itself a smooth function of another variable — here, of concentration): keep the Donnan transport model but promote one parameter from a scalar to a smooth function of concentration. Because χ is a property of the membrane in equilibrium with the *local* external solution, it is evaluated **at each face separately**. For NaCl (1:1) the membrane co-ion concentration (11) becomes

$$c_{\text{co}}^{\text{m}}(c_s; \chi(c_s)) = \frac{1}{2} \left[\sqrt{\chi(c_s)^2 + 4\delta^* c_s^2} - \chi(c_s) \right], \quad \widehat{\Delta c_{\text{m}}} = c_{\text{co}}^{\text{m}}(c_{\text{int}}; \chi(c_{\text{int}})) - c_{\text{co}}^{\text{m}}(c_{\text{p}}; \chi(c_{\text{p}})), \quad (28)$$

and for $\text{CaCl}_2/\text{LaCl}_3$ the charge coefficient in the cubic/quartic (12)–(13) is likewise $\chi(c_s)$; the transformed response Δc_{m} (built from c_{p} and J_{w}) is *unchanged*, so $\chi(c)$ only shifts the predictor. We expand $\chi(c) = \sum_k \beta_k B_k(c)$ in a **cubic B-spline** basis (a flexible curve built by blending a handful of simple local pieces; the coefficients β_k , called **control points**, are essentially the curve’s height at a few anchor concentrations, and the curve smoothly interpolates between them) with a small number of interior knots at quantiles of the observed c_{int} (no extrapolation beyond the data). Two properties make this clean: the basis is non-negative and sums to one, so constraining the control points $\beta_k \geq 0$ guarantees $\chi(c) \geq 0$ without an exp link; and a spline with all β_k equal reproduces the constant- χ model pointwise, so the constant fit is the exact *zero-interior-knot nested submodel* (i.e. the simpler constant- χ model is a special case of the flexible spline model, obtained by forcing all control points equal — which is what makes a direct statistical comparison between them valid) and an F -test/AIC comparison against it is valid. Estimation is joint nonlinear least squares in $(\beta, \delta^* \text{ or } K)$; the χ - δ^*/K confounding is broken by concentration *range* (high- c pins the partition constant, low- c curvature pins the charge), so a wide-range dataset is essential — the same identifiability-needs-range lesson as §5.2. A **pointwise band** on $\chi(c)$ — an error bar around the fitted curve’s height at each individual concentration, analogous to a Wald interval but recomputed at every point along the curve — follows from the Gauss–Newton covariance, $\text{Var}\{\hat{\chi}(c)\} = \mathbf{B}(c)^\top \text{Cov}(\hat{\beta}) \mathbf{B}(c)$. (This first cut uses an unpenalized few-knot spline and plain `scipy.least_squares` (`analysis/chi_spline/chispline_cacl2.py`). The natural robustness upgrades — a **roughness penalty (P-spline)** (an extra term in the fitting objective that discourages a wiggly curve unless the data really demand it, trading a little fit quality for a more stable, less overfit shape), the Pyomo/ParmEst nonlinear-constraint formulation of §4.1, and dataset pooling — are deliberately deferred (future work below). A spline is preferred over a global polynomial (local

support, tame extrapolation) and over committing to a specific **adsorption isotherm** (a mechanistic chemistry equation for how much of an ion sticks to a surface as a function of its concentration, e.g. Langmuir) up front: it lets the data reveal the shape, which can *then* motivate a mechanistic isotherm.)

6.2 Preliminary result: CaCl_2 MC5 05.27.26

We fit the widest-range CaCl_2 dataset (Estrada’s MC5 05.27.26, $n = 1645$, c_{int} to 116 mM; CP-corrected, §3) with 1 and 2 interior knots, against the constant- χ baseline (Table 14, Fig. 16).

Model	params	SSE	pseudo- R^2	AIC	multi-start % global
constant χ (= 2.95 mM)	2	418.3	0.959	−2248	—
1-knot $\chi(c)$ spline	6	52.6	0.9948	− 5653	100%
2-knot $\chi(c)$ spline	7	65.7	0.9936	−5283	13%

Table 14: CaCl_2 MC5 05.27.26 (CP-corrected): constant χ vs. a $\chi(c)$ B-spline. The 1-knot spline reduces SSE eight-fold and is overwhelmingly favored over the constant model (nested $F(4, 1639) = 2851$, $p \approx 0$; $\Delta\text{AIC} = -3405$) with 100% multi-start convergence. The 2-knot fit is *under-converged* (13% global; its SSE exceeds the 1-knot’s, impossible at a true optimum) — so the 1-knot model is the reliable first cut and the “does a second knot help?” question is not yet answerable here.

CaCl₂ $\chi(c)$ spline, first cut (MC5 05.27.26, CP-corrected)

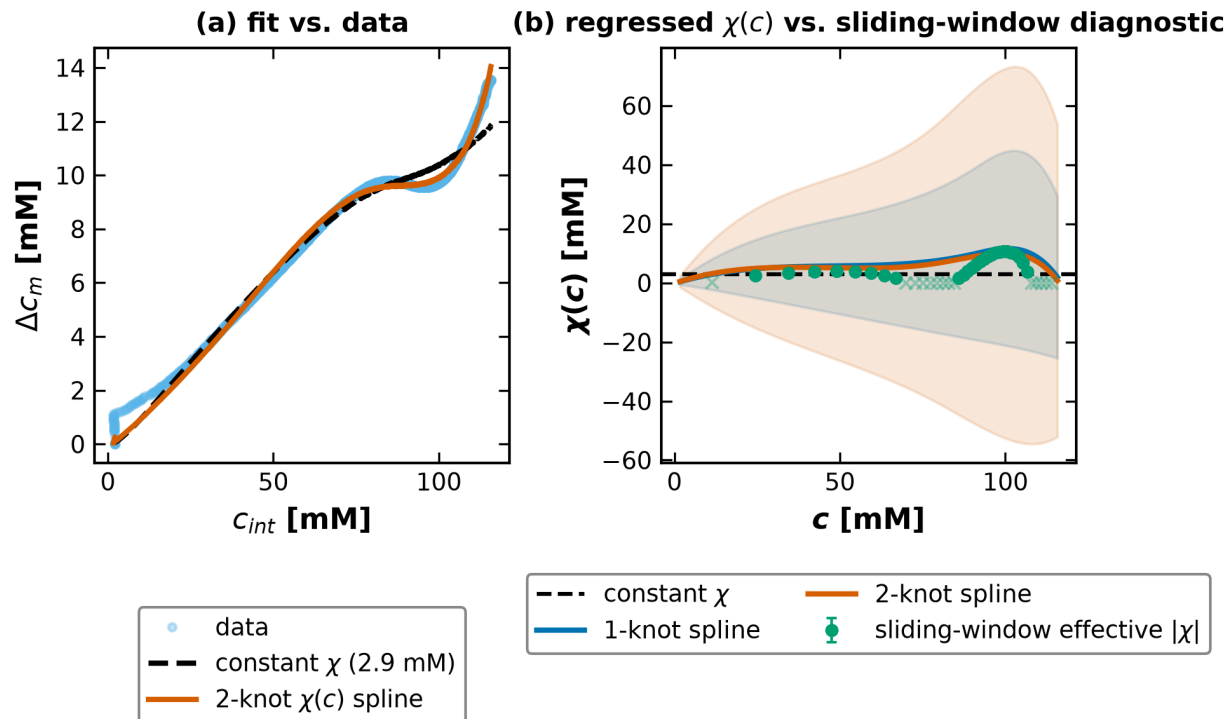


Figure 16: CaCl₂ $\chi(c)$ spline first cut. (a) The constant- χ model (dashed) misses the data’s smooth wave, which the spline follows. (b) The regressed $\chi(c)$ (with ± 1.96 pointwise SE bands) against the model-free sliding-window effective- $|\chi|$ (green). The point estimate tracks the diagnostic (correlation 0.83, RMSE 3.5 mM), but the bands are enormous — $\chi(c)$ is only weakly identified pointwise, especially at high c where the co-ion is insensitive to charge (the bands are a linearized Gauss–Newton approximation and ignore the $\chi \geq 0$ bound, hence stray negative).

What this does and does not show. The constant-charge model is **clearly inadequate** for CaCl₂: an eight-fold SSE reduction, an overwhelming F -test, and 100%-robust convergence say the data genuinely demand concentration-dependent structure, and the regressed $\chi(c_{int})$ tracks the independent windowed diagnostic. **But the $\chi(c)$ shape cannot yet be read as the adsorption isotherm**, for two reasons. First, $\chi(c)$ is poorly identified *as a function* — the pointwise bands span tens of mM (Fig. 16b), the fit is far better constrained than the underlying $\chi(c)$. Second, and more important, a **null check on NaCl** — the well-identified 1:1 case with *no* expected adsorption drift — fails: the same 1-knot machinery, refitting δ^* jointly, still improves the NaCl fit significantly (SSE 52.4 $\rightarrow$ 19.6, $F = 286$, $\Delta AIC = -671$) with a decidedly non-flat $\hat{\chi}(c)$ (control-point spread 62% of the mean). The likely mechanism is a **time–concentration confound within a single run**: under concentrating-lag diafiltration the feed concentration rises monotonically with time, so a smooth function of c is nearly interchangeable with a smooth function of *time*. An unpenalized $\chi(c)$ can therefore absorb any slowly-varying temporal structure in the residuals — in particular

NaCl’s strong AR(1) autocorrelation ($\hat{\phi} \approx 0.74$, §2.3) — whether or not it reflects adsorption chemistry, and the i.i.d. F -test/AIC used here do not know about that autocorrelation (the same anti-conservative trap §2.3 flagged for the constant- χ regions). The same flexibility likewise absorbs the calibration-source and transformed-vs-raw mean-model questions (§1.7, §2.4). So the CaCl_2 improvement is real — it robustly beats constant- χ and matches an independent local estimator — but this first cut **cannot yet rule out that the drift is partly or wholly a time-correlated-noise artifact**, the “clean mean model first” risk this extension carries. It is suggestive, not yet a green light for the full treatment.

AR(1)-whitened follow-up (outcome). The cheap, decisive test proposed above has now been run (`analysis/chi_spline/chispline_ar1_whitened.py`): re-estimate $\hat{\phi}$ from the constant- χ model’s time-ordered residuals (Cochrane–Orcutt, §2.3’s Method A convention), whiten *both* the constant and 1-knot-spline residual functions with that *same* common $\hat{\phi}$ (a fair, one-noise-model comparison), and redo the nested F -test on the whitened SSE with the actual $n - p$ df.

Dataset	Regime	$\hat{\phi}$	F	p	ΔAIC	$\chi(c)$ spread
NaCl	i.i.d.	—	286.0	≈ 0	−671	—
NaCl	whitened	0.747	27.8	≈ 0	−96	53% of χ_0 (not flat)
CaCl_2	i.i.d.	—	2851.5	≈ 0	−3404	—
CaCl_2	whitened	0.999	57.6	≈ 0	−208	$563 \times \chi_0$ (degenerate)

Table 15: AR(1)-whitened constant-vs-1-knot-spline comparison, common $\hat{\phi}$ per dataset (from the constant model, Cochrane–Orcutt-converged), actual $n - p$ df. NaCl: $n = 692$, $\hat{\phi}_{\text{OLS}} = 0.743 \rightarrow 0.747$ CO-converged (matches §2.3’s 0.743 for the same primary dataset almost exactly, as expected — same residual series). CaCl_2 : $n = 1645$, $\hat{\phi}_{\text{OLS}} = 0.995 \rightarrow 0.999$ CO-converged — a **new** number (the autocorrelation analysis of §2.3 only characterized NaCl).

Neither branch of the proposed decision rule cleanly holds. NaCl’s spurious drift is substantially *weakened* by whitening ($F : 286 \rightarrow 28$, $\Delta\text{AIC} -671 \rightarrow -96$) but does **not evaporate** ($p \approx 0$ still, control-point spread still 53% of χ_0) — AR(1) alone does not fully explain the confound, consistent with §2.3’s own finding that AR(1) is a large first-order correction but *not fully sufficient* for this residual series (16 of 30 ACF lags still exceed the band after whitening). CaCl_2 ’s side of the test is worse than inconclusive — it is **numerically degenerate**: $\hat{\phi}$ is already 0.995 from the raw OLS residuals (*before* any Cochrane–Orcutt iteration), i.e. far closer to a unit root than NaCl’s 0.743, and the whitened $\chi(c)$ blows up to $\sim 500\text{--}1700\text{ mM}$ (vs. a 2.95 mM constant estimate) — physically meaningless. The likely reason: CaCl_2 ’s constant- χ model is not just noisy but *wrongly shaped* (that is the entire premise of this section), so its OLS residuals are dominated by the smooth, large-amplitude systematic wave that the spline is built to capture, not by stationary AR(1) noise — fitting an AR(1) process to residuals from a badly mis-specified mean model is close to circular, and whitening at the resulting near-unit-root $\hat{\phi}$ over-differences the signal along with any noise. That $\hat{\phi}$ contrast is itself weakly informative, though: CaCl_2 ’s constant- χ residuals are far more autocorrelated than NaCl’s ($\hat{\phi}_{\text{OLS}} = 0.995$ vs. 0.743) — dominated by systematic structure rather than stationary noise, consistent with (but not proof of) genuine concentration-dependence, since a time-smooth calibration or temperature artifact would leave the same signature. **Verdict: the confound is not resolved by simple AR(1) whitening.** For NaCl it survives (weakened, not gone); for CaCl_2 the test cannot even be run cleanly, because the very mis-specification under investigation contaminates the noise-model estimate needed to correct for it. This rules out the

“AR(1) alone fixes it” branch and points — as anticipated — toward **pooling across differently-paced CaCl_2 runs** (time and concentration decorrelate *across* runs even though they are collinear within one) as the next step, rather than a richer single-run noise model.

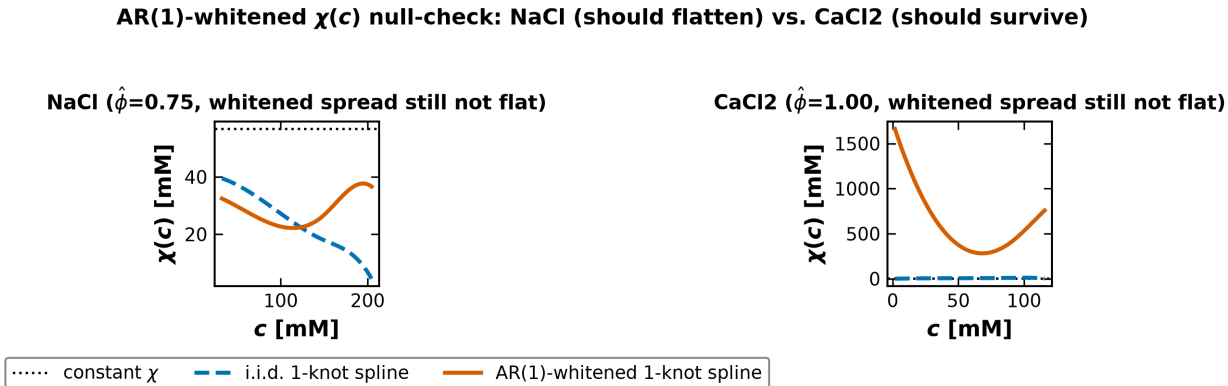


Figure 17: AR(1)-whitened $\chi(c)$ (orange) vs. the i.i.d. fit (blue dashed) and the constant estimate (black dotted), common $\hat{\phi}$ per dataset. **NaCl** (left): whitening changes the shape but does not flatten it — still not the flat line a true null result would show. **CaCl_2** (right): the whitened curve is off by two to three orders of magnitude from the constant estimate, the numerical signature of whitening at $\hat{\phi} \approx 0.999$ described above — not a usable estimate of $\chi(c)$, and not evidence for or against real adsorption structure.

6.3 Recommended future work

- **Regularize.** Add a roughness penalty (P-spline) with a cross-validated smoothing parameter; the 1-knot wiggle and the 2-knot non-convergence both signal an under-constrained fit. This tames the pointwise bands and makes “how many knots” well-posed.
- **Disentangle before interpreting — outcome of the cheap, decisive test.** The AR(1)-whitened re-run (§6.2, “AR(1)-whitened follow-up”) has now been done, and **neither branch of the original either/or held**: NaCl’s spurious drift is weakened but does not evaporate under whitening, and CaCl_2 ’s own AR(1) diagnostic turns out to be numerically degenerate (its constant- χ residuals are dominated by the systematic mis-specification under test, not stationary noise, so $\hat{\phi} \rightarrow 0.999$ and the whitened $\chi(c)$ blows up). A single-run AR(1) correction is therefore **not sufficient** to settle this either way. The confound must instead be broken by **pooling across differently-paced runs** (time and concentration decorrelate *across* runs even though they are collinear within one), the eventual multi-dataset plan — now the load-bearing next step, not an alternative to a richer single-run noise model. The calibration-source and errors-in-variables questions (§1.7, §2.4) should be settled in the same spirit before the shape is read as adsorption.
- **Generalize the numerics and UQ.** Move to the Pyomo/ParmEst nonlinear-constraint formulation (§4.1) for the quartic LaCl_3 case and for proper profile/bootstrap intervals on $\chi(c)$; pool the three CaCl_2 datasets for a shared $\chi(c)$.
- **Then, mechanism.** If a robust, penalized, multi-dataset $\chi(c)$ survives, fit a physical adsorption isotherm (e.g. Langmuir) to it for a few-parameter mechanistic interpretation — the spline is the discovery tool, the isotherm the explanation.

7 Analysis plan with Claude Code

This document and the accompanying code were produced with Claude Code using a deliberate two-model division of labor.

Orchestration (Opus). A single “coordinator” session (Claude Opus) built scientific context (reading the DATA1/DATA2/Soft-Sensor papers [1, 2, 3], Laurianne’s scanned Donnan notes [5, 6, 7], Bill’s MATLAB [8], and the group’s Word-document derivations [9, 10, 11, 12, 20, 21, 14] — the last extracted to L^AT_EX via `pandoc`), inventoried the shared Box folder, and *reconciled* the mathematics across these sources (sign conventions, the δ vs. δ^* notation, and the factor-of-2 issue). It maintains this consolidated writeup, decides modeling conventions, and drafts well-scoped task prompts (saved under `docs/prompts/`).

Implementation (Sonnet). Separate “worker” sessions (Claude Sonnet) executed the well-scoped tasks, growing well beyond the original Step-2/3 scope: the `scipy` port and ParmEst cross-validation (multi-start diagnostics, nonlinear confidence regions); the raw→processed preprocessing pipeline (§1.6) and its validation against six previously raw-only NaCl runs; the CaCl₂ and LaCl₃ analyses (cubic/quartic root-solves, nonlinear-constraint ParmEst formulations, adsorption-hypothesis diagnostics); a whole-report figure-compliance audit (font scale, bold math, boxed legends, §1.2-adjacent presentation fixes); the AR(1) autocorrelation correction and its follow-up double-counting fix (§2.3); the raw-measurement reformulation (§2.4); and this cross-document consistency/synthesis pass. The PI runs these sessions and passes their output back to the coordinator for review and integration — keeping a human checkpoint between steps.

Reproducibility. All analysis runs in a documented conda environment (`environment.yml`, `data3-regression`); code lives under `analysis/step*/` with per-step `README.md` files; figures are regenerated into `docs/reports/figures/`. Project status has also been presented to both groups [22, 23, 24].

Lessons learned (so far).

- Having the stronger model *reconcile multi-source mathematics* before any coding caught real discrepancies (factor of 2, sign conventions) that would otherwise have propagated into the regression.
- Read-only “inventory” subagents efficiently mapped a large (~ 683 file) shared drive and extracted equations from Word documents.
- Delegating implementation with a *self-contained* prompt (context + data paths + exact deliverables + what to report) let the worker sessions run end-to-end and return paste-ready numbers.
- Cross-validating two independent implementations (`scipy` vs. ParmEst) surfaced subtle tool traps (e.g. ParmEst’s `measurement_error` covariance scaling) and gave confidence in the estimates.
- An Opus audit of the AR(1) work (§2.3) caught a **real statistical error** before it reached collaborators: an earlier draft combined GLS/whitening with an n_{eff} degrees-of-freedom correction, double-counting the same autocorrelation and inflating the reported profile CI to a spurious $\sim 8\times$ (vs. the correct $\sim 2.7\text{--}2.9\times$ non-CP at the time, now $\sim 2.5\text{--}2.7\times$ CP-corrected).

Catching this required recognizing that two statistically valid corrections for the *same* effect are alternatives, not complements — a subtlety easy to miss when each piece is individually correct.

- Iterative report-refinement and figure-compliance rounds (stale table/figure fixes, font-scale/legend/bold-math audits across every figure) were as important as the initial analysis for producing a document a collaborator can actually read and trust — worth budgeting for explicitly, not treating as an afterthought.

8 Appendix: Per-Dataset NaCl Detail

(Not yet refreshed to the CP-corrected convention: every number in this appendix is still the non-CP value (predating the CP-corrected regression re-run); §2.5.1’s summary table and figures ARE CP-corrected (primary). Refreshing all nine individual write-ups below is flagged as remaining work, not silently stale.) Full preprocessing and fit/residual detail for every one of the nine NaCl runs summarized in §2.5.1, in the same order as Table 9. This appendix is intended as a complete reference for each individual run (favoring completeness over brevity, per this document’s purpose as an internal working record) — §2.5.1 is the place to read for the cross-dataset synthesis. For the three originally-processed runs, only the fit + residuals panels apply (no reconstruction was needed); for the six reconstructed raw-only runs, the calibration fit, $J_w(t)$, and reconstructed $c_{\text{int}}(t)$, $c_p(t)$ panels are also shown.

8.1 MC2 (05.07.24)

Arrived pre-processed (Src [A], Table 9); $n = 482$. Corrected model: $|\chi| = 18.46$ (95% CI 18.06–18.87) mM, $\delta^* = 0.2888$ (CI 0.2857–0.2920), SSE = 4.47, 100% multi-start convergence. This is the narrow-concentration-range run ($c_{\text{int}} \lesssim 44$ mM) discussed in §2.5.1 as a likely case of weak curvature/identifiability rather than a different membrane.

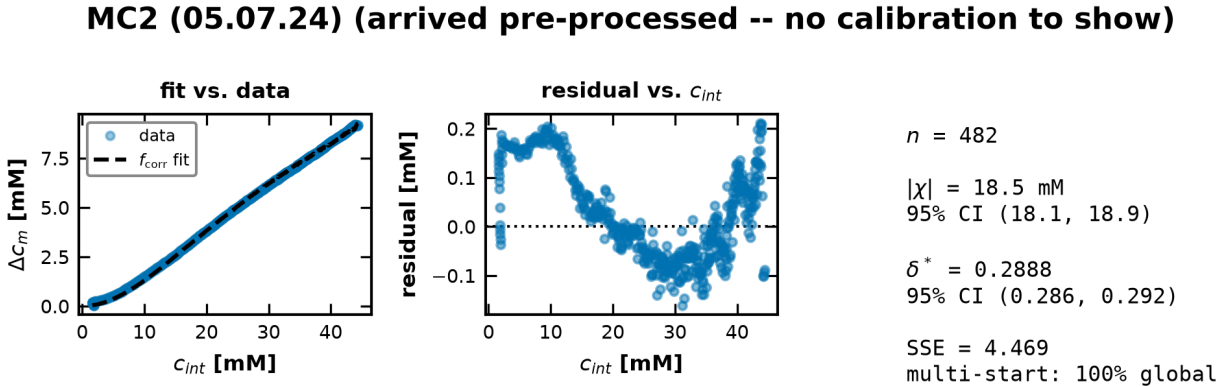


Figure 18: MC2 (05.07.24): fit vs. data, residual vs. c_{int} , and parameter summary.

8.2 MC5 (05.27.26)

Arrived pre-processed (Src [A]); $n = 747$. Corrected model: $|\chi| = 49.56$ (CI 48.58–50.55) mM, $\delta^* = 0.3251$ (CI 0.3232–0.3270), SSE = 130.07, 100% multi-start convergence. Matches its repeat (§8.3) almost exactly.

MC5 (05.27.26) (arrived pre-processed -- no calibration to show)

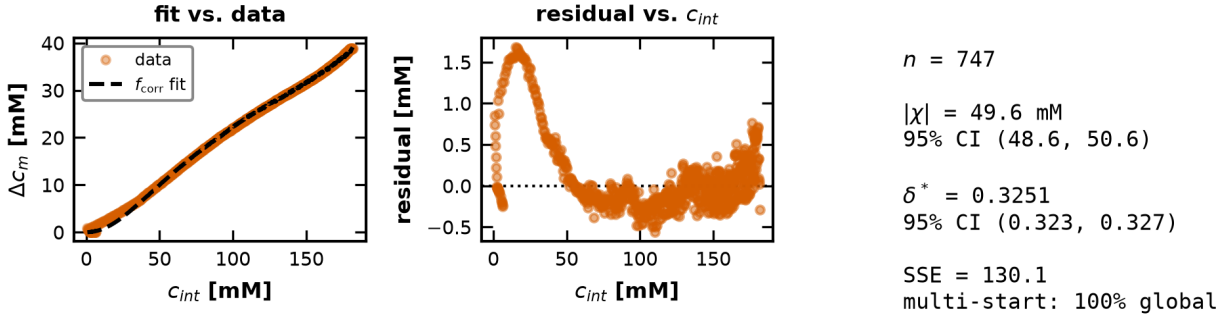


Figure 19: MC5 (05.27.26): fit vs. data, residual vs. c_{int} , and parameter summary.

8.3 MC5 (05.27.26) (2)

Arrived pre-processed (Src [A]); $n = 747$. This is the **primary** NaCl dataset analyzed in §2.1–2.2. Corrected model: $|\chi| = 49.55$ (CI 48.58–50.54) mM, $\delta^* = 0.3251$ (CI 0.3232–0.3270), SSE = 129.62, 100% multi-start convergence.

MC5 (05.27.26) (2) (arrived pre-processed -- no calibration to show)

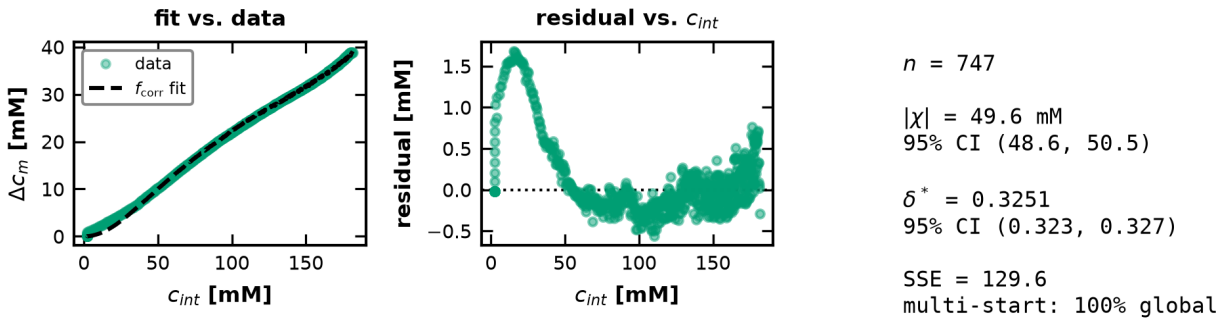


Figure 20: MC5 (05.27.26) (2): fit vs. data, residual vs. c_{int} , and parameter summary.

8.4 MC3 (07.09.24)

Reconstructed from raw (§1.6); per-experiment calibration $\text{cond.} = 145.0 + 98.0c$ ($R^2 = 0.9951$, $n = 14$ vial points), consistent with the other five reconstructed runs (§1.6 item 3). $n = 391$ valid rows after preprocessing. **This run does not converge:** the reconstructed c_{int} , c_p show a near-zero apparent rejection over most of the run (visible in panel (c) below, where the two curves nearly overlap), which starves the 2-parameter Donnan model of curvature and lets **least_squares** wander to a nonphysical optimum ($|\chi| \rightarrow 2.1 \times 10^6$ mM). This is flagged (not silently reported) as an ill-conditioned fit in Table 10.

MC3 (07.09.24)

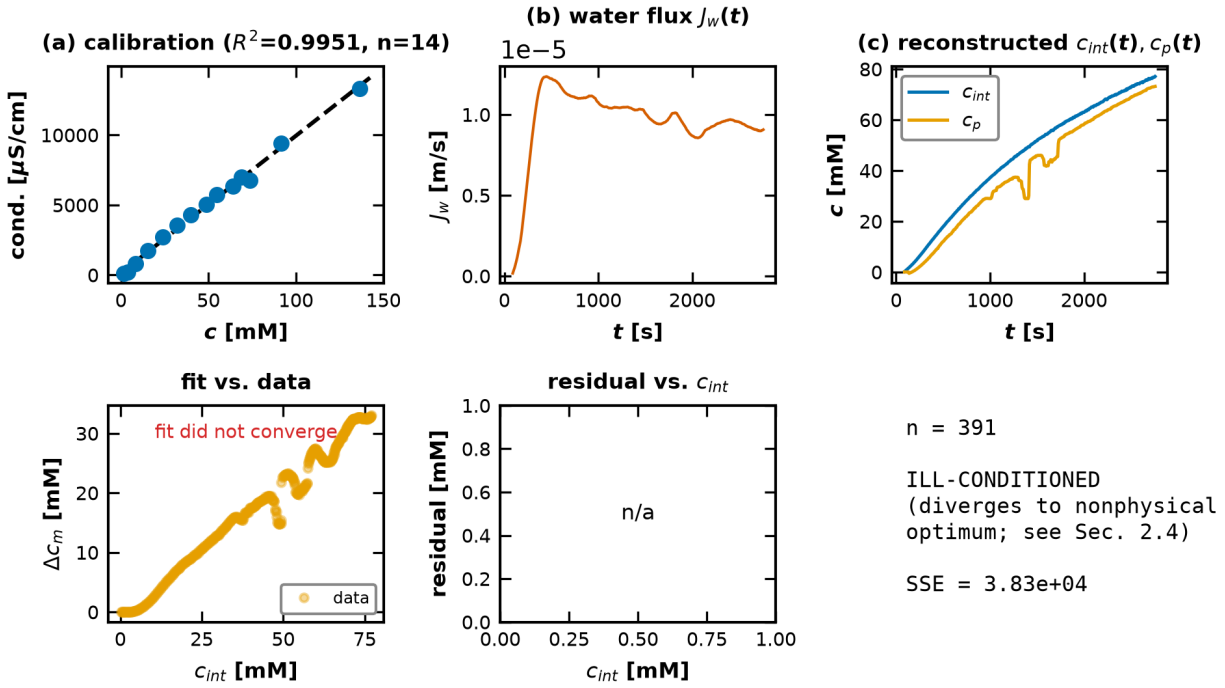


Figure 21: MC3 (07.09.24): calibration, $J_w(t)$, reconstructed $c_{int}(t), c_p(t)$, fit vs. data (does not converge), residual vs. c_{int} (n/a), and parameter summary.

8.5 MC3 (07.22.24, S)

Reconstructed from raw; per-experiment calibration $\text{cond.} = 88.8 + 96.6 c$ ($R^2 = 0.9995$, $n = 14$). $n = 1312$ valid rows (a long sequential/reverse “diluting” run). Corrected model: $|\chi| = 20.33$ (CI 12.44–31.91) mM, $\delta^* = 0.3000$ (CI 0.2484–0.3651), SSE = 101995, 99% multi-start convergence. Poorly identified (95% CI spans nearly $2\times$); the fit-vs-data panel below shows the residual structure at high c_{int} that later shows up as extreme startup-trim sensitivity (§2.5.1).

MC3 (07.22.24, S)

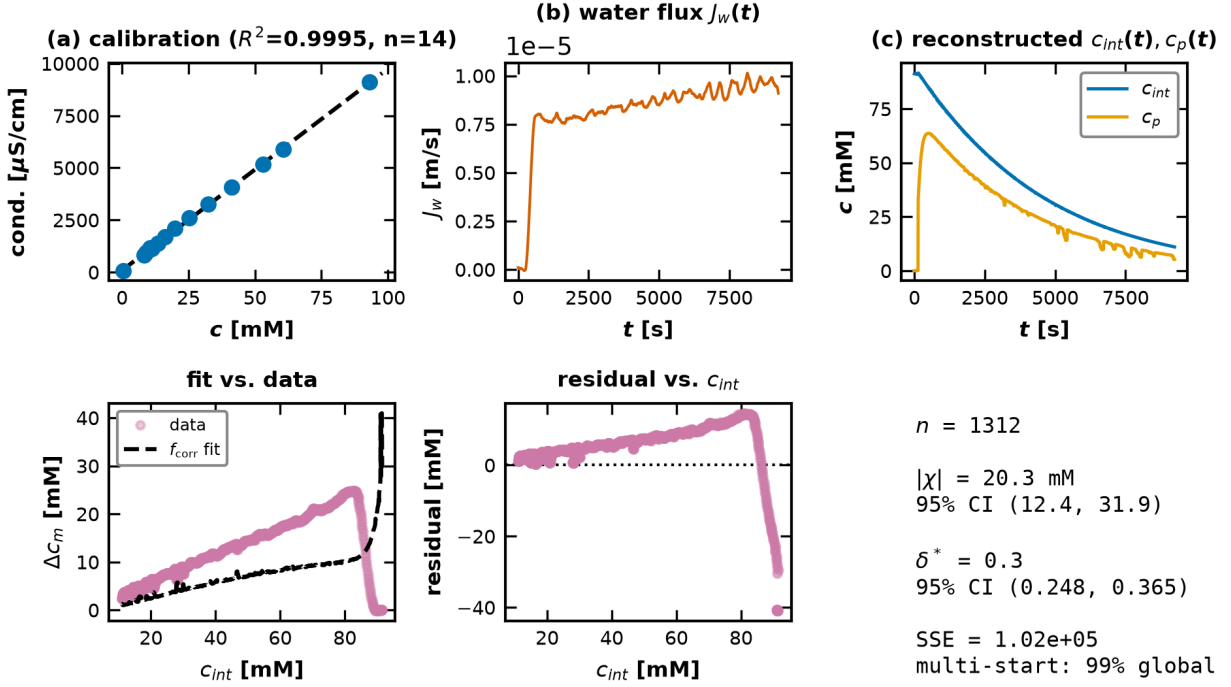


Figure 22: MC3 (07.22.24, S): calibration, $J_w(t)$, reconstructed $c_{int}(t), c_p(t)$, fit vs. data, residual vs. c_{int} , and parameter summary.

8.6 MC4 (07.11.24, S)

Reconstructed from raw; per-experiment calibration $\text{cond.} = 17.4 + 95.4c$ ($R^2 = 0.9971$, $n = 13$ vial points – one fewer than the other five runs, since this run’s “Final Tube” vial is missing a conductivity reading, but the fit quality is unaffected, §1.6). $n = 317$ valid rows. Corrected model: $|\chi| = 106.63$ (CI 94.46–120.06) mM, $\delta^* = 1.9313$ (CI 1.8520–2.0214), SSE = 447.0, 100% multi-start convergence. Both parameters are far outside the range of every other NaCl run (Fig. 9b) – a real outlier among the reconstructed runs, not just a wide-CI one.

MC4 (07.11.24, S)

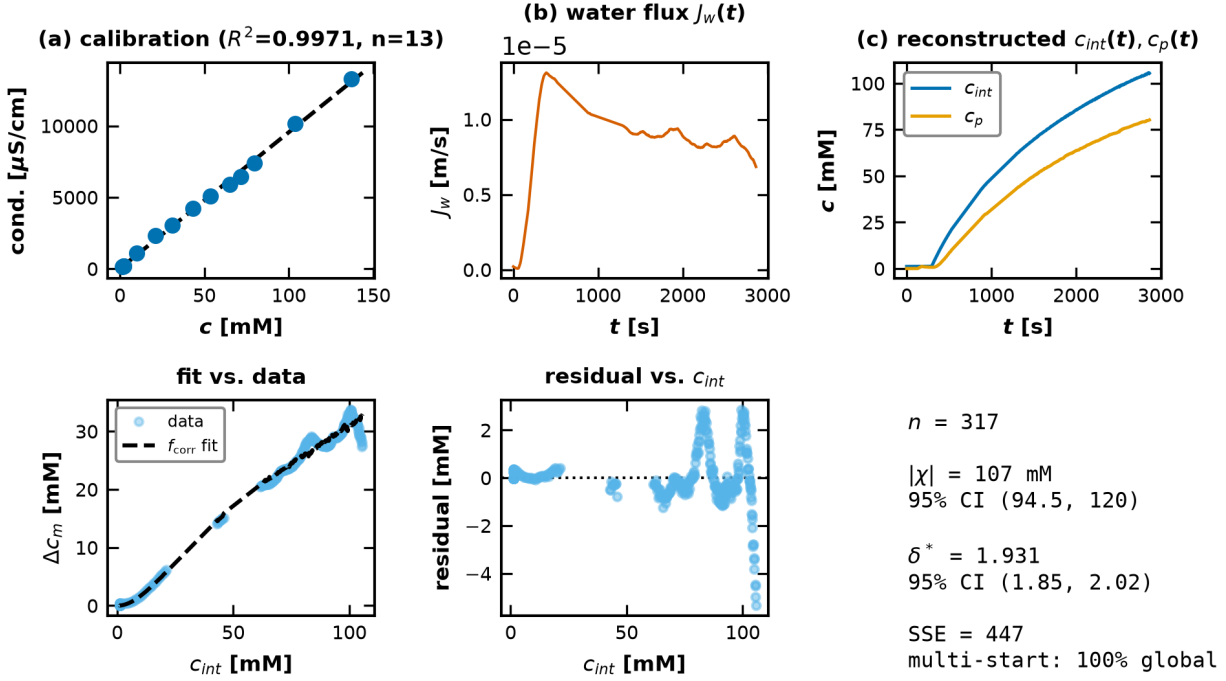


Figure 23: MC4 (07.11.24, S): calibration, $J_w(t)$, reconstructed $c_{\text{int}}(t), c_p(t)$, fit vs. data, residual vs. c_{int} , and parameter summary.

8.7 MC5 (07.23.24)

Reconstructed from raw; per-experiment calibration $\text{cond.} = 111.3 + 93.0 c$ ($R^2 = 0.9962$, $n = 14$) – this is the calibration shown as the representative example in Fig. 1a. $n = 496$ valid rows. Corrected model: $|\chi| = 61.99$ (CI 60.53–63.49) mM, $\delta^* = 0.7035$ (CI 0.6966–0.7106), SSE = 52.9, 100% multi-start convergence. One of the better-behaved reconstructed runs (95% CI width < 5%, hence included in the well-identified zoom of Fig. 9a) but still $\sim 25\%$ off MC5 (05.27.26)'s $|\chi|$ – a reconstruction-pipeline discrepancy, not a data-quality one for this specific run.

MC5 (07.23.24)

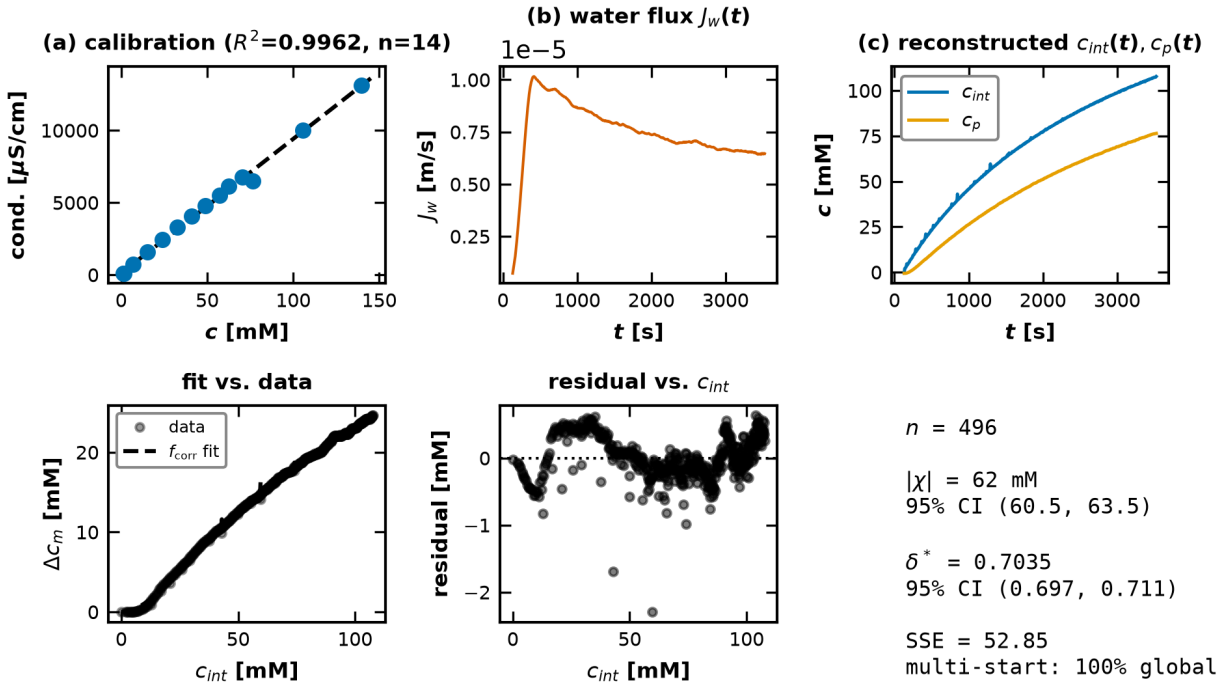


Figure 24: MC5 (07.23.24): calibration, $J_w(t)$, reconstructed $c_{\text{int}}(t), c_p(t)$, fit vs. data, residual vs. c_{int} , and parameter summary.

8.8 MC5 (07.23.24, S)

Reconstructed from raw; per-experiment calibration $\text{cond.} = 177.0 + 93.2 c$ ($R^2 = 0.9965$, $n = 14$). $n = 500$ valid rows. Corrected model: $|\chi| = 169.15$ (CI 88.07–534.92) mM, $\delta^* = 0.7147$ (CI 0.4415–1.9529), SSE = 11690, 100% multi-start convergence. **The worst-identified run in the whole set:** the 95% CI on $|\chi|$ spans a factor of $6\times$. The fit-vs-data panel below shows why – a pronounced high- c_{int} divergence (the reverse/diluting tail) that the 2-parameter model cannot track.

MC5 (07.23.24, S)

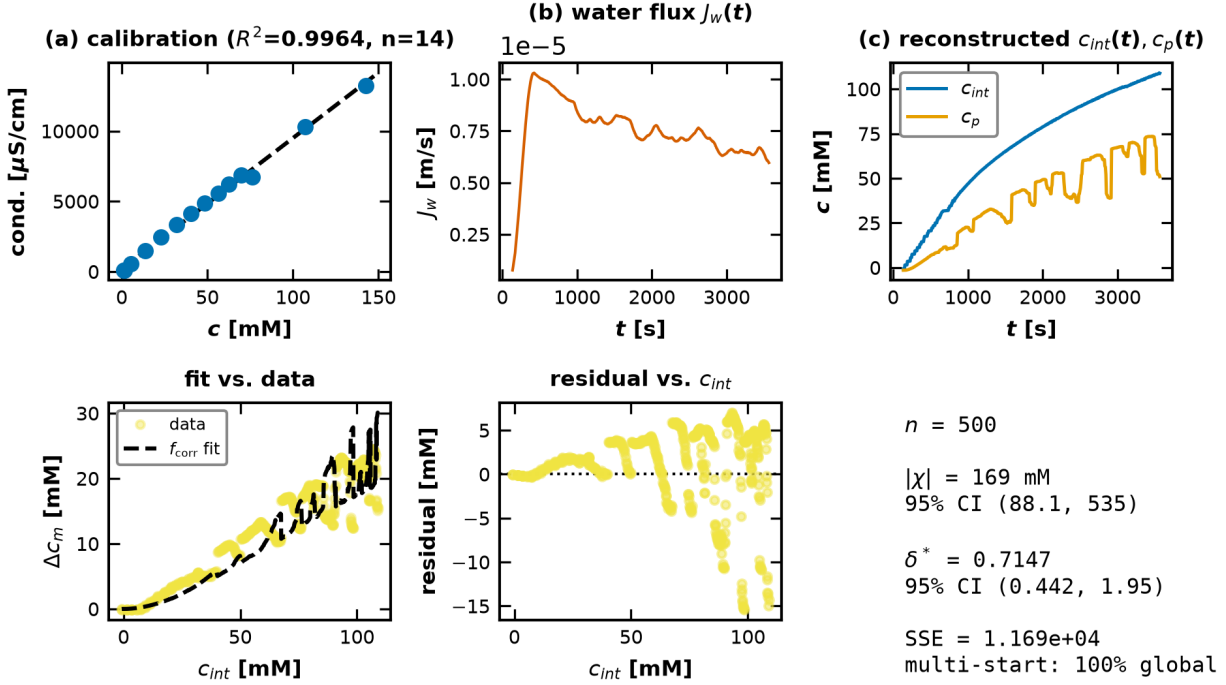


Figure 25: MC5 (07.23.24, S): calibration, $J_w(t)$, reconstructed $c_{\text{int}}(t)$, $c_p(t)$, fit vs. data, residual vs. c_{int} , and parameter summary.

8.9 MC5 (07.23.24, S2)

Reconstructed from raw; per-experiment calibration $\text{cond.} = 136.7 + 98.4 c$ ($R^2 = 0.9991$, $n = 14$). $n = 1324$ valid rows (the longest run, another sequential/reverse “diluting” protocol). Corrected model: $|\chi| = 35.63$ (CI 24.45–52.10) mM, $\delta^* = 0.3434$ (CI 0.2801–0.4312), SSE = 68556, 100% multi-start convergence – the only reconstructed run whose 95% CI overlaps MC5 (05.27.26)’s. Like MC5 (07.23.24, S) above, the fit-vs-data panel shows a high- c_{int} “hook” where the reverse/diluting protocol’s data deviates from the corrected model; this is the shared failure mode of both S/S2 runs (§2.5.1).

MC5 (07.23.24, S2)

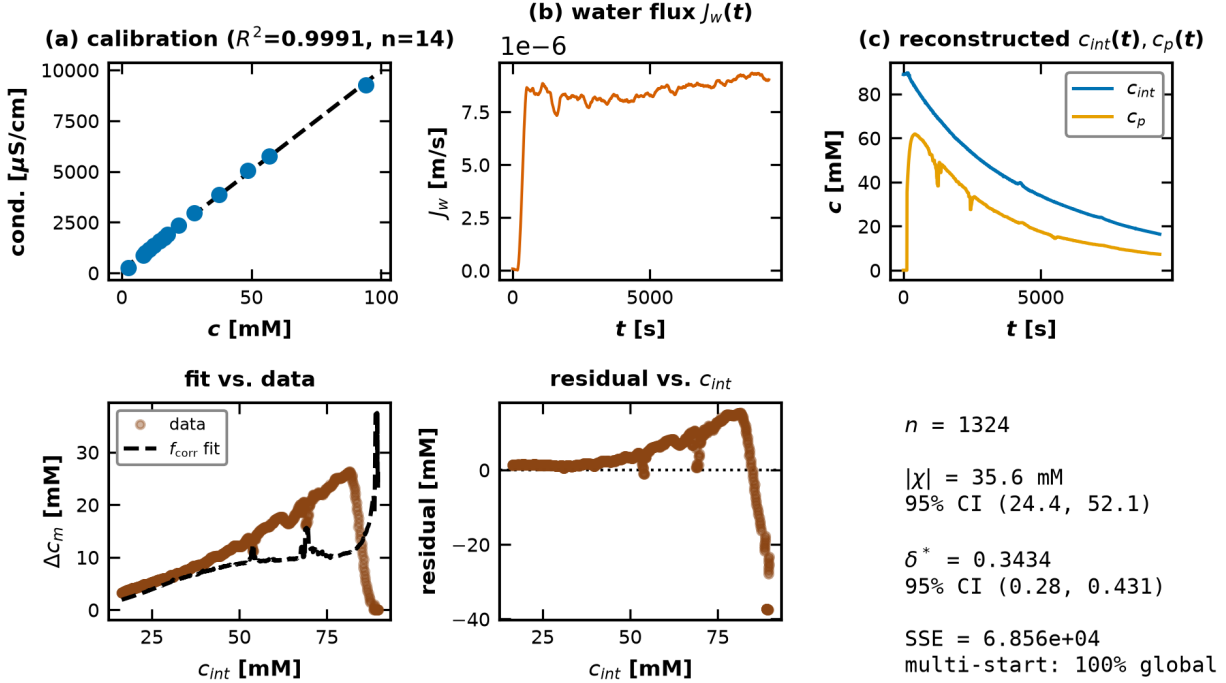


Figure 26: MC5 (07.23.24, S2): calibration, $J_w(t)$, reconstructed $c_{int}(t)$, $c_p(t)$, fit vs. data, residual vs. c_{int} , and parameter summary.

9 Appendix: Per-Dataset CaCl_2 Detail

(Not yet refreshed to the CP-corrected convention: every number in this appendix is still the non-CP value (predating the CP-corrected regression re-run); §3.2's summary table and figures ARE CP-corrected (primary). Refreshing all three individual write-ups below is flagged as remaining work, not silently stale.) Full fit/residual/drift detail for every one of the three processed CaCl_2 runs summarized in §3.2, in the same order as Table 11. As with §8, this appendix is a complete per-run reference (favoring completeness over brevity); read §3.2–3.3 for the cross-dataset synthesis. Each figure shows (a) the corrected-model fit vs. data, (b) the residual vs. c_{int} (with its sign-runs z -score), (c) the sliding-window effective $|\chi|(c_{int})$ diagnostic, and (d) the parameter/CI/fit-quality summary.

9.1 MC3 (07.11.24, S)

$n = 609$, $c_{int} \in [1.0, 47] \text{ mM}$ (the narrowest-range CaCl_2 run). Bounded fit: $|\chi| \approx 0$ (95% CI 0–204.9 mM – very poorly identified, the narrow concentration range gives weak curvature), $K = 0.02374$ (CI 0.02318–0.02432), SSE = 388.3, pseudo- $R^2 = 0.945$, 100% multi-start convergence. Residual sign-runs $z = -23.6$ (strongly non-random). Of its 14 sliding windows, only 5 are reliably identified (large-SE/boundary-pinned windows excluded from the trend), rising from ~ 4 to $\sim 18 \text{ mM}$ over $c_{int} = 35\text{--}47 \text{ mM}$ – consistent with sitting on the rising leg of the same shape the wider-range MC5 runs show in full (§3.3).

MC3 (07.11.24, S)

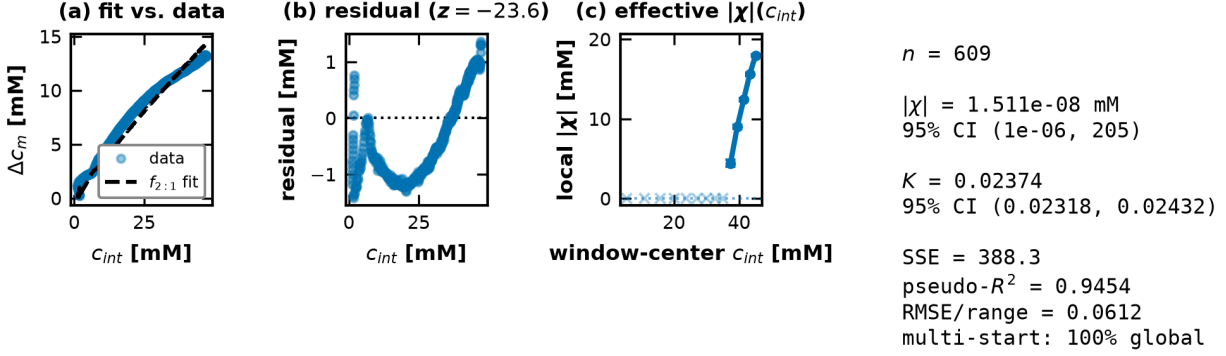


Figure 27: MC3 (07.11.24, S): fit vs. data, residual vs. c_{int} , effective $|\chi|(c_{int})$, and parameter summary.

9.2 MC5 (05.27.26)

$n = 1645$, $c_{int} \in [1.6, 108]$ mM. Bounded fit: $|\chi| = 0.0176$ (95% CI 0–0.369 mM), $K = 0.00080$ (CI 0.00079–0.00081), SSE = 643.2, pseudo- $R^2 = 0.937$, 96% multi-start convergence. Residual sign-runs $z = -39.8$. The effective- $|\chi|$ window fit is well identified (17 of 40 windows reliable) and shows the inverted-U rise-to- ~ 13 mM-then-fall pattern in full (§3.3). Matches its near-repeat (§9.3) closely in point estimate and diagnostic shape.

MC5 (05.27.26)

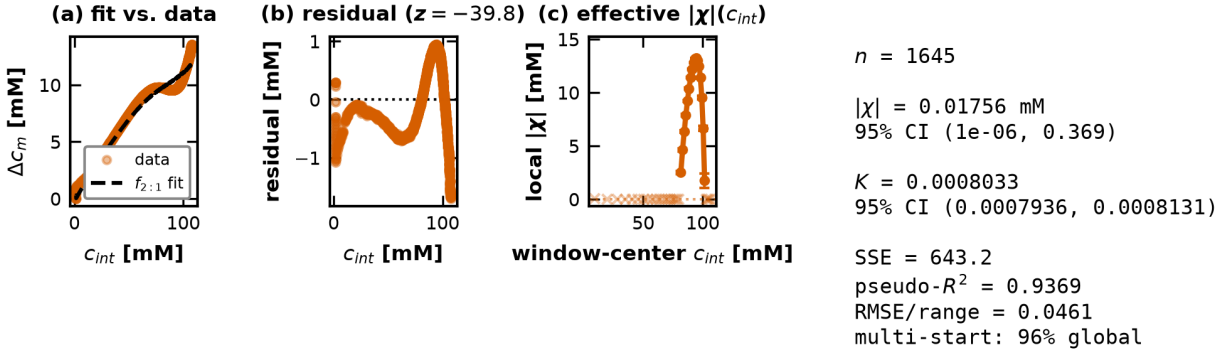


Figure 28: MC5 (05.27.26): fit vs. data, residual vs. c_{int} , effective $|\chi|(c_{int})$, and parameter summary.

9.3 MC5 (06.27.26)

$n = 1885$, $c_{int} \in [1.3, 103]$ mM; a near-repeat of MC5 (05.27.26) (§9.2) on the same cut. Bounded fit: $|\chi| = 0.0121$ (95% CI 0–0.284 mM), $K = 0.00032$ (CI 0.00031–0.00032), SSE = 1003.8, pseudo- $R^2 = 0.838$ (visibly worse than 05.27.26's 0.937), 99% multi-start convergence. Residual sign-runs $z = -42.7$. The effective- $|\chi|$ window fit (21 of 46 reliable) shows the same qualitative inverted-U shape as 05.27.26 but with a sharper high- c_{int} drop, mirrored in panel (a)'s fit-vs-data – worth

a follow-up look at whether this run has more noise or a genuinely different high-concentration behavior (e.g. a different membrane fouling state on that date; §3.3).

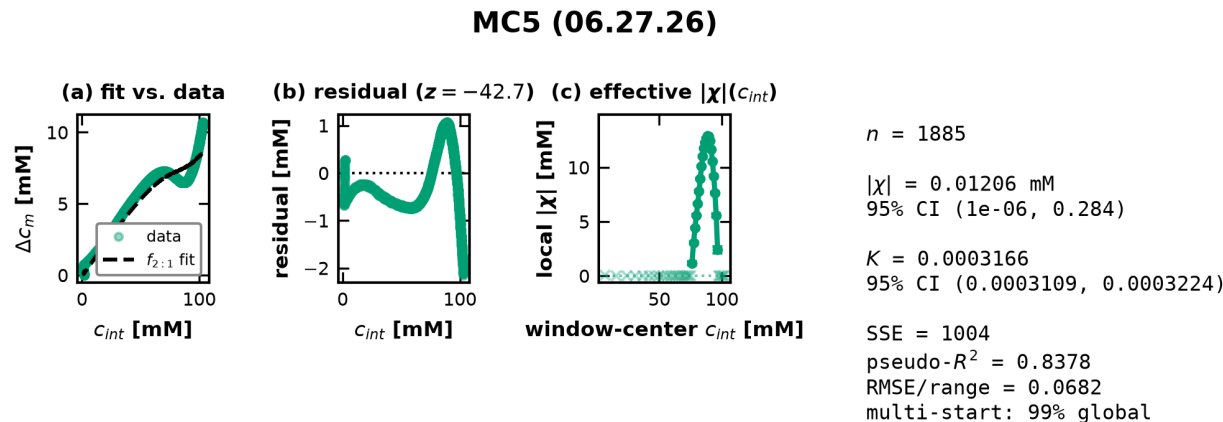


Figure 29: MC5 (06.27.26): fit vs. data, residual vs. c_{int} , effective $|\chi|(c_{int})$, and parameter summary.

10 Appendix: Per-Dataset LaCl_3 Detail

Full fit/residual/drift detail for the single processed LaCl_3 run summarized in §4.2. As with §8/ 9, this appendix is a complete reference (favoring completeness over brevity); read §4.2–4.3 for the synthesis. The figure shows (a) the corrected-model fit vs. data, (b) the residual vs. c_{int} (with its sign-runs z -score), (c) the sliding-window effective $|\chi|(c_{int})$ diagnostic, and (d) the parameter/CI/fit-quality summary.

10.1 MC2 (05.21.24)

$n = 623$, $c_{int} \in [2.0, 27.8]$ mM (a narrow range, the same identifiability headwind flagged for NaCl ’s MC2 and CaCl_2 ’s MC3). Bounded fit: $|\chi| = 0$ (95% CI 0–94.4 mM – pinned at the lower bound), $K = 8.758 \times 10^{-8}$ (CI (8.298, 9.236) $\times 10^{-8}$), SSE = 9.677, pseudo- $R^2 = 0.803$, 100% multi-start convergence. Cross-validated against Method B (ParmEst nonlinear-constraint quartic, §4.2): K agrees to 5 significant figures, SSE to $< 0.01\%$. Residual sign-runs $z = -24.9$ (strongly non-random, the most extreme of any single-salt dataset in this report). Of 19 sliding windows, only 7 (all at $c_{int} \gtrsim 21$ mM) are reliably identified, rising from ≈ 0.06 to ≈ 0.85 mM – LaCl_3 ’s effective $|\chi|$ never approaches even CaCl_2 ’s already-small estimates.

MC2 (05.21.24)

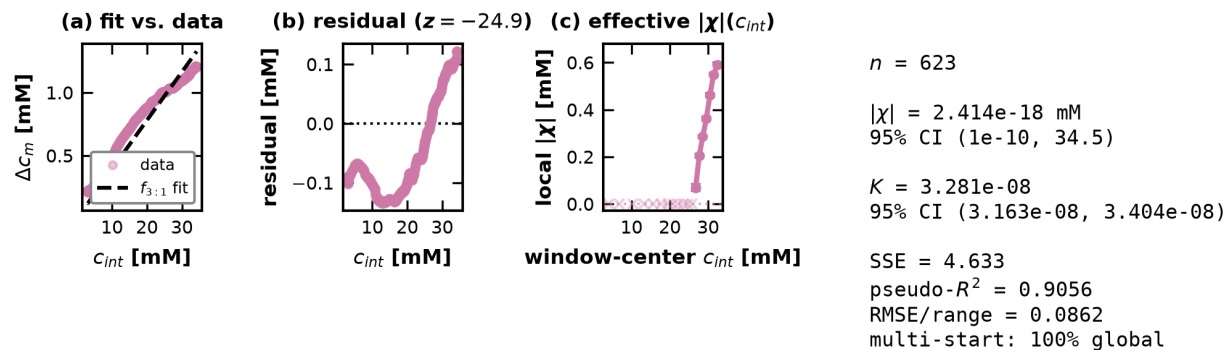


Figure 30: MC2 (05.21.24): fit vs. data, residual vs. c_{int} , effective $|\chi|(c_{int})$, and parameter summary.

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